ONTOLOGY OF NOTHING

Monograph

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ABSTRACT
Background: the study addresses the fundamental aporia between the absolute immutability of the unconditioned and the changeability of encountered reality.
Problem: existing philosophical and scientific models offer no logically consistent derivation of distinction and conditionedness free of hidden assumptions and of a second principle.
Method: ontological reduction without symbolic formalization, which fixes the dependent structure without introducing becoming or transitions into the unconditioned. The course of the derivation: encountered reality as conditioned → unconditionedness as its limit → the givenness of the unconditioned in the conditioned → Nothing (absolute unconditionedness), immutability, uniqueness, transparency → self-understanding (Selfhood) → information as the conditioned in which the unconditioned is given → the initiation of Selfhood → distinction, which is change → consciousness → the directedness of the removal (programme) and the limit of the path → elementary changes and measures → interaction and bond → aggregated forms → emergence → matter → life → reason and intellect → stages and sifting.
Results: a structure of being is obtained in which changeability, multiplicity and consciousness belong to dependent information and proceed by the removal of opacity from Selfhood without violating the absoluteness of the unconditioned. The unconditioned is given in the conditioned as limit and is itself determined by nothing. Derived are: the finiteness of distinctions and the attainability of self-transparency; the absence of an uncaused outcome; the identity of distinction and lived experience; the transition to the living; the stages of the path and the sifting of the intellectual *I*. The necessity of the structure is asserted on a single accepted position about reality, named explicitly in the text: encountered change; also accepted explicitly are the definitions, the finitist understanding of the determinacy of number and the rule of method.
Contribution: the classical aporia of transition is removed; distinction is derived as knowledge of difference proceeding by the removal of opacity, and the dependent categories are derived from it; the consequences that physics and biology can test only in part are stated, as well as what the system does not derive.
Implications: the monograph offers a system of ontological levels that allows physics, cosmology, consciousness and the limit to be rethought as forms of an informational structure dependent on the unconditioned.

KEYWORDS:

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Nothing; Unconditionedness; Conditionedness; Givenness; Dependence; Immutability; Uniqueness; Transparency; Self-understanding; Selfhood; Information; Initiation; Change; Distinction; Consciousness; Programme; Elementary changes; Measures; Interaction; Aggregated forms; Emergence; Matter; Life; Reason; Intellect; Stage; Sifting.

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PREFACE

Throughout its history philosophy has faced a fundamental question: how can changeability, multiplicity, consciousness and reality arise if there exists an absolutely immutable unconditioned?

All the schools — ancient, religious, idealist, materialist, dualist, monist, processual — answer differently, but none removes the principal contradiction: every change requires conditions, and absoluteness excludes them.

The scientific picture of the world faces the same. Quantum indeterminacy, entanglement, the absence of classical substantiality in elementary states, the informational reading of descriptions and the dependence of the result on measurement are hard to reconcile with classical materialism; they are compatible with an informational ontology, although they are not its ground.

The aim of the monograph is an ontology in which information is derived as the conditioned itself, taken from the side of what is given in it of the unconditioned — the mode of being of the unconditioned's knowledge of itself. Through the initiation of Selfhood the conditionedness of reality proceeds in it without introducing becoming into the unconditioned. On this basis the philosophical foundations of the fundamental sciences are refined.

The work consists of four parts: analytical-philosophical, metaphysical, scientific-integrative and worldview.

The language is strict, without imagery or hidden assumptions. The text is built as a derivational exposition: categories are introduced successively, each from what has already been derived; there is no symbolic formalization in it.

This is necessary: explaining fundamental categories through illustrations, analogies or intuitive self-evidence leads to hidden assumptions that cannot be checked for consistency. In ontology such assumptions become implicit second principles: they destroy the uniqueness of the unconditioned and sever the tie of the conditioned with that by which it is determined. Therefore each new category is derived as an inevitable consequence of the preceding one; there are no examples in the text on which the derivation would rest. The structure of being here is not a hypothesis but a conclusion.

INTRODUCTION

The monograph solves the central problem of philosophy: how a changeable reality, a multiplicity of forms, sequences, interactions, consciousness and reason can exist if the unconditioned is conceived as absolutely immutable. No philosophical school — from ancient metaphysics and religious ontologies to German idealism, analytic philosophy and physicalism — has provided a consistent model that removes this aporia.

The problem requires not a revaluation of concepts but a strict ontological fixation: the relation of the immutable unconditioned to changeable reality must be expressed without a logical gap and without hidden conditions. What is at issue is not transition as a process or as becoming, but a structure of dependence that preserves the immutability of the unconditioned and excludes the alternatives without loss of consistency.

The method of negative testing sets the limiting conditions under whose violation the construction loses its consistency: distinction within the unconditioned, information as an

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independent principle, change proceeding in the dependent not by the removal from Selfhood, or a second independent source.

Necessity is understood as the exhaustion of the space of alternatives. Determinacy can lie in three places: on the side of the unconditioned, on the side of a second independent principle, or on the side of the dependent; there are no other places (chapter 4). The first two violate the conditions of a consistent ontology; the third observes them. This proves where the solution can lie. That the series of categories derived below is its realization is asserted to the extent that each category is derived from what has already been derived.

The method applies only to fundamental categories. It does not extend to empirical descriptions and does not explain particular manifestations: it fixes only the conditions of their ontological possibility.

I introduce a strict ontological structure. Its fundamental elements are as follows. Nothing is absolute unconditionedness: that to which no determinacy is applicable, and with it no limitation and no conditionedness; it is obtained as the limit of the conditioned. Immutability, uniqueness and transparency are the same unconditionedness taken from the side of change, of a second, and of opacity. Self-understanding is the inapplicability of non-understanding of oneself; Nothing in self-understanding is Selfhood. The unconditioned is given in the conditioned as limit: the determinacy of the conditioned can be neither given nor stated without it. Information — in the full sense of the Latin *informatio* — is the conditioned itself, taken from the side of what is given in it: the mode of being of Nothing's knowledge of itself; it is the first dependent, determined by that whose mode of being it is, and therefore conditioned. The initiation of Selfhood — the removal of opacity in the dependent — is that by which information has knowledge of its difference not by itself but from Selfhood. Something is information taken from the side of the succession of states, that is, changeable reality. This structure explains changeability without violating the absoluteness of the unconditioned.

Consciousness is self-knowledge accrued by information with every distinction and beginning with the first. Elementary changes are the smallest changes of distinguished information. Time, space, localization, speed and energy are not primary entities but measures — magnitudes derived from a single fact: the working moment of an elementary change. Aggregated forms are stable configurations of distinctions. Life is a form in which the *I* distinguishes the disturbance of its own bond and restores it from the external. Reason is the level of consciousness that knows the bonds (interactions that have held and set a common algorithm) between distinctions and chooses within the conditions in which it stands. Intellect is the level that knows, beyond this, its own place in the whole and the programme, and makes choices concerning programmes.

Particular attention is given to fundamental physics: quantum nonlocality, the nature of the wave function, entanglement, the absence of volumetric substance in elementary states, the holographic principle and the connection of energy with information (Bell, 1964; Wheeler, 1990; Zeilinger, 1999; Pusey, Barrett, & Rudolph, 2012). I show that the data of physics are compatible with the ontology and do not contradict it, but are not the ground of its conclusion, and I state the consequences of the system that physics can test.

The monograph concludes with the teleological structure of reality: the path of distinctions proceeds in stages, each of which ends with the exhaustion of the new for the universe's knowledge of itself as a whole and with the sifting of instances of the intellectual *I*, and at the limit leads to the self-transparency of information, in which it remains dependent and conditioned. The concluding doctrine gives an integral worldview that explains reality as the successive unfolding of informational structures.

LITERATURE REVIEW

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Philosophy and science offer several classes of models of the origin of distinctions, conditionedness and the structure of being. The review pursues a single aim: to identify where each model meets the aporia of transition from the immutable to changeable reality and how it resolves it. Each position is reconstructed from the primary source — or, where this is stated, from the scholarly literature — together with the author's own answer to the aporia; it is then shown where the aporia remains for that author and where the present system agrees and diverges. The review does not seek historical completeness: it is limited to the traditions closest to the proposed ontology or directly opposed to it.

The Neoplatonic Tradition: Plotinus

The closest structural analogy is Plotinus's scheme of emanation. The One is absolutely simple, devoid of predicates and of inner differences. Being perfect, it as it were overflows, and its superabundance produces another (Plotinus, 1991, V.2.1). From it proceeds Nous (Intellect) as the first distinction, then Soul as the bearer of the temporal dimension, then the material world as the lowest level of being (Plotinus, 1991, V.1–3; III.7.11; II.4). The similarity is that everything dependent is determined by the first and has no independent status. There is no correspondence of levels: Selfhood is not a second level like Nous but the name of Nothing taken in self-understanding (see below).

Plotinus himself sees the aporia and answers it with the doctrine of two activities. Every thing has an activity of its essence, which is the thing itself, and an activity from its essence, which proceeds from it, as heat proceeds from fire while the fire remains fire (Plotinus, 1991, V.4.2). What proceeds neither diminishes nor alters its source, as the light around the sun does not alter the sun (Plotinus, 1991, V.1.6). The One is "the power of all things" (Plotinus, 1991, III.8.10); in producing, it is not moved, does not will and does not act.

The aporia remains in two places. First, procession is determined by the nature of the One — it proceeds because the One is such — and determinacy by nature is a determination of the unconditioned. This is the same knot as in Ibn Sīnā (see below). Second, by Plotinus's own account the distinction between the activity of the essence and the activity from the essence is inapplicable to the One, since in it essence and activity are not distinguished. Hence the transition from the One to Nous is conveyed by the image of fire and sun, not reconstructed. And Nous, standing between the One and the world, is an intermediate principle.

There is no emanation here. Selfhood does not proceed from Nothing as a separate level: it is the name of Nothing taken in self-understanding. Information does not "flow out" of essence; it is the mode of being of Nothing's knowledge of itself — a dependent, determined by that whose mode of being it is. Dependence here is determinacy by a limit, not production, and no doctrine of two activities is needed: nothing proceeds, and there is nothing to be diminished or left undiminished.

Advaita Vedānta: Śaṅkara

Advaita starts from the Upaniṣadic "one without a second" (Chāndogya Upaniṣad 6.2.1). Brahman is immutable and non-dual; the multiplicity of the world arises through superimposition (adhyāsa): ignorance superimposes on Brahman attributes that do not belong to it, as the look of silver is ascribed to mother-of-pearl (Śaṅkara, 1890, Introduction). The world is not a transformation of Brahman (pariṇāma) but its appearance (vivarta): Brahman does not become the world; the world only appears upon it.

Śaṅkara's solution is radical: change exists only in appearance, and Brahman is untouched by it, as mother-of-pearl does not become silver. Later Advaita defines māyā as indeterminable either as being or as non-being (anirvacanīya): were it being, there would be a second principle; were it non-being, there would be no appearance either (Deutsch, 1969).

The aporia remains in the question whose ignorance this is. Rāmānuja raises it among his objections to the doctrine of māyā (Rāmānuja, 1904, I.1.1). If ignorance belongs to Brahman, there is limitation in the unconditioned. If it belongs to the individual soul, then the soul itself

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is a product of ignorance, and ignorance presupposes itself. Defining māyā as indeterminable gives it a third status — neither being nor non-being — and thereby introduces a condition not derived from the ground.

The agreement with the present system lies in the immutability of the unconditioned and in change belonging only to the dependent. The divergence lies in the status of the dependent. Here reality is not appearance: the single accepted position is precisely that it is (chapter 4). Dependence is not a superimposition requiring a bearer of ignorance: the conditioned is determined by unconditionedness as by a limit, and no third status between being and non-being is required.

Madhyamaka: Nāgārjuna

Nāgārjuna begins by holding that nothing arises from itself, from another, from both, or without a cause (Nāgārjuna, 1995, 1.1). Whatever arises dependently he calls emptiness; nothing exists that has not arisen dependently, and hence nothing non-empty exists (Nāgārjuna, 1995, 24.18–19). From total dependence Nāgārjuna concludes that beings have no intrinsic nature (svabhāva) and no independent support.

His own solution is the doctrine of the two truths. The Buddha's teaching rests on a conventional truth and an ultimate truth; without reliance on the conventional, the ultimate cannot be taught (Nāgārjuna, 1995, 24.8–10). Madhyamaka does not deny the conventional world: it denies that the world has a ground outside the web of dependencies. Emptiness itself is also empty and serves as no ground.

Here the aporia is not resolved but dissolved by renouncing the ground. The conditioned holds by the mutual dependence of the conditioned, and the chain of dependencies terminates in nothing. The whole of the conditioned then holds on itself, and the question why there is distinction and change at all remains unanswered.

The present ontology draws the opposite conclusion from the same total dependence — the unconditioned as the limit of the conditioned. The divergence comes down to one position: to abolish unconditionedness is to abolish the conditioned. Madhyamaka does not accept it. I derive it: the conditioned is determined by the absence of unconditionedness and by nothing besides, and therefore does not stand without it (chapter 4). The unconditioned here is neither an intrinsic nature nor a support among things — remotion ascribes nothing to it — and therefore Nāgārjuna's objection against svabhāva does not apply to it.

The Kyoto School: Nishida and Nishitani

In the essay "Basho" (1926/27) Nishida Kitarō introduces place — that in which beings are already placed prior to any division into subject and object (Nishida, 2012, 49–102). The ultimate place is absolute nothingness: it does not stand over against being but encompasses both being and relative nothingness. In his late works Nishida describes the world as the self-determination of absolute nothingness and as the self-identity of absolute contradictories (Nishida, 1987).

His student Nishitani sets absolute nothingness against nihilism: relative nothingness — nihilistic emptiness — is overcome only by being radicalized to emptiness, that is, to absolute nothingness (Nishitani, 1982). Here the solution of the aporia is existential: the passage to the ground is accomplished by the conversion of the existing person.

The aporia remains in both cases. In Nishida absolute nothingness determines itself, and self-determination is activity in the ground; the self-identity of contradictories takes contradiction into the ground rather than removing it. In Nishitani absolute nothingness is a standpoint that the existing person takes, that is, it is reached by experience and thereby depends on the one who reaches it.

The similarity with the present system is that nothing is not an empty negation of beings and that the unconditioned is conceived prior to distinction. The divergence lies in the structure of the unconditioned and in its status. Here container and contained are inapplicable to the

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unconditioned: the dependent is not placed in the unconditioned but determined by it as by a limit. The unconditioned determines itself by nothing: determinacy lies on the side of the dependent. And Nothing is neither a standpoint nor an experience, but that by which everything conditioned, every experience included, is determined; it is not reached but derived as a limit.

The Apophatic Tradition: Dionysius the Areopagite and Nicholas of Cusa

The method of the present ontology — to speak of the unconditioned only by what is inapplicable to it, that is, by remotion — goes back to the apophatic tradition. In Dionysius the Areopagite the cause of all is above every affirmation, as the perfect and unique cause of all, and above every negation, being free of every limitation (Pseudo-Dionysius, 1987, Mystical Theology 5). The ascent ends in silence: speech falls entirely silent (Pseudo-Dionysius, 1987, Mystical Theology 3).

Dionysius resolves the relation of God to the world with the doctrine of the divine processions: unity proceeds into multiplicity without losing unity; he himself calls this procession a divine differentiation, and God participated in "unparticipatingly" (Pseudo-Dionysius, 1987, Divine Names 2). The path of the soul leads back — to union with the One who is beyond all being and knowledge (Pseudo-Dionysius, 1987, Mystical Theology 1). The aporia remains in the very word "differentiation": a procession named divine differentiation introduces difference into the absolutely simple, and "unparticipating participation" names the transition but does not reconstruct it.

Nicholas of Cusa holds that the infinite as infinite is unknowable, since it escapes all comparative relation, and that there is no comparative relation of the infinite to the finite (Nicholas of Cusa, 1981, I.1, I.3). The infinite for him is God, not an unbounded magnitude — the same understanding of the infinite as in the present system. He resolves the relation of God to the world with the pair enfolding and unfolding: God enfolds all things in himself, the world is the unfolding of what is enfolded, and the universe is a "contracted" maximum (Nicholas of Cusa, 1981, II.3–4). The aporia remains in the positive determinations: God is conceived as the absolute Maximum in which opposites coincide, and unfolding is a relation of the enfolded to the unfolded, that is, a relation with two terms.

Here remotion is not a statement about the unconditioned but an indication of what, from the conditioned, is inapplicable to it; it requires no cancellation and serves as the ground of derivation. Neither a maximum, nor a coincidence of opposites, nor a procession is applicable to the unconditioned: opposites are distinguished and therefore belong to the conditioned, and two terms with a relation between them are the structure of the conditioned (chapter 4). The unconditioned pours out nothing and unfolds nothing, and contains no difference: every difference lies on the side of the dependent. The path ends not in union but in self-transparency within the category of the dependent: information remains conditioned even at the limit.

The Abrahamic Scriptures and Theology

The Abrahamic Scriptures hold to one central thesis: there is the unconditioned, and everything else depends on it. In the Qur'an this thesis is given by the formula "lā ilāha illā llāh". The root ʔil- originally designates not a "god" but an initiator; the meaning "god" is the outcome of later sacralization (Ryszhanov, 2026). Accepted lexicography derives "ilāh" from the meaning "the one who is worshipped"; here the conclusion of the study cited is adopted. The derivation of the present ontology does not depend on this reading: the Scriptures are compared here with what has been derived, not used as its premise. The formula then reads: there is no initiator but the Initiator — and it admits no other unconditioned. Generation is removed from the Initiator by the text itself: "He did not beget and was not begotten" (Qur'an 112:3). The Initiator is therefore neither a parent nor a maker, but the one from whom initiation proceeds.

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In the Torah the same thesis is given by the self-naming “eh’yeh asher eh’yeh” (Exod. 3:14). Read isomorphically, it is: “I am that I am”. “Am” is the same word by which the unconditioned is spoken of here: it does not ascribe existence to it as a property but removes non-being (chapter 8). The person expressed in “eh’yeh” by the form of the verb is kept in English by the form “am”. On the aspectual reading, the prefix form “eh’yeh” expresses not tense but imperfective aspect — being not closed by completion. The relative “asher” introduces neither person nor gender; the second member repeats the first. The unconditioned is named here in the first person, that is, in self-understanding, and no predicate is applied to it. The root h-y-h also admits the meaning “to become”, but becoming is inapplicable to the unconditioned, and the meaning “to be” is chosen here by the ontology, not by the grammar. The renderings “I am the One who is” and “I will be what I will be” bring into the formula, respectively, a predicate and a tense that are not in it.

Everything else in the Scriptures is subordinate to the central thesis and expressed figuratively. It cannot be otherwise: language distinguishes, the distinguished is conditioned, and differences are inapplicable to the unconditioned (chapter 4). Therefore the deeds the Scriptures ascribe to God — created, spoke, commanded — are images. Here they are read as designating the initiation of the removal of opacity, which is not an action of the Absolute upon information and not an event in its presence (chapter 5). This reading is an interpretation; it does not harmonize Scripture with philosophy after the fact but is derived from the ontology and correlated with the images of the Scriptures.

The aporia arises not in the Scriptures but in their theological interpretation. There is no religion without interpretation: doctrine and community exist only through it, otherwise systematic theology would not have arisen at all. In all three traditions it has tried to reconstruct the relation of the unconditioned to the created and has arrived at four types of solution. None of them removes the aporia.

The first type is creation as a free act of will: Saadia Gaon, Augustine, Thomas Aquinas, the Ash‘arites. So that the act should not become a change in God, Thomas declares creation not a change but a relation of dependence, real only in the creature (Thomas Aquinas, 1948, I, q. 45, a. 2 ad 2; a. 3; q. 13, a. 7). In this he is close to the present system. But the aporia moves into the will: God could have not created, and the distinction “creates — could have not created”, that is, possibility, is introduced into the unconditioned. In kalām the same knot is the question why the eternal will determines the world to one moment rather than to one like it; al-Ghazālī answers that will is precisely what singles out one of two like things (al-Ghazālī, 2000, First Discussion). But singling out one from the indiscernible is a distinction in the unconditioned itself. Maimonides purifies the doctrine of attributes more consistently than anyone: only negations are true of God (Maimonides, 1963, I.50–60). But he keeps creation as an act of will (Maimonides, 1963, II.13, II.25), and the aporia remains in the same place.

The second type is necessary emanation. Ibn Sīnā removes will: the world proceeds from the Necessary Existent eternally and necessarily, and from the one only one proceeds (Avicenna, 2005, IX.4). But then procession is determined by the nature of the first principle, and determinacy by nature is a determination of the unconditioned. The chain of intellects mediating between the first principle and the world is a series of intermediate principles. The same objection applies to this type as to Plotinus (see above).

The third type is distinction within the absolute itself. Gregory Palamas distinguishes in God the unparticipable essence and the participable energies (Gregory Palamas, 1983, III.2): distinction is introduced into the simple. Lurianic Kabbalah introduces tsimtsum, the “contraction” of the Infinite (Ein Sof) that clears a place for the world (Scholem, 1941, lecture 7). Taken literally, contraction is a change in the Infinite; taken figuratively, it leaves the relation unreconstructed.

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The fourth type is the unity of being. The school of Ibn ʿArabī (Chittick, 1989) and Meister Eckhart remove the aporia by holding that the dependent is not other than the unconditioned; Eckhart speaks of a breakthrough beyond God to the groundless ground (Eckhart, 2009, Sermon 87 [Pr. 52]). They are close to the present system in their understanding of the unconditioned. But the goal of the path for them is the annihilation of the self in the absolute (fanāʾ in the Sufi tradition), followed by subsistence in it (baqāʾ), or identity with it. This diverges both from the distinction the Scriptures hold and from the present system: here consciousness directed towards the limit remains dependent and conditioned, and the self-transparency attained is not unconditionedness (chapters 6, 30).

Thus the Scriptures keep the distinction between the unconditioned and the dependent but do not reconstruct the relation between them. Theology reconstructs it and in doing so either introduces possibility into the unconditioned, or determines it by a nature, or introduces distinction into it, or removes the distinction itself. The present ontology, derived independently of the Scriptures, keeps the same distinction and removes the aporia without these losses: the dependent is not created by an action and does not proceed from essence but is the mode of being of Nothing's knowledge of itself; initiation is neither an action nor an event; the Initiator is the one from whom it proceeds.

Hegel: The Beginning of the Science of Logic

Hegel begins logic with pure being — indeterminate immediacy in which there is nothing to distinguish. Precisely for this reason pure being is the same as pure nothing, and their truth is becoming, in which each passes over into the other (Hegel, 1969, 82–83). The beginning presupposes nothing but itself, and everything further is the immanent development of the concept from this beginning.

Hegel's own solution of the aporia is that the absolute does not stand before change but is the movement itself taken in its fullness: the transition from the immutable to the changeable need not be explained from outside, for the ground is development. The aporia is thereby not removed but carried into the ground: change becomes a change of the absolute itself, and absoluteness is violated.

The present ontology starts not from the identity of being and nothing but from the encountered, and Nothing is obtained as the limit of the conditioned (chapter 4). "Pure being" already bears a positive predicate: it is distinguished from non-being. And the "absence of determinacy" in Hegel is a characteristic of thought that has nothing to distinguish, whereas Nothing here is an ontological characteristic: determinacy is inapplicable to it in principle. Therefore Nothing does not pass over into being, and change is localized in the dependent — in information, not in the unconditioned.

Heidegger: The Question of Nothing and the Ontological Difference

Heidegger is one of the few major thinkers of the twentieth century for whom "nothing" became a positive ontological theme. In the lecture "What Is Metaphysics?" the Nothing is neither the negation of beings nor an empty concept: it "nothings" (das Nichts nichtet), disclosing itself in anxiety as the horizon on which beings as a whole become manifest (Heidegger, 1998, 82–96). In Being and Time a similar role is played by the difference between Being and beings, later called the ontological difference: Being is not a being, is not predicated and is not localized, yet makes possible the understanding of every being (Heidegger, 1962, §§ 2–4).

Heidegger's solution is that the ground of beings is not sought in a being: Being is not a first being, and therefore the question of a transition from an immutable being to changeable ones does not arise. The point of contact with the present system is that Heidegger's Being, and the Nothing as its horizon, make possible the understanding of every being; here a similar role is played by Nothing — not as a condition, however, but as the limit by which everything conditioned is determined.

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The aporia remains in two places. First, the Nothing is disclosed through the existence (Existenz) of finite Dasein, that is, through an already distinguished consciousness. Heidegger does not assert an ontic dependence of the Nothing on Dasein: what is at issue is the mode of disclosure. But the criterion of unconditionedness excludes every structure in which the unconditioned discloses itself only through a finite subject: this places the condition of manifestness before the unconditioned. Second, the analytic of Dasein temporalizes being: Being is understood through the temporal horizon (Heidegger, 1962, § 5; § 65). Here, by contrast, time is derived as a measure from the working moment of an elementary change and is a derivative category.

Bergson: Nothing as a Pseudo-Idea

Bergson gives the most direct objection to any ontology of Nothing. The idea of nothing, for Bergson, is a pseudo-idea. To think the annihilation of everything, we abolish objects one after another, but each time we remain as an observer, that is, as something. The idea of nothing contains not less but more than the idea of something: the idea of the object, its negation, and the motive of the negation — an expectation that has not found what it sought. Therefore the question why there is something rather than nothing is a pseudo-problem (Bergson, 1911, ch. 4).

Bergson's own solution is the primacy of duration: reality is continuous becoming, and change needs no immutable bearer. The aporia of transition from the immutable to the changeable is thereby dissolved, but at the price of the ground: duration is accepted as first and is not itself derived. This is the same defect as in Heraclitus and Whitehead: changeability is primary and remains without ground.

Bergson's objection is directed against nothing as the absence of everything — as a void obtained by abolishing things. The present system rejects such a nothing as well. Nothing here is not obtained by abolishing objects and is not a void in their place: it is unconditionedness, obtained as the limit of the conditioned by the remotion of every determinacy, including the void, which is already something distinguished (chapter 4). Therefore Bergson's objection does not apply to it, and the question "why something rather than nothing" is dissolved here as well, but for another reason: Nothing is not an alternative to something but that by which something is determined.

Sartre: Nothingness through Consciousness

Sartre distinguishes being-in-itself and being-for-itself. Being-in-itself is what it is: it is full, devoid of negation, and uncreated (Sartre, 1956, Introduction, VI). Nothingness comes into the world through human reality: man is the being through whom nothingness comes to the world, and nothingness "lies coiled in the heart of being — like a worm" (Sartre, 1956, Part One, ch. 1, V).

Sartre's solution is the reverse of the present system. There is no transition from an immutable ground to the changeable in him: being-in-itself is not a ground but a contingent given without ground, and negation, distinction and change are brought in by consciousness. The aporia is thereby not resolved but split in two: being-in-itself remains unexplained facticity, and nothingness depends on consciousness, which is itself grounded in nothing. Here the relation is the reverse. Nothing is not produced by consciousness and does not depend on it: consciousness is self-knowledge accumulated by dependent information, and is itself determined by Nothing as by a limit (chapter 12). Distinction is not brought by consciousness into ready-made being: it proceeds by the removal of opacity and is change taken from the side of the knowledge of difference (chapter 11). The one point of agreement with Sartre is that negation and distinction do not belong to the fullness of the ground; but Sartre endows being-in-itself with this fullness, whereas here even fullness as a determination is inapplicable to the unconditioned.

Priest: Nothing as a Contradictory Object

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In contemporary logic the position closest to the present system is Priest's. For him nothing is the absence of all objects, defined as a mereological sum; it is a contradictory object that is at the same time not an object and, in a certain sense, the ground of reality (Priest, 2021). Priest's own solution is dialetheism: contradiction in the ground is accepted as true, and the aporia is removed by the ground both being and not being.

The similarity lies in Nothing determining reality. The divergence lies in the means, and the aporia remains in the solution itself. A ground that both is and is not the source of distinction does not decide whether distinction has a source, and therefore does not do the work of a ground (see "Discussion"). Moreover, mereology makes Nothing a term in the relation of part and whole. Here part and whole are inapplicable to the unconditioned (chapter 4), and there is no contradiction in the unconditioned: Nothing is spoken of not by an assertion together with a negation but by remotion, which ascribes no property to Nothing.

Why There Is Something Rather Than Nothing

The title of the present ontology places it in relation to the question why there is something rather than nothing. Leibniz poses this question as the first of those that follow from the principle of sufficient reason and answers it with a necessary being that bears the reason of its existence within itself: the sufficient reason must lie outside the series of contingent things (Leibniz, 1989, §§ 7–8). With this agrees the argument of the present system that the whole of the conditioned does not hold on itself. But in Leibniz God chooses the best from an infinite multitude of possible worlds (Leibniz, 1985, § 8), and choosing introduces into the ground intention and the possibility of otherwise. The aporia here is the same as in the first type of theological solution: possibility is introduced into the ground. Here the unconditioned does not choose: the dependent is not the chosen but the mode of being of Nothing's knowledge of itself.

The contemporary defence of the principle of sufficient reason holds that every contingent fact has an explanation, and defends the principle against the argument that it entails modal fatalism (Pruss, 2006). The hypothesis of a necessarily existent being, able to be the ultimate ground of everything, is defended as well (Pruss & Rasmussen, 2018). The similarity with the present system lies in the conditioned not holding on itself. The divergence is that their necessary being is a being, whereas here even being as a predicate is inapplicable to the unconditioned. The modal conclusion here differs from Pruss's. Relative to the completeness of the conditions what has set in is one (chapter 11), and in this sense the system accepts for the conditioned what is called modal fatalism. But neither the necessity of the conditioned itself follows from this — it is determined by unconditionedness, not deduced from it as a theorem — nor the futility of choice: choice is one of the conditions by which what sets in is determined (chapter 21).

An analytic formulation of the same question is given by van Inwagen. The probability that there would be nothing is zero: there are infinitely many possible worlds, there is at most one world in which there is nothing, and all worlds are equiprobable (van Inwagen, 1996). Parfit poses the question as a choice among cosmic possibilities. One of them is the null possibility: nothing ever exists; of all possibilities it would have required the least explanation, being the simplest and least arbitrary (Parfit, 1998). Since it did not obtain, Parfit looks for a "selector" — a highest law by which precisely this possibility obtained.

In both cases nothing is an empty possible world or a null possibility — an alternative to a world with beings — and the question is decided by a measure of probability or by a law of selection. The present ontology rejects this very formulation. A world, even an empty one, is already something distinguished, that is, conditioned, and possibility belongs only to the conditioned (chapter 11). Nothing is not an alternative to something but that by which something is determined as conditioned. What obtained was chosen neither by God nor by a law: the dependent is not the chosen but the mode of being of Nothing's knowledge of itself.

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There are no alternatives between which probability or a selector could choose, and the question is not resolved but dissolved.

Whitehead: Process Philosophy

Whitehead's Process and Reality is the most elaborate twentieth-century attempt to build an ontology in which changeability is primary (Whitehead, 1978, 18–30). The unit of reality is the "actual occasion": an indivisible event of prehension in which former states are synthesized into a new unity. God in Whitehead is the "primordial nature", the principle of possibility, which provides the "eternal objects" as unrealized potentials.

Whitehead, like this system, explains multiplicity and changeability without a substrate of substance. But if becoming is primary, the source of becoming itself remains without ground. Whitehead makes creativity the "category of the ultimate" (Whitehead, 1978, 21), but does not derive it: he accepts it as ultimate. Moreover, the "consequent nature" of God changes as the world is actualized (Whitehead, 1978, 345–351): change enters the ground itself and destroys its absoluteness. Here all change is localized in information — the dependent mode of being of Nothing's knowledge of itself — and the essence of Nothing remains immutable.

A Nothing above Being: Damascius, Eriugena, Schelling

Closest to the name "Nothing" stand the doctrines in which the highest is called nothing not by deficiency but by excellence. Damascius places above the One the Ineffable, of which one cannot even say that it is a principle (Damascius, 2010). Eriugena calls God "nothing by excellence" (*nihil per excellentiam*): nothing of what is can be said of Him, for He is above all that is (Eriugena, 1987, bk. III). Schelling introduces the Groundless (*Ungrund*) — that which precedes every opposition and in which there are no differences (Schelling, 2006). The similarity with the present system lies in the remotion of every determinacy from the highest. The divergence lies in the transition. In Damascius and Eriugena a hierarchy proceeds from the highest, and the procession is determined by its nature — the first exit (chapter 2); in Schelling a split into ground and existent comes forth from the Groundless, that is, distinction arises from the groundless itself. Here the unconditioned brings forth nothing: it is given in the conditioned as limit, and difference lies only on the side of the dependent.

Informational Ontology: Wheeler and Floridi

Wheeler formulates the informational ontology of physics as "it from bit": every "it" — a particle, a field, spacetime itself — derives its existence from yes-or-no answers, from bits, and the universe is participatory: observation enters its very constitution (Wheeler, 1990). The unit of description here is the bit of distinction. The aporia remains in the presuppositions: a yes-or-no answer is already a distinction, and a question presupposes a questioner. Distinction and observer are accepted, not derived. More broadly, quantum theory, information theory and thermodynamics admit an informational reading, but physics fixes the regularities of the distinguished and does not explain its ontological origin (chapter 22).

The most systematic informational ontology in contemporary philosophy is Floridi's "informational structural realism" (Floridi, 2008; 2011). For him reality is structurally informational: the world is the totality of informational objects, understood through relations rather than as substances. Floridi defines a datum as a lack of uniformity — a difference between states (Floridi, 2011). He deliberately restricts his realism by the method of levels of abstraction: ontological claims are made at a given level of description, and the question of the origin of the structure itself is not posed by this method.

The charge that Floridi does not explain the origin of the structure therefore needs a qualification: he does not seek to. But then difference is the initial primitive of his ontology, and this is precisely the point the present system derives. The similarity lies in treating information as primary and structural, not as a property of material bearers. The divergence lies in the status of difference: here it is not a primitive but the knowledge of difference

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proceeding by the removal of opacity, and information is the first dependent, the mode of being of Nothing's knowledge of itself (chapters 9, 11).

The Metaphysics of Ground and Dependence

Analytic metaphysics of recent decades has made dependence the basic relation between levels of reality. Lowe distinguishes existential dependence and identity dependence: x depends on y if x cannot exist, or cannot be what it is, without y (Lowe, 1998). Fine understands dependence as essential: the dependent is such by its essence (Fine, 1995), and introduces ground as a distinctive relation of explanation, distinguishing full and partial ground (Fine, 2012). A survey of these distinctions is given by Tahko and Lowe (Tahko & Lowe, 2020). Schaffer takes relations of grounding to be the main subject of metaphysics (Schaffer, 2009) and regards the whole as fundamental: the cosmos is prior, and explanation descends from the whole to the parts (Schaffer, 2010). Sider defends realism about structure — a privileged way of describing the world (Sider, 2011).

A separate debate concerns whether the chain of grounds must terminate. Cameron examines the intuition that chains of ontological dependence must terminate and finds the arguments in its favour unconvincing; he also admits a model with a single fundamental ground and an infinite chain of the dependent (Cameron, 2008). Bliss asks why a regress of grounds is vicious and examines the proposed criteria of viciousness (Bliss, 2013).

In these theories the aporia of transition is not resolved because it is not posed: dependence is described within an already given changeable and distinguished reality, and the ground is a fact or a being — that is, something determined (chapter 4). The relation of ground has two terms and an order between them, and this is the structure of the conditioned (chapter 4).

The agreement with the present system lies in recognizing dependence as the basic ontological relation. The divergence lies in the structure of dependence. Here it is not a relation of two terms: the unconditioned is not a term of a relation but the limit by which the conditioned is determined (chapter 4). Dependence is full: the conditioned is determined by the absence of unconditionedness and by nothing besides. The question of the length of the chain is answered in chapter 4: a whole that contains the limited bears limitation within itself and does not remain absolute, and a chain of the conditioned, however long, consists of the conditioned; the limit is reached not by the end of the chain but by that by which each of its links is conditioned. The finiteness of the path of distinctions itself is derived separately (chapter 13).

Thus the literature offers no consistent ontology that derives distinction, the measures, multiplicity and consciousness from a single unconditioned. Neoplatonism and Arabic Peripateticism determine the ground by its nature. Advaita carries change into appearance and leaves ignorance without a bearer. Madhyamaka renounces the ground. The Kyoto School introduces self-determination into the ground or makes it something reached by experience. The apophatic tradition names the transition but does not reconstruct it; the doctrines of a Nothing above being derive a hierarchy or a split from the highest. Abrahamic theology carries the aporia over from the Scriptures into interpretation: it introduces possibility, a nature or distinction into the unconditioned, or removes the distinction between the unconditioned and the dependent. Hegel and Whitehead place change in the ground itself. Heidegger temporalizes the ground. Bergson and Sartre reject nothing as a ground: the former as a pseudo-idea, the latter as a product of consciousness. Priest accepts contradiction in the ground. Wheeler and Floridi accept difference as a primitive. The metaphysics of ground describes dependence within an already given reality. Removing this common defect is the task of the monograph.

METHODOLOGY

The Research Framework

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The method of the study is ontological reduction: a derivational exposition without symbolic formalization. Its task is to fix the structure of the conditioned without genetic transitions: through the dependence of the conditioned on the unconditioned, which does not affect the unconditioned and introduces neither hidden presuppositions nor additional principles.
The principal device is the successive fixation of necessary conditions: the conditionedness of the encountered and of consideration itself; unconditionedness as the limit of the conditioned; the givenness of the unconditioned in the conditioned; the immutability and uniqueness of the unconditioned; transparency as the inapplicability of opacity; self-understanding as the inapplicability of non-understanding of oneself; information as the conditioned taken from the side of what is given in it, the mode of being of the unconditioned's knowledge of itself; the initiation of Selfhood as the removal of opacity by which knowledge of difference opens in information; distinction as knowledge of difference, the appearance of which is change; consciousness as self-knowledge accrued with every distinction; the directedness of the removal and the limit of the path; elementary changes and the measures derived from them; interaction, aggregated forms, emergence and matter; life; the development of consciousness up to reason and intellect; stages and sifting.
Every category declared derivable must remove an identified contradiction and have an explicitly indicated ground in what precedes it. In this way hidden assumptions and a plurality of grounds are excluded.

Principles and Procedure

The methodology rests on a logical reconstruction of fundamental categories in which what I accept without derivation is named explicitly and reduced to a minimum. The argument takes one position as given: encountered change. That to abolish unconditionedness is to abolish the conditioned is derived from the definition of limitation. Everything else is derived from what is accepted and from what has already been derived; derived from them is also the identity of distinction — as that by which knowledge of difference is given — with lived experience, set out in chapter 12. Besides the accepted position, definitions are accepted explicitly (Appendix, Df1–Df14): they fix the meanings of words and contain no other assumptions; where a definition names what is derived (Df5 — T10, Df8 — T14), its applicability is established by derivation, not by definition. Also accepted explicitly are the finitist understanding of the determinacy of number (chapter 13) and the rule of method: of the encountered only what is derived from the encountered is accepted. No hidden presuppositions are introduced in the work.
The following are applied:
ontological reduction;
remotion: the unconditioned is spoken of only by what is inapplicable to it;
the principle of transparency as the inapplicability of opacity and of self-understanding as the inapplicability of non-understanding of oneself; the dependent status of information;
the principle of fixing change as the succession of states of dependent information, without ontologizing information as an independent principle;
a strict sequence of categories;
comparison with fundamental physics and neuroscience, which does not serve as a premise of the derivation;
the distinction between the limit and proximate grounds;
the distinction between what is derived and what the system does not derive.
The methodology excludes metaphor, hidden principles, dogmatic premises and logical gaps.
Ontological reduction is the reduction of all categories to one accepted position about reality and explicitly accepted definitions, understanding of the determinacy of number and rule of

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method, admitting a strictly derived structure of being without hidden presuppositions or underived entities.
The steps of the procedure: the conditionedness of the encountered and of consideration itself; unconditionedness as the limit of the conditioned; the givenness of the unconditioned in the conditioned; Nothing as absolute unconditionedness; immutability, uniqueness and transparency — by the remotion of limitation; self-understanding as the inapplicability of non-understanding of oneself, and Selfhood as the name of Nothing in self-understanding; information as the mode of being of Nothing's knowledge of itself — the first dependent; the initiation of Selfhood as the removal of opacity by which information has knowledge of its difference; change as the succession of states of information, proceeding by this removal; distinction and the levels of the conditioned from logically connected dependences.
Consistency is checked successively at each level: whether there are hidden transitions, whether the level admits an ambiguous reading without violating the conditions, whether all derivatives follow from what has already been derived. The criterion of the derivation is not empirical: it is logical and ontological — the absence of contradictions, gaps and hidden assumptions in deriving the dependent forms. The consequences of the system, however, are partly testable by experience: they are stated in chapters 22 and 26 and in the section "Limitations".
Concordance with science is the compatibility of the ontology with the data at the level of reality, not its justification by the means of physics or neuroscience, and not the derivation of empirical laws from it. It means the absence of logical incompatibility with established scientific models and the compatibility of the data with the derived structure, in places only partial (chapter 22). The system therefore retains a metatheoretical status and does not compete with science in empirical explanation.
The deduction here is ontological, not formal-logical. Each category is introduced as a necessary condition of the consistency of the preceding one, not as a theorem of a calculus. The criterion of rigour is not formal derivability but stability under negative modification: removing an element violates the initial conditions. Formal ontologies exist (Zalta, 1983), but formalization is not applied here: the unconditioned is spoken of only by remotions, while a calculus presupposes objects with a determinacy that is inapplicable to the unconditioned. Rigour here is therefore the successive introduction of categories with a justification of each and a demonstration that the alternatives violate the conditions of consistency (chapter 4).

Conditions of a Correct Reading

Here the conditions of a correct reading of the whole construction are fixed. This is not a prohibition of other readings — a philosophical theory cannot and should not prohibit interpretations. Below it is stated which interpretations of the listed concepts violate the conditions of consistency and how; this can be checked and contested.
1. Information is the mode of being of Nothing's knowledge of itself, a dependent category determined by that whose mode of being it is. It has no ontological initiative of its own: it is not a source, not an agent, not a process and not an independent entity; initiation proceeds from Selfhood. Its determinacy lies on its own side and establishes nothing in Nothing. An interpretation that ascribes to information an active, generative or governing function is inadmissible.
2. Consciousness is self-knowledge accrued by information with every distinction; reason and intellect are its levels: reason knows the bonds between distinctions, intellect knows, beyond this, its own place in the whole and the programme. These are not epistemic subjects with respect to essence and not sources of knowledge about it: no positive knowledge of the unconditioned is given to them; only remotions are accessible. A reading that endows consciousness or reason with cognitive subjecthood with respect to essence is incorrect.

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3. Limit completion is fixed as the attainment of the self-transparency of information — self-understanding in its own category. It is attained as opacity is removed: every distinction removes a blind spot, and those removed do not return. This transparency is not unconditionedness: information that has attained it remains dependent and conditioned.
4. None of the subsequent categories describes change, becoming or process in the essence of the unconditioned. All changes are localized exclusively in the dependent. Any assumption of inner difference, dynamics or becoming in the unconditioned contradicts the conditions of a consistent ontology.
5. Transparency is the inapplicability of opacity; self-understanding is the inapplicability of non-understanding of oneself; neither is a separate ontological level. Measures, forms, consciousness and reason have no autonomous ontological status and cannot be regarded as independent principles. Their autonomization in any form is equivalent to introducing a second principle and is therefore logically inadmissible.
6. The present ontology is not intended to explain particular empirical processes, physical mechanisms or observed phenomena. It fixes exclusively the conditions of the ontological possibility of changeability, distinction and conditionedness. Any expectation of empirical explanation lies beyond the scope of this work.

On the Inadmissibility of an Asymmetric Ground and on the Language of Remotions

Within the unconditioned an asymmetric structure is impossible without the return of the initial aporia. To single out the “primary” and the “derivative” within the unconditioned is to introduce a hidden multiplicity or to suppose its change. This is said of the structure of the unconditioned, not of the relation of the dependent to it: dependence determines the dependent and does not determine the unconditioned, and therefore the asymmetry of dependence does not affect the unconditioned. Hence a criterion for every statement about the unconditioned, presented to any reader regardless of agreement with the conclusions.
Inapplicable to the unconditioned is everything by which it would be determined, that is, would be “this and not that”: two and the relation between them, part and whole, bearer and states, subject and the action it performs, container and contained, before and after.
Applicable is that by which the dependent is determined: nothing is thereby established about the unconditioned. What is distinguished is not one’s own and another’s, but the side on which determinacy is laid. The positions of the work itself are checked by the same criterion: “information is the mode of being of Nothing's knowledge of itself” lays determinacy wholly on the side of information and establishes nothing in the unconditioned.
In the present system information is not defined through distinction and is not identified with it. It is the mode of being of Nothing’s knowledge of itself. Distinction and change are identical with each other (T15) and proceed by the removal from Selfhood; information does not enter this identity: its basic determinacy — its difference from Nothing — exists without succession, while distinction gives its states (chapter 9).
Information is not introduced as a domain, an entity or a process. It is neither a second principle nor an independent place of what happens. Its dependence fixes the mode of existence of that which is determined by that whose mode of being it is, and it has no unconditionedness.
This ontology does not claim knowledge of the unconditioned as an object. The description of Nothing and of Selfhood is not cognition in the objectual sense: they are spoken of by remotions. This is consideration by the means of the conditioned from within distinguishedness: the unconditioned is obtained in it as the limit of the conditioned, not as an object. Language here performs not a descriptive but a limit-indicating function: it marks the boundary of the sayable, not properties of the absolute. Wittgenstein says something close of the boundary of the sayable: whereof one cannot speak, thereof one must be silent

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(Wittgenstein, 1922, 7). Here the boundary does not cut speech off: remotions say of the unconditioned what is inapplicable to it, and thereby indicate the limit without making it an object.

Therefore demands for asymmetry in the unconditioned, a causal explanation or objectual knowledge of the absolute are inapplicable to this ontology. They presuppose a thinking that already contains distinguishedness and, applied to the unconditioned, return the initial aporia.

PART I. THE ANALYTICAL-PHILOSOPHICAL BLOCK

Chapter 1. The Principal Aporia of Philosophy

From its very beginning philosophy runs up against one aporia: how a changeable reality can depend on the unconditioned, if differences and transitions are inapplicable to the unconditioned. This holds for every school, epoch and culture. Parmenides keeps the immutable and loses change; Heraclitus keeps change and loses that on which it would depend (chapter 2). Hence the aporia does not belong to a single tradition: it stands before every attempt to explain the changeable from the immutable.

The Statement of the Aporia

The aporia is set by three requirements. The first is unconditionedness: differences and transitions are inapplicable to that on which everything else depends. The second is changeability: reality is a multiplicity of distinguishable states succeeding one another. The third is uniqueness: there is no second, independent unconditioned. Taken separately, none of the requirements raises doubt. The difficulty is to hold them together.

In the traditional statement the difficulty takes the form of a dilemma. Changeability requires distinction. Distinction either belongs to the unconditioned or has another source. If it belongs to the unconditioned, there is distinguishability in the unconditioned, and unconditionedness is violated. If the source is another, a second independent principle is introduced, and uniqueness is violated.

The dilemma rests on an assumption that it does not name: the unconditioned is thought of as the source of distinction — as that from which distinction proceeds or in which it is placed. Only under this assumption is the question put in the form “inside or outside”: distinction is assigned a place either in the source or in another source. But the division “inside — outside” is a division by container and contained, and container and contained are divided by the limitation of one by the other. Such a division does not apply to the unconditioned (chapter 4). By accepting it, the dilemma thinks the unconditioned as conditioned from its very first step.

What Is Excluded and What Remains

Once the hidden assumption is dropped, the question is no longer about the place of the source of distinction but about the side on which determinacy lies. Then there are not two cases but three. Determinacy may lie on the side of the unconditioned — this is the first horn of the dilemma, and it violates unconditionedness. It may lie on the side of a second, independent of the unconditioned — this is the second horn, and it violates uniqueness. And it may lie on the side of the dependent: of what is not the unconditioned and not independent of it. The dilemma does not see the third case: thinking the unconditioned as a source, it does not distinguish the dependent from a second principle.

The third case is not proved here but only not excluded. The first two are excluded: an ontology that lays determinacy on the side of the unconditioned or on the side of an independent second does not remove the aporia but reproduces it. That there are no other places for determinacy and that the third place does not contradict the conditions is derived in chapter 4; how the dependent is constituted, in chapter 5.

The aporia is therefore structural rather than material. It cannot be removed by replacing terms or by complicating concepts as long as the unconditioned and the changeable are

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placed in one domain — so that change is either introduced into the unconditioned or given a status of its own beside it.

Historical Forms of One Error

Ancient, religious-metaphysical, rationalist and materialist systems reproduce one error; what differs is only the language in which it is formulated. A detailed analysis is given in chapter 2 and in the Literature Review; here it is indicated only which horn of the dilemma is chosen. Parmenides keeps unconditionedness and denies becoming: the world of experience loses its ontological status. Heraclitus makes becoming primary and leaves it without that on which it would depend: change is accepted as a given without a ground, that is, as an independent principle. Plato places beside the immutable idea the demiurge and the receptacle, on which changeability rests. Aristotle keeps the Prime Mover unmoved but rests the changeability of the moved on potency and matter — a second principle beside it.

Religious-metaphysical systems hold the unconditioned and at the same time admit creation, emanation or procession. The transition in them is either not reconstructed, or determines the unconditioned by a nature or an action and thereby makes it conditioned, or, if creation is thought of as an external action, introduces a second principle.

Modern rationalism transfers the problem into substance, attributes, monads or dialectical becoming and either admits development within the unconditioned or relies on an underived logical principle — a hidden condition. Materialism and physicalism transfer changeability into matter and the laws of nature and accept matter's capacity to differ as a given without a ground. Analytic philosophy and the scientific picture of the world describe connections between already distinguished states and accept distinguishability itself as an irreducible axiom (chapter 3).

The structure is the same in all variants: changeability is either explained at the same level to which unconditionedness is ascribed or placed in an independent second. This exhausts the first two cases. The third — determinacy on the side of the dependent — is reconstructed by none of these systems.

The Purpose of the Monograph

Hence the task: to construct an ontology that strictly separates the unconditioned and the dependent; lays all determinacy on the side of the dependent; introduces no distinction into the unconditioned; and admits no second principle in either explicit or hidden form. Changeability is thereby neither derived from nothing nor accepted as independent: it is encountered, and what is derived is that it is dependent, where it proceeds and by what it is determined.

This requires a minimal and complete sequence of categories in which each is derived from the accepted position and from what has already been derived. A category introduced without derivation would have its ground by itself and would become a hidden second principle; a missing link would leave the dependence between levels unexplained. Therefore the exposition proceeds by derivation rather than by building a model: what is accepted without derivation is named explicitly (chapter 4), and everything else is derived.

Chapter 2. A Critical Survey of Philosophical Schools

Every school faces one difficulty: how a changeable reality comes out of an absolute ground. This holds already for the earliest systems. Method, language and epoch differ, but the exits from the aporia are few. Chapter 1 named them: on the assumption that the ground is the source of distinction, distinction is either placed in the ground itself or set outside it as a second principle (chapter 1); the completeness of the places of determinacy is derived in chapter 4. Adjoining them is a hidden form of the second exit: the transition rests on a condition not derived from the ground — on possibility, will, appearance, an observer; such a condition is a hidden second principle (chapter 4). Below it is called the third exit.

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Renouncing the ground — recognizing change as a primary fact — is not a separate exit. It is equivalent to the second: change is accepted as a given without ground, that is, as an independent principle. The present system also takes change as given — as the encountered (chapter 4); but it does not take it as independent: that it is dependent, and by what it is determined, is derived. The difference lies not in whether the fact is accepted but in whether it is accepted as primary. Therefore every school, whatever it calls its support — idea, substance, process, emptiness, matter or nothing — takes one of these exits, and each of them violates either the absoluteness or the uniqueness of the ground.

The survey therefore makes no claim to historical-philosophical completeness. Its aim is to indicate, for each school, which exit it takes. Positions requiring detailed analysis — with a reconstruction of the author's own answer to the aporia — are analysed in the "Literature Review"; here the result is given. What is assessed is the logical form of the solutions, not the completeness of the traditions. These exits must be excluded before a consistent ontology is built.

The initial failure is seen most clearly in ancient philosophy.

Parmenides.

If being is absolutely one and immutable, multiplicity is impossible. Parmenides's model shows the limit: by preserving absoluteness, it makes every form of manifestation inadmissible (Parmenides, 2011, fr. 2–8).

Heraclitus.

If becoming is primary, the source of change remains unexplained (Heraclitus, 2011, fr. 12, 30, 50). In Heraclitus logos and measure set the order of change but do not derive it. Process becomes the initial category, and the conditions of its possibility remain outside the model. This is a renunciation of the ground, that is, the second exit: change is accepted as a given without ground. This is not a solution of the aporia but a refusal to pose it.

Plato.

Plato separates the world of ideas and the world of becoming (Plato, 2008, 27d–28a). He explains the transition between them by the demiurge, who makes the changeable after an immutable pattern (28a–29a). But Plato himself admits that two kinds are not enough and introduces a third — the receptacle (chōra), in which everything that comes to be comes to be, and which is neither the pattern nor its likeness (48e–52d). Alongside the immutable idea stand an active principle and a receiving principle. This is the second exit: changeability rests on principles not derived from the idea.

Aristotle.

The transition from potency to act requires a mechanism of change. If the mechanism is inside the first principle, absoluteness is violated. If outside, a second principle arises. The Unmoved Mover moves as an object of love, itself remaining unmoved (Aristotle, 1924, XII.6–7). The contradiction is thereby fixed but not removed: the changeability of what is moved remains with potency and matter — a second principle alongside the Prime Mover.

The Stoics.

The Stoics posit two principles of all things — the active and the passive: the passive is unqualified substance, matter; the active is the logos in it, god (Diogenes Laertius, 1925, VII.134). The logos shapes matter and pervades it, and in the world cycles the world arises from fire and returns into fire. This is the second exit in explicit form: changeability is secured by matter not derived from the logos, while the logos itself, present in the changing world and renewing its cycles, is drawn into change.

Ancient systems show the first level of structural failure. Parmenides keeps the ground and loses change; the others keep change at the price of a second principle — matter, the receptacle, potency — or by renouncing the ground.

Religious-Metaphysical Models

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Neoplatonism determines procession by the nature of the One: it proceeds because the One is such, and determinacy by nature is a determination of the unconditioned — the first exit (see "Literature Review").
The Abrahamic Scriptures hold the distinction between the unconditioned and the dependent but do not reconstruct the relation between them. The aporia arises in theological interpretation: it either introduces possibility into the unconditioned, or determines it by a nature, or introduces distinction into it, or removes the distinction itself (see "Literature Review").
Vedanta distinguishes Brahman and māyā. Change for it is mere appearance, but māyā receives a third status — neither being nor non-being — and ignorance is left without a bearer: belonging to Brahman, it would limit the unconditioned; belonging to the soul, it would presuppose itself. This is a hidden condition, the third exit (see "Literature Review").
Madhyamaka derives from total dependence the absence of a ground: multiplicity has no support, and there is nothing from which to explain the transition to it. This is a renunciation of the ground (see "Literature Review").
Daoism derives the world from the Dao through generation: being is born of non-being; the Dao gives birth to the one, the one to two, two to three, and three to the ten thousand things (Laozi, 1963, ch. 40, 42). Laozi's non-being (*wu*) is closer than any other tradition to the name "Nothing": the Dao is nameless, and it is said of it that the Dao that can be named is not the constant Dao (Laozi, 1963, ch. 1). The similarity lies in the remotion of name and determinacy. The divergence lies in the transition: the Dao "gives birth", generation is a series of transitions proceeding from the first, and the Dao appears as an active principle. This is the first exit: distinction is placed in the ground itself. Here the unconditioned gives birth to nothing: it is given in the conditioned as limit.
The Kyoto School conceives the ground as absolute nothingness. In Nishida it determines itself, and self-determination is activity in the ground — the first exit; in Nishitani it is reached by the conversion of the existing person, that is, it depends on the one who reaches it (see "Literature Review").
The common defect is as follows. The transition is either not explained, or hidden in symbolic form, or becomes a dogmatic presupposition. If change becomes a change of the absolute itself, absoluteness is violated. If it is set outside, a second principle arises. If it rests on an underived condition, that condition is itself a hidden second principle.

Rationalist Systems

Descartes and Spinoza.

Substance is immutable, and the world is changeable. Descartes connects them by the conservation of the world, which requires the same action as creation (Descartes, 1996, Med. 3): the world is held by God's continuous action, and action is introduced into the ground — the first exit. Spinoza answers otherwise: from the necessity of the divine nature there follow infinitely many things in infinitely many modes (Spinoza, 1994, I, prop. 16), and things are modes of the one substance (Spinoza, 1994, I, def. 3–6). Here there is neither will nor a second principle, but the procession of modes is determined by the nature of substance. This is the first exit — the same as in Ibn Sīnā: determinacy of the ground by its nature.

Leibniz.

Leibniz seeks the sufficient reason outside the series of contingent things and finds it in a necessary being (Leibniz, 1989, §§ 7–8). But God chooses the best from an infinite multitude of possible worlds (Leibniz, 1985, § 8), and the agreement of monads is a pre-established harmony set up by this choice (Leibniz, 1965, §§ 51–56). Choice introduces possibility and intention into the ground — the first exit: the choice of "this and not another" is determinacy on the side of the ground (see "Literature Review").

Kant.

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Kant offers no solution of the aporia but contests the task itself. The separation of phenomena and noumena (Kant, 1998, A235–260/B294–315) restricts cognition to possible experience, and the antinomies show that reason, inferring the unconditioned from a series of conditions, arrives at assertions that contradict one another (Kant, 1998, A426–461/B454–489). This is the most direct objection to any ontology of the unconditioned. The answer to it is as follows. The unconditioned here is neither an object of possible experience nor a member of a series of conditions: it is not cognized positively, and it is spoken of only by remotion (chapter 4). The antinomies arise when the unconditioned is thought as the first member of the series or as its completed whole; here it is neither, but the limit by which each link is determined (chapter 4). Kant's main argument remains: the limit by which reason completes the series is a regulative idea, not knowledge of an object. The answer lies in where the limit is obtained from. A regulative idea is a demand of reason applied to experience from outside. Here, however, the limit is not applied to the encountered but contained in it: the determinacy of the conditioned can be neither given nor stated without the unconditioned (chapter 5). Therefore the assertion of the limit is not knowledge of an object beyond experience but an analysis of what is given in the encountered itself; it neither gives nor requires positive knowledge of the unconditioned.

Hegel.

The development of the Absolute begins with the transition of pure being and nothing into becoming (Hegel, 1969, bk. 1, sec. 1, ch. 1). It introduces change into the ground itself — the first exit (chapter 4; see "Literature Review").

Rationalist schemes replace ontological necessity by logical constructions. But logic is the structure of distinction and is itself dependent (chapter 6); it cannot explain how the dependent depends on the unconditioned.

Philosophy of Existence and of Process

Heidegger ties the disclosure of the Nothing to the existence of finite Dasein and understands Being through the temporal horizon; thereby the condition of manifestness is placed before the ground. Bergson declares the idea of nothing a pseudo-idea and accepts the primacy of duration — a renunciation of the ground. Sartre derives nothingness from consciousness and leaves being-in-itself an unexplained given — the second exit in two forms at once. Whitehead makes creativity the ultimate category and introduces change into the consequent nature of God — the first and second exits together (see "Literature Review").

Materialism and Physicalism

Matter is defined through properties, and properties require distinctions. Materialism does not ground distinction but accepts it as a given. Consciousness is declared a product of structure (Dennett, 1991; Kim, 1998). But structure itself presupposes distinguishable states. Causality is held to be fundamental. But a cause is that without which a change would not have set in (chapter 23), and therefore presupposes distinguished states, which materialism does not derive.

Materialism derives the dependent from the dependent, as if from an ultimate ground. This is the renunciation of the ground in the form most widespread today: distinguishedness and change are accepted as a primary fact of matter.

Analytic Philosophy

Analytic metaphysics has restored the standing of ontology. Lowe returns to metaphysics the place of the most fundamental form of inquiry, on whose conclusions all other disciplines rest (Lowe, 1998); Maudlin opposes Humean reductionism and derives ontology from physics rather than from the analysis of language (Maudlin, 2007). The metaphysics of ground and dependence describes the relations of levels of reality, but within an already given distinguished reality: its ground is a fact or a being, that is, something determined. Priest accepts contradiction in the ground (dialetheism). Van Inwagen and Parfit pose the question

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“why something rather than nothing” as a choice among possibilities — the third exit (see “Literature Review”). Every description already presupposes distinction, and linguistic, semantic and functional models do not explain the source of distinctions.

The Scientific Paradigm

Quantum mechanics shows the fundamental role of information and distinction but does not explain their origin. Entanglement yields correlations not explicable by physical parameters. In 1961 Wigner held that the laws of quantum mechanics cannot be formulated in a fully consistent way without reference to consciousness (Wigner, 1961); later, in the late 1970s, he abandoned this position (Esfeld, 1999). Stapp holds that including consciousness in science requires treating the mind–brain problem together with the measurement problem of quantum theory (Stapp, 2009, 119–146). The nature of the observer is not defined within the physical model itself. Informational physics makes the bit of distinction the unit of description (Wheeler, 1990), but accepts distinction and the observer rather than deriving them. Science describes distinguished states, not the conditions of their possibility.

Scientific theories fix distinguishedness as a given but do not derive its ontological status.

The General Conclusion of the Critical Survey

All philosophical and scientific approaches run up against one problem: how changeable reality arises from an immutable ground. If the source of change is made a property of the absolute — its nature, activity, self-determination or development — absoluteness is destroyed. If it is set outside — as matter, the receptacle, appearance or a primary fact — uniqueness is destroyed. If it rests on an underived condition — possibility, will, an observer — that condition is a hidden second principle.

The problem lies neither in historical limitation nor in a lack of concepts. It is the logical limit of the very scheme in which the absolute and the changeable are placed on one level, or in which the ground is rejected. As long as the scheme is retained, the aporia is inevitable. Hence the necessity of a new ontology. In it the absolute and the changeable are strictly separated. Change, taken as given, is derived only as a dependent form that does not affect essence.

The analytical block performs a critical function. It shows that the aporia cannot be solved in any model with hidden conditions, multiple principles or inner differences in the absolute. Only after these errors are removed is the transition to a positive ontology possible. In it there is change, and the absoluteness of the unconditioned is not violated.

Chapter 3. The Limits of the Scientific Picture of the World

Science describes phenomena with exceptional precision. But its concepts do not explain the fundamental categories: consciousness, information, causality, time, space and the nature of elementary objects. The chapter indicates where the scientific picture of the world meets the aporia of chapter 1. Physics does not serve here as a premise of the ontology: the ontology is derived from encountered change (chapters 4, 5) and is compared with physics (chapter 22).

The limit of science is the level of its own categories. It fixes the behaviour of already distinguished states but does not explain distinguishability itself. Every model rests on measurement, and measurement already presupposes distinctions. The concepts “state”, “interval” and “observation” presuppose conditions that science does not analyse. Therefore, however its instruments develop, it does not remove the aporia: it describes the structure of Something, but not that by which Something is determined.

Fundamental physics does not take matter in the habitual sense to be primary. Electrons, quarks and gluons are described as excitations of quantum fields. They have no classical extension, density, hardness, volume or form — in the sense in which macroscopic bodies have these properties. The formalism of quantum theory does not ascribe to them definite values of all quantities prior to measurement. The interpretation of these states as

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informational is one of the interpretations accepted in physics, not the content of the theory itself (chapter 22).

Hence a limitation essential for ontology. If matter is not primary, it cannot serve as the ground of the picture of the world. Describing stable forms of the distinguished, it does not explain by what the distinguished is determined. But replacing matter by information does not remove the aporia either: information taken as self-sufficient becomes a second principle, and taken as a mathematical quantity it explains nothing. What is required is a level at which both distinction and information have a strictly dependent status.

The wave function describes the state of a system, and the Schrödinger equation its regular evolution. Physicists interpret differently what stands behind this description, and ontology does not choose between interpretations that differ only in description (chapter 22). What is essential for it is something else: the description fixes the regularities of the succession of already distinguished states, but does not explain that there is succession and by what it proceeds.

In measurement the system exhibits a definite value of the measured quantity. In the standard interpretation the act of measurement is fundamental, while what measurement is and how it differs from other interactions the theory does not define. Distinction appears in it as an unexplained function, although it is precisely with it that the transition from an indefinite value to a definite one is connected.

In the present ontology measurement requires no special status: it is an interaction in which one configuration distinguishes the state of another with respect to the measured quantity, and the outcome of measurement is the onset of knowledge of difference in this respect (chapter 22). The distinguishing one is not a biological subject but any configuration in which distinction in the given respect proceeds. The measurement problem therefore points not to a special nature of the observer but to the fact that distinction is not derived in physics.

Quantum entanglement is a state of a composite system that does not decompose into the states of its parts. The correlations of the outcomes of measurements on the parts violate Bell's inequalities and are not explained by local causes; yet one cannot control the outcome on one side by changing the measurement on the other, and there is no superluminal signalling (chapter 23). In an ontology in which objects are primary and space precedes them, such concordance requires a special explanation.

Here space is the totality of localizations and of the distances between them, and distance is measured by the number of steps of elementary changes (chapter 15); space does not precede distinctions. Therefore that by which an outcome is determined need not lie in the same localization as the outcome. Entangled parts are distinctions of one configuration, and that by which their outcomes are determined is in this configuration as a whole, not in an action of one object on a remote other (chapter 23). That entanglement points to the failure of the classical ontology of objects has been noted by physicists as well (Bohm, 1980, ch. 6; D'Espagnat, 2006). This concordance does not prove an informational ontology, but it agrees with it.

In the standard interpretation of quantum mechanics randomness is accepted as fundamental. The Born rule, according to which a measurement yields one of the possible results with the probability given by the wave function, is a postulate of the theory, not a consequence of it (Born, 1926). What is a fact here is the distribution of results over a large number of measurements; that an individual outcome is determined by nothing is an axiom of the interpretation. Experiment does not establish it: Bohm's deterministic mechanics (Bohm, 1952) reproduces the same distributions given an assumption about the initial distribution of particles (quantum equilibrium), and the many-worlds interpretation, in which there is no randomness (Everett, 1957), derives them, although this derivation remains contested.

Science relies on randomness as fundamental: nature allows us to calculate only probabilities

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(Feynman, 1985, 19). For ontology this means that what has no ground is accepted as fundamental.

The cause of a change is here the name of that in the conditioned without which the change would not have set in — the proximate ground (chapters 8, 10). An individual outcome has a cause, but that by which it is determined may be opaque to the one considering; randomness is this opacity of the one considering, not the absence of a cause. That the axiom of an uncaused outcome contradicts itself is shown in chapter 23.

Space in the contemporary picture of the world is a geometric description that does not explain its own origin. Time is a parameter by which changeability is described, but changeability itself is not explained by it. Both categories require distinction: without distinctions there is neither position nor sequence. Here time is derived as a measure of change — duration is measured by the number of steps of elementary changes — and space as the totality of localizations (chapter 15). A theory that takes them as primary explains differences by parameters that themselves depend on differences.

Consciousness is the principal unsolved problem of science. Materialism takes consciousness to be a function of the brain. The neurosciences study correlations between brain states and experience but do not explain why there is experience at all. Functional models explain consciousness through structures that themselves require a ground: a function does not explain what makes it possible. Correlations show correspondence, not that by which experience is determined. The “hard problem” arises under the assumption that a physical process and subjective experience are different objects (Chalmers, 1996).

In the present ontology consciousness is self-knowledge accrued by information with every distinction, and distinction is knowledge of difference whose appearance is change (chapters 11, 12). If matter is an aggregated form of information (chapter 19), consciousness cannot be a function of matter: it does not arise from it but is given already in distinction. Therefore consciousness is explicable in an ontology of distinctions, not by a description of physical processes; this ontology is compared with neuroscience in chapter 26.

Information is one of the fundamental categories of contemporary physics. The holographic principle: the area of a surface bounds the information content of the adjoining region of spacetime (Bousso, 2002, 825–874). Landauer’s principle: the erasure of information — a logically irreversible operation — requires the dissipation of energy (Landauer, 1991, 23–29). Quantum information theory: through entanglement with the environment, information about a system is redundantly recorded in it, and classical objectivity arises from these records (Zurek, 2009, 181–188). But information has no ontological definition: it is used as a mathematical quantity, and its status is not defined.

Without such a definition the scientific picture of the world is incomplete. If information is an independent entity, it becomes a second principle and violates the uniqueness of the unconditioned. If it is merely a mathematical quantity without being of its own, it explains nothing in the structure of reality. In physics information is always information about a system. Here, however, information is not data about something else: it is the conditioned itself, taken from the side of the givenness of the unconditioned in it (chapter 5). Therefore its status is strictly dependent, neither primary nor independent.

Thus the scientific picture of the world faces limitations that are not removed by the old conceptual apparatus. Matter does not explain distinguishedness. Information, taken as independent or as a quantity, does not explain distinction. Time does not explain change. Probability does not explain the outcome. Space does not explain itself. Entanglement does not agree with a space that precedes distinctions.

These limitations are not a set of separate problems of different sciences. They are consequences of one thing: every scientific category presupposes distinction but does not explain it. Without distinction there is neither state, nor time, nor space, nor causality, nor

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consciousness. Therefore these difficulties come not from a lack of data but from the absence of an ontological level at which distinction has dependent status. Science fixes the regularities of the distinguished, but not that by which the distinguished is determined; without such a level there is no passage from description to explanation. The present ontology derives this level; it does not compete with science and does not correct its conclusions (chapter 22).
A number of contemporary metaphysical positions do not accept the unconditioned at all and build ontology from relations, processes or structures. These are not solutions of the aporia but a refusal to pose it. Relational ontology sees the world as a network of relations without a ground but accepts the distinguishedness of the elements of the network as a given. Process ontology declares change primary but likewise accepts the difference of "before" and "after" as a given. In both cases distinction is accepted without a ground, that is, determinacy has its ground by itself — the second exit (chapter 2). A detailed analysis of these positions and of the metaphysics of grounding is given in the Literature Review.

Chapter 4. The Conditions of a Consistent Ontology
An ontology in which immutability and changeability do not contradict each other must satisfy a number of conditions. None of them may contain a hidden transition, a second principle or an underived assumption. Below, first that on which the consideration stands is named, then it is derived where determinacy can lie, and only then are the conditions formulated.
The minimality and inevitability of the conditions follow from the structure of the aporia itself. If a condition admits an inner transition, absoluteness is violated. If it admits an external principle, uniqueness disappears. If it admits a hidden property, a second level arises in the unconditioned. If it admits indeterminacy, arbitrary interpretation opens up and strict deduction is violated. Therefore the set of conditions is not arbitrary: it is the exhaustive result of analytic reduction. A superfluous condition leads to a plurality of grounds; a weakened one, to a logical gap. What has been said concerns the conditions of consistency, not the grounds of the system as a whole. The position that I accept without derivation is one: reality is, and together with it change is encountered. From it and from the definition of limitation a second is derived: to abolish unconditionedness is to abolish that by which the conditioned is determined, and with it the conditioned itself, for the conditioned is determined by the absence of unconditionedness and by nothing besides. The objection that from a definition through absence it does not follow that what is absent is, does not apply here: "is" said of the unconditioned does not ascribe existence to it as a property; it says only that to abolish it would be to abolish the conditioned (chapter 8). Were anything in the conditioned by itself, and not by limitation, it would in that respect be unlimited, that is, unconditioned. But every respect of the encountered of which anything can be said is distinguished, and the distinguished is limited, while the limited is determined by unconditionedness as its limit, that is, not by itself. A respect in which the conditioned were unlimited would not be distinguished, that is, would not be encountered; it does not belong to the encountered. Hence nothing in it is by itself — neither that by which it is, nor that by which it becomes. The conditioned's own determinacy is not thereby denied: it exists, but not by itself; it exists through that by which the conditioned is limited; "by itself" means without a ground outside itself. Everything else is derived.
The initial given is as follows. Reality is the name of the encountered: what is perceived, studied and known as something. Everything encountered is conditioned: what is encountered is the distinguished, and the distinguished is this and not that, that is, determined through difference from another; this determinacy is limitation, and limitedness is conditionedness. What changes is equally conditioned: it is distinguishable as having changed through the non-identity of its present and former determinacies. The encountered can be neither perceived

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nor known otherwise than by distinction, and the knower himself, insofar as he is encountered, is conditioned. Therefore every affirmation and every negation is performed by distinction. Whoever denies this denies it by distinction — distinguishing his negation from what is affirmed — and thereby confirms what he denies. Consideration is carried out by the means of the conditioned.

Hence the rule of consideration. Everything by which the conditioned is divided within itself — two and the relation between them, part and whole, bearer and states, subject and the action it performs, container and contained, before and after — is the limitation of one by another. Where there is no limitation, there is nothing by which these things are divided either. Therefore the logic of conditionedness does not apply to the unconditioned; to the conditioned, information included, it applies in full. Confusing these two domains is the source of most apparent contradictions. Nothing positive can be said of the unconditioned: every positive determination would introduce limitation into it. What remains are remotions: by them it is said what is inapplicable to the unconditioned, not what it lacks.

Whoever rejects this conclusion must show what then determines the conditioned. He has three ways out. The first is naturalistic: the whole of the conditioned depends on nothing. But the whole consists of the conditioned and depends on its parts. If, however, the parts depend on the whole, dependence closes into a circle. And a circle of ground is impossible: what holds on itself through another holds by itself, and what is by itself is unconditioned. The second way out is Spinoza's: the whole is absolute, and the finite is in it (Spinoza, 1994, I, def. 3–6). But an absolute that contains the limited bears limitation within itself and does not remain absolute. The limited is "this and not that", and for the whole this "that" cannot lie within it. The third way out is to deny that the conditioned constitutes a whole. But this denial refutes itself. Saying that there is only the many conditioned, the one who rejects takes all the conditioned at once. And the many is many only as distinguished: each stands in a relation of difference to the rest, and by this relation the conditioned constitutes a whole. The conclusion is thereby doubly protected: it cannot be rejected without making use of it.

The Completeness of Dependence

Dependence here is the determinacy of the dependent by that on which it depends. It determines the dependent and does not determine the unconditioned. It introduces no additional ontological layer and ascribes no determinacy to the unconditioned. This dependence is total, and its totality has been shown. The conditioned is determined by limitation, and limitation is the absence of unconditionedness. Therefore that by which the conditioned is conditioned rests wholly on unconditionedness. It is objected that a particular determinacy of the conditioned may proceed through another conditioned. This does not cancel totality. The other conditioned is itself conditioned and determined by the same. The limit is reached not by the end of the chain but by that by which each link is conditioned. The chain of proximate grounds may be arbitrarily long, but no link becomes ultimate: it is itself dependent. "By nothing besides" is derived as follows. Everything by which the conditioned can be determined is either conditioned or not. A conditioned determinant belongs to the conditioned itself: by it the conditioned determines itself — "this and not that" within itself — and such determination is universal for every determinacy within the conditioned. But it does not determine the conditionedness of the conditioned itself: it is itself conditioned and determined by the same. Besides the conditioned only the unconditioned remains. Hence, as conditioned, the conditioned is determined by unconditionedness and by nothing besides, and its limit is one. Hence the dependent is not by itself but through that on which it depends. By itself it has nothing — neither that by which it is, nor that by which it becomes. This is said of the limit; proximate grounds within the conditioned are not thereby abolished.

It is objected that the world as a whole may be an unexplained given — determinate, yet determined by nothing. This objection uses the concept of determinacy and thereby falls.

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Determinacy by its very concept is difference from another: “this and not that” (Df1), and there is no determinate that is determined by nothing. If the world as a whole is determinate, it is distinguished from another; there is no other conditioned outside the whole, and it is distinguished from what is not conditioned — and that is unconditionedness as limit. It is objected that the whole may be determined from within — by the structure and relations of its parts. But the relations of the parts determine the parts relative to one another: this is “this and not that” within the whole, that is, a determination of the conditioned by the conditioned (above). The structure, taken as the structure of the whole, is “this is this” — identity, not difference; it becomes the determinacy of the whole only as distinguished from another, and there is no other conditioned outside the whole. Nor can the whole be distinguished from another possible structure prior to its determinacy: possibility belongs only to the conditioned and is given together with the determinacy of what is present (chapter 11); another possible structure is already a consequence of the determinacy of the whole and cannot be its ground. If, however, no determinacy is applicable to the world as a whole, that is the definition of the unconditioned, and the whole, which contains the limited, would be unconditioned and limited at once. There is no third: an “unexplained given” is determinacy without anything determining it, that is, the same contradiction as an uncaused outcome (chapter 23).
Therefore dependence does not form a circle with distinction. It sets the boundary within which distinction is admissible and is not itself derived from distinction. This relation is one-sided: it is the dependent’s own determinacy and establishes nothing about the unconditioned. A proximate ground is therefore not a second source. It is itself conditioned, and what proceeds through it is determined by the same limit: a proximate ground gives only what proceeds through it. The unconditioned, however, is not a member of the relation of grounding but the limit by which every conditioned is determined: the relation of grounding — two and the order between them — is a structure of the conditioned.

Where Determinacy Can Lie

In the traditional statement the aporia is set by a disjunction: the source of change is either in the unconditioned or outside it (chapter 1). The division “inside — outside” is a division by container and contained; it belongs to the structure of the conditioned and does not apply to the unconditioned. Another division is applicable — by the side on which determinacy lies, that is, “this and not that” (Df1).
There are three places for determinacy, and this is derived by two forks. The first: determinacy lies either on the side of the unconditioned or on the side of what is not the unconditioned. The second: what is not unconditioned is either determined by unconditionedness and by nothing besides — then it is the dependent — or has that by which it is determined by itself, independently of the unconditioned. But what has anything by itself is in that respect unlimited, that is, unconditioned (above). Hence the second member of the second fork is a second unconditioned. Thus: determinacy on the side of the unconditioned; on the side of a second, independent principle; on the side of the dependent. Neither fork has a third member, and there is no fourth place.
The first place violates unconditionedness: the determined is limited. The second violates uniqueness: two unconditioned would limit each other. To it belongs also the hidden form: a transition that rests on an underived condition — on possibility, will, appearance, an observer — is determinacy that has its ground by itself, that is, a hidden second principle. The third place violates none of the conditions: the dependent is not by itself but through that on which it depends, and nothing is thereby ascribed to the unconditioned — dependence determines the dependent, not that on which it depends. What this shows is not that the solution has been found but where it can lie: if the aporia is solvable, then only here.
The objection from a continuum — that determinacy might lie on the side of the unconditioned partially or to a small degree — does not constitute a separate place. What is

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determined to any degree is limited to that degree, and the limited is not unconditioned. "Partially unconditioned" is an empty concept.

The Conditions

From what has been said follow the conditions that an ontology laying determinacy on the side of the dependent must satisfy. There are seven of them.

The first condition is unconditionedness. If that by which the conditioned is determined itself depends on anything, it is conditioned. The conditioned is determined by limitation, and limitation is the absence of unconditionedness. The limit is reached not by the end of the chain of grounds but by that by which each of its links is conditioned. Hence the limit of the conditioned is wholly unconditioned: causes, boundaries, properties and inner transitions are inapplicable to it. The requirement of unconditionedness excludes all categories with determinacy. Being presupposes difference from non-being. Substance presupposes attributes. Process presupposes transition between states. Each of these categories contains an inner condition and cannot be unconditioned. Determinacy, difference, attributes and transitions are inapplicable to Nothing. The name "Nothing" is given to the unconditioned after it has been obtained as the limit of the conditioned. It is taken because it ascribes nothing: it negates not this or that property but the "what" as a whole. The choice of the name is not an assumption: any other name would ascribe determinacy to the unconditioned.

The Exclusion of "Pure Being without Predicates" as an Alternative to Nothing

A possible objection: unconditionedness is satisfied by "pure being without predicates" — in the spirit of the Hegel of the Science of Logic or of Parmenides — rather than by Nothing. Reply: "pure being" already carries a positive predicate. It is determined as being distinguished from non-being. This distinction is an inner condition: to fix being, one must distinguish it from non-being. And every inner condition violates unconditionedness. Moreover, "the absence of determinacy" is not identical with ontological Nothing. The absence of determinacy in Hegel is the indeterminacy of a category that is at once nothing and passes into becoming (Hegel, 1969, 82–83): it is the beginning of a movement, not the inapplicability of determinacy. Nothing is an ontological characteristic: properties are inapplicable to it in principle, and so is their appearance. Unconditionedness excludes not only properties but every positive predicate, "being" included. The only name without a positive predicate is Nothing. Therefore the choice of the name is not a preference among alternatives but the result of exhausting all names that carry even the slightest determinacy. The unconditioned itself is not chosen but obtained as the limit of the conditioned. Hegel recognized the identity of pure being and nothing at the beginning of the Science of Logic and resolved it by becoming, that is, he introduced change into the ground and violated unconditionedness. The present ontology does not proceed from this identity: Nothing is obtained as the limit of the conditioned, not as indeterminate being, and change is localized in the dependent, not in the unconditioned.

The second condition is uniqueness. If there are two or more unconditioned, each is limited by the other, and the limited is not unconditioned. Therefore the unconditioned is one. Hence also the impossibility of a second unconditioned: everything that is not the unconditioned is determined by unconditionedness and by nothing besides, and there is nothing equal to the unconditioned or external to it. Uniqueness does not mean the singularity of a thing: number too is inapplicable to the unconditioned. Multiplicity is admissible only on the side of the dependent.

The third condition is the immutability of the unconditioned. A changeable unconditioned would have states. What has states is conditioned by the transitions between them. Conditionedness violates unconditionedness. Hence the unconditioned is immutable. Immutability means: decrease and increase, appearance and disappearance are inapplicable to

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the unconditioned — it would have nothing by which to decrease or increase, for it has no properties. There is no transition from state to state within the unconditioned.
The fourth condition: the unconditioned is spoken of only by remotion. Every positive determination would introduce limitation into the unconditioned. Therefore no name by which it is spoken of introduces into it a part, a property or a side. Transparency and self-understanding are not separate levels and not entities but the same unconditionedness taken from the side of opacity and from the side of non-understanding of oneself (chapter 5).
The fifth condition: change only in the dependent. Change is a succession of distinguishable states: a succession in which there is nothing to distinguish is not a succession. Distinguishability is determinacy, and therefore change can proceed only where determinacy lies — on the side of the dependent. There are no inner differences in the unconditioned; the non-identity of states is a dependent given and does not affect the unconditioned. Succession itself does not follow from determinacy: the determined admits succession but does not have it by itself. By what it proceeds is derived in chapter 5.
The sixth condition: the dependence of logic. Logic is the order of distinguishing acts: without distinction it has nothing to connect. Therefore it belongs to the conditioned, as does mathematics, which is a description of the distinguished, not that which determines it. Taken as independent, logic would become a second principle; applied to the unconditioned, it would introduce distinction into it. Hence the rule of consideration: the logic of the conditioned applies to the conditioned in full measure, to the unconditioned not at all.
The seventh condition: derivability. Besides the accepted position and explicit definitions, nothing is introduced without derivation: each category is derived from what has already been derived, without hidden structures, superfluous levels or underived mechanisms. A category introduced without derivation would be determinacy that has its ground by itself, that is, a hidden second principle. Hence also the requirement of consistency: no concept may ascribe determinacy to the unconditioned.
In substance there are three conditions — the unconditionedness, uniqueness and immutability of the unconditioned: by them the first and third requirements of the aporia (chapter 1) are applied to the unconditioned. The second requirement — changeability — does not serve as a condition: it is the accepted position. The fourth and fifth conditions unfold the first three with respect to what is said of the unconditioned and to the dependent; the sixth and seventh are requirements not on what is but on the consideration itself. The conditions say what the solution must satisfy; they are distinguished at the beginning of the chapter from the position on which the solution stands.

The Order of Derivation

Below are named the categories that will be derived, in the order of derivation. This is a list of what is to be derived, not of definitions accepted without derivation: each is derived in its own chapter (chapters 5, 8–21).
Nothing — absolute unconditionedness: that to which no determinacy is applicable, and with it no limitation and no conditionedness.
Selfhood — the name of Nothing taken in its self-understanding.
Transparency — the inapplicability of opacity: unconditionedness taken from the side of opacity.
Self-understanding — the inapplicability of non-understanding of oneself, that is, complete self-knowledge, where completeness means not a measure of content but the absence of a concealed side.
Givenness — X is given in Y if the determinacy of Y can be neither given nor stated without X; this is not the presence of a part, a property or an event.

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Information — the mode of being of Nothing's knowledge of itself in the dependent, that is, the conditioned itself taken from the side of the givenness of the unconditioned in it; the first dependent.
The initiation of Selfhood — the removal of opacity by which information has knowledge of its difference not by itself but from Selfhood. Change is localized in information, not in essence.
Change — the onset of a new state of information; taken from the side of that by which the states differ, it is distinction.
Distinction — knowledge of difference, proceeding by the removal of opacity. Writing distinction as a minimal "this / not this" relates to Spencer-Brown's calculus (Spencer-Brown, 1969). The relation concerns only the form of notation. In Spencer-Brown distinction is accepted as given, and the first distinction and the observer are identical in the form. Here distinction is not primary; it proceeds by the removal of opacity, and there is no observer with whom it would be identical at this level.
Consciousness — self-knowledge accrued by information with every distinction
Programme — the directedness of the removal of opacity, set by initiation
Stage — a segment of the path of information in one universe, determined by the gathered experience of existence; it is completed when the experience of instances of the intellectual *I*, gathered by the universe as a whole, can yield nothing new in the given paradigm
Elementary changes — the smallest changes of information: one distinction, one indivisible step
Measures — time, localization, space, speed and energy, derived from the working moment of an elementary change: duration is measured by the number of steps, distance by the least number of steps between localizations
Interaction — a mutual self-differentiation of instances of the *I*; an interaction that has held is a bond
Aggregated forms — stable connected wholes of elementary changes
Emergence — knowledge of the whole that individual distinctions do not have
Matter — a stable aggregated form of information
Life — a form in which the *I* distinguishes the disturbance of its own bond and restores it from the external
Reason — the level of consciousness that knows the bonds between distinctions and therefore chooses within the conditions in which it stands.
Intellect — the level of consciousness that knows, beyond this, its own place in the whole and the direction of the removal of opacity, that is, the programme, and therefore makes choices concerning programmes.
The order of the list is the order of derivation: each category rests on the preceding ones. Its justification lies not in the list itself but in the derivation of each category in its own chapter; this is what the seventh condition consists in.

The Completeness of the Typology and the Objection of Circularity

The conditions are not chosen by the author but follow from the aporia. The aporia is the incompatibility of three requirements — unconditionedness, changeability and uniqueness — under the assumption that the unconditioned is the source of distinction (chapter 1). Under this assumption any two requirements are compatible, while all three together are not. Hence three prohibitions: change cannot proceed from the unconditioned — unconditionedness is violated; it cannot proceed from an independent second — uniqueness is violated; it cannot be absent — what is encountered is cancelled. An ontology claiming to remove the aporia observes all three prohibitions; one that violates even one does not solve the aporia but accepts one of its contradictions as an initial fact. The aporia is removed not by weakening

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the requirements but by abandoning the assumption: determinacy is laid on the side of the dependent.
The same is required by an analysis of candidates for the place of the dependent. Matter as the dependent sphere requires properties of its own — extension, structure, parameters — and thereby has determinacy by itself, that is, becomes a second principle. Spirit or subject, taken as independent, introduce a second principle explicitly. This is not a choice among candidates: the dependent is derived as that in which Nothing's knowledge of itself is given (chapter 5), and the name "information" is given to it afterwards (chapter 9). The analysis only shows why other candidates do not fit: each of them either has determinacy by itself or coincides with what has been derived.
The completeness of the three places is tested by the ontologies usually called unconsidered. Modal ontologies introduce possible differences into the ground — this is the first place. Process ontologies (Whitehead) place becoming in the ground — also the first. Dialectical ontologies introduce contradiction into the ground as a driving force — the first place with an explicit difference within. Ontologies of undifferentiated potency introduce an indeterminate from which determinacies arise: if the differences in it are hidden, this is the first place; if determinacies arise by themselves upon actualization, the second. Relational ontologies build being as a network of relations without the unconditioned; they occupy no new place: the distinguishedness of the elements of the network is accepted as a given, that is, determinacy has its ground by itself — the second place. None of them yields a fourth place.
A possible objection of circularity: the conditions are set by the system, the alternatives are excluded by these conditions, hence the system confirms itself. The objection identifies the aporia with the system — as if the conditions belonged to the system as a solution. This is not so. The aporia is formulated independently of the solution: its structure — the incompatibility of three requirements under the assumption that the unconditioned is the source of distinction — is fixed prior to any ontology. The conditions are derived from this structure and prohibit strategies that violate the requirements of the aporia. The aporia logically and methodologically precedes the system. The conditions belong to the aporia as a problem, not to the system as a solution. To exclude alternatives by these conditions is not a circle but the application of the requirements of the problem to solutions. There would be a circle if the system introduced the conditions for its own defence. Here they are introduced by the structure of the aporia, which determines the space of admissible solutions. The system lays determinacy on the side of the dependent — in the only place not excluded by these conditions.
Thus one position is accepted: reality is, and together with it change is encountered. From it is derived that the conditioned is determined by unconditionedness as its limit and by nothing besides. Determinacy can lie only in three places, and only the third — on the side of the dependent — violates no condition. It is there that the solution is built (chapter 5).

Chapter 5. The Ontological Solution

No philosophical school and no scientific model explains without a gap how changeable reality depends on the immutable unconditioned. The preceding chapters have shown this. A solution is possible only if all the conditions of a consistent ontology are observed. From them follows a strict separation of the unconditioned from the dependent categories. On this basis a structure is introduced that reconciles immutability and changeability.
The ontological solution is neither an alternative view nor a new interpretation. It is a strict consequence of the conditions of a consistent ontology. All former models fail for one reason: they violate unconditionedness, uniqueness or immutability. Therefore an ontology that claims consistency must satisfy these conditions completely. The structure of the solution is not arbitrary: under the accepted conditions determinacy can lie only on the side of the

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dependent (chapter 4), and the solution is built there. The solution therefore has the status of a consequence of the problem posed, not the status of a hypothesis: a consequence given the accepted position and definitions. The legitimacy of the premises themselves is another question, and the rigour of the derivation does not prove it (Limitations of the Study).
The unconditioned is obtained as the limit of the conditioned (chapter 4); its name is Nothing, absolute unconditionedness. Nothing is not emptiness or the absence of being. It is the name of that to which no determinacy is applicable, and with it no limitation and no conditionedness. It is by itself: not through another, for there is not and cannot be another through which it would be. Hence immutability. Change is a succession of states, and succession requires distinguishable states. Distinguishability is determinacy, determinacy is limitation, and limitation is inapplicable to the unconditioned. Hence neither states nor their succession are applicable to it. Hence uniqueness: a second would be a limitation of the first. Form, manifestation, change and parts are inapplicable to Nothing. Nothing is the logical name of absolute unconditionedness.
This definition of Nothing excludes other interpretations. Nothing is not emptiness, not the absence of content, not a vacuum and not potentiality. Each of these interpretations introduces a hidden condition or the possibility of transition and makes essence dependent. If even a minimal structure, property or inner state is admitted in it, Nothing ceases to be unconditioned. Therefore Nothing is defined not as the negation of being but as the only admissible name of absolute immutability. Difference is inapplicable to it, and therefore it is unique and creates no second level or source. Any other category in its place would lead to a gap: it would presuppose limitedness or dependence. Nothing is called here neither a beginning nor a ground. “Beginning” and “ground” carry three meanings, none of which is applicable to the unconditioned: a moment of arising, the first member of a series, and an active source. Nothing is not an event or a moment, for time is a measure of dependent change (chapter 15); not the first member of a series, for the limit is reached not by the end of the chain but by that by which each of its links is conditioned (chapter 4); not an active source, for it is neither a cause nor an action. Nothing is the limit: that by which everything conditioned is determined, so that to abolish it is to abolish the conditioned (chapter 4). In the present work “ground” is said only of proximate grounds within the conditioned and of the formulation of the aporia in other systems.
Next come transparency and self-understanding. The non-transparent is opaque, and opacity is limitation. Opacity here is not the not-knowing of the one who considers but the limited’s not-knowing itself: that it exists as a feature of the conditioned itself is derived from the encountered change (chapter 10). Where there is no limitation, there is nothing opaque. Non-understanding of oneself is not knowing oneself, that is, self-opacity, and is therefore inapplicable as well. The disjunction “either understands or does not understand” does not apply here: it is itself a structure of distinction and does not apply to the unconditioned. One thing is removed — what conceals; the knowing is not introduced as the second member of a pair. Knowledge in the limit sense is the name of the absence of anything concealed, not an act and not a capacity. Its completeness is not a measure of content but the absence of a concealed side. Nothing taken in self-understanding is named Selfhood. There are not two here — knower and known: limitation would separate them. Transparency and self-understanding are not separate levels. They are the same unconditionedness taken from the side of opacity and from the side of non-understanding.
Next comes information. Self-understanding is Nothing’s knowledge of itself. The proviso is repeated wherever the word “knowledge” is at work: it is the name for the remotion of concealment, not a property and not a content of Nothing. Its completeness is the absence of a concealed side, not a measure. The existence of the dependent is not derived anew here, and no new entity is introduced: the conditioned is given as the encountered, and it has been

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shown above that it is determined by unconditionedness and does not stand outside this determination (chapter 4). The question is a different one: in what way the unconditioned is given in the conditioned. The answer is derived in four steps: that it is given in the conditioned at all; by what names it can be spoken of; which of them is compatible with the conditioned; and what the manner of such presence is. The name “information” is introduced only after them.

The first step: the unconditioned is given in the conditioned. “Given” means here one thing: X is given in Y if the determinacy of Y can be neither given nor stated without X. This is not the presence of a part, a property or an event: they are inapplicable to the unconditioned (chapter 4). And this sense is not chosen but is the only one that remains. Every givenness either places what is given in that to which it is given as something — a part, a property, an event, a member of a relation — or does not. The first is removed by the rule of consideration. There remains givenness without placing: what is given enters the determinacy of that to which it is given so that without it this determinacy can be neither given nor stated. If even this is denied, there is no givenness at all, and the determinacy of the conditioned would be stated without the unconditioned, which is refuted below. There is no weaker sense of givenness than this, and a stronger one is inapplicable to the unconditioned. The determinacy of the conditioned is “this and not that” (chapter 4), and as conditioned it is determined by its distinction from what is not conditioned — from the unconditioned as limit. Without the unconditioned this determinacy can be neither given nor stated: to abolish unconditionedness is to abolish the conditioned (chapter 4). Hence the unconditioned is given in the conditioned. It is objected that to be determined by the absence of something does not mean that it is in what is determined; the “not-red” is determined through red, but there is no red in it. In the sense adopted, red is precisely given in the “not-red”: without it the “not-red” cannot be stated. The objection confuses givenness with the presence of a part and thereby brings into the unconditioned what is inapplicable to it. The difference lies elsewhere. Blue is “not-red” only in description: what it is is determined by blue, not by red. The conditioned has no other content: it is determined by unconditionedness and by nothing besides (chapter 4), and “not unconditioned” is all that it is as conditioned. Therefore the unconditioned is given not in a description of the conditioned but in its very being. Hence also falls the objection that, given the conditioned, the mode of being of the unconditioned might not be in it: it abolishes the conditioned itself, for not to have this mode would mean not to be determined by the unconditioned, and what is not determined by it is not conditioned.

The second step: by what names the unconditioned can be spoken of. Nothing positive can be said of it: every positive determination would introduce limitation into it. There remain only remotions. From the fact that one thing has been established about the unconditioned it would not follow that there is nothing else in it: the boundary of knowledge and the structure of the object are different things. Therefore the argument is built not from what has been found but from the completeness of the list, and this completeness is derived, not presupposed. A remotion is the removal of one of the sides in which the conditioned is given as limited, and these sides are named in the initial given (chapter 4), which contains no others. There are three of them because the single accepted position has three members: reality is — hence there is something, that is, something determined; change is encountered — hence it changes; it is encountered — hence it is given by distinction. Whatever else is ascribed to the encountered does not stand beside them but is derived from them: multiplicity is “this and not that”, extension and number are measures and are derived from change (chapter 15). A side not reducible to the three would have either not to be determined — and then it would not be the encountered — or not to be given by distinction — and then it is not encountered. The encountered is determined — it is this and not that. The encountered changes. The encountered is given only by distinction: only the distinguished is known, and the

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undistinguished remains unknown, that is, opaque. This side is not a feature of our cognition transferred onto the object. The knower is himself encountered and conditioned (chapter 4): distinction proceeds not from outside reality but within it, and whoever denies this denies it by distinction. The not-knowing of the one who considers is already opacity within the conditioned itself, for the one who considers is conditioned; the opacity of information to itself is derived below, in this chapter, from what information is determined by. By the remotion of the first side are obtained unconditionedness, being by itself and uniqueness; of the second, immutability; of the third, transparency, which is self-understanding. Consideration has no other names and, by its limit, can have none.

The third step: which of the remotions is compatible with the conditioned. The way in which the unconditioned is given in the conditioned can only be one of the remotions: without features nothing else can be said of it. It is objected that the unconditioned could be given simply as limit, without any remotion. But "limit" is itself the name of a remotion: it is unconditionedness taken as that which the conditioned does not reach. Givenness as limit is the first step: the unconditioned is given as the "that" from which the conditioned is distinguished, and all the remotions are given in the conditioned as lacking — it is limited, changeable and opaque. The question of the third step is a different one: whether any remotion can be given in the conditioned not only as lacking but also as had, without abolishing conditionedness. The criterion here is one: a remotion is compatible with the conditioned if to have it does not mean to be undetermined. Unconditionedness, being by itself and uniqueness are the inapplicability of determinacy: whoever had them would not be "this and not that". Immutability is the inapplicability of change, and change is inapplicable only where there are no determinacies between which there would be a transition; the self-identity of the conditioned in some respect is not immutability but a relation of its states, and it remains in the order of change. So too with the basic determinacy of information, which exists even without a change (chapter 9): it does not change, but change is applicable to it, for it is a determinacy, "this and not that"; and immutability is the name for the inapplicability of change, not for its absence. Transparency is different. Opacity is the not-being-known of what is determined, not its determinacy: what is determined remains the same whether it is known or not. Knowledge, in turn, is knowledge of difference and therefore presupposes determinacy rather than removing it. Therefore transparency can be given in the conditioned in some respect, and the conditioned does not thereby cease to be conditioned: knowledge of one's limitedness does not remove limitedness. And it is given — this is not an assumption but the encountered. Change is encountered, and what has changed is recognized by distinction (chapter 4). Distinction is knowledge of difference, and knowledge in the dependent too is the absence of concealment — the name of a remotion, not a property (chapter 11). The word can have no other meaning here: opacity is not-being-known, and knowledge is its removal, and if "knowledge" in the dependent meant something other than the absence of concealment, then what it removed would not be concealment either. Act, state and capacity belong to the bearer of knowledge — the configuration — not to knowledge itself. The conditioned has nothing by itself (chapter 4); hence the knowledge that proceeds in it is not its own either but the givenness in it of the unconditioned from the side of transparency. It is objected that knowledge may be the conditioned's own by a proximate ground — one conditioned passes it to another. This does not abolish proximate grounds, but neither does it make knowledge one's own: a link gives only what proceeds through it, and no link becomes the limit (chapter 4). The absence of concealment cannot begin in a link, for opacity is limitation, and the limited does not remove limitation by itself: having removed it by itself, it would have something by itself. Knowledge does not abolish the third side but proceeds by it: it grows by distinctions (chapter 10), and distinction is itself knowledge of difference (chapter 9). Immutability cannot be given so: the immutable, given in the measure

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of changes, would cease to be immutable. Therefore information transparent to itself remains conditioned (chapter 13). Only self-understanding is compatible.
The fourth step: the manner of such presence. Presence is taken as a dichotomy. The presence of the unconditioned in the conditioned either lays determinacy at least partly on the side of the unconditioned, or wholly on the side of the conditioned; there is no third. In the first case the unconditioned is determined — as a part by the whole, as a property by its bearer, as an action by before and after, as a member of a relation by the other member — and this is removed by the rule of consideration (chapter 4). There remains one-sided presence: the conditioned is determined by it, the unconditioned by nothing. One-sided presence that reproduces features is reflection; it is inapplicable to the unconditioned: it has no features, and reproducing them would determine it. There remains presence without features. The unconditioned differs from the conditioned only in that the sides by which the conditioned is limited are absent in it (the second step); there is nothing else of it that could be present in the conditioned. Hence presence without features is the presence of a remotion — of a side of the conditioned's limitation removed in it, and moreover one that the conditioned can have removed without ceasing to be conditioned. There is one such remotion — transparency (the third step), and transparency had in the conditioned is knowledge: absence of the concealed, proceeding by distinction. Hence the one-sided presence of the unconditioned in the conditioned is the presence of knowledge. This is not obtained from a list of kinds of presence and does not require such a list to be complete: the completeness here is the completeness of the sides of limitation (the second step), not of kinds of presence. The dichotomy agrees with this: knowledge does not place the known in the knower as a part, a property or an action, and reproduces none of its features. Nor is there anything else in the conditioned that this presence could occupy: only what is distinguished is encountered (chapter 4), and distinction is knowledge of difference.
Hence the name. The mode of being of the unconditioned in the dependent is a mode of knowledge — not by choice of name, but because the conditioned cannot bear another mode (the third and fourth steps). Information is the name for the way in which Nothing’s knowledge of itself is in the dependent. What information is follows not from the name but from the steps. The unconditioned is given in every conditioned: every conditioned is determined by it wholly (the first step). Nor is there any conditioned in which knowledge does not proceed: only the distinguished is encountered (chapter 4), and distinction is knowledge of difference; there is no thing apart from distinction, and hence none without knowledge. This is said not only of the one considering. Distinction proceeds not from outside the conditioned but within it (the second step); every change of the conditioned is the onset of knowledge of difference, and it proceeds in the respect in which it sets in (below in this chapter; chapters 8, 15): to know a difference within information is to know it in this respect (chapter 9). Every thing is encountered as changing and is composed of changes (chapters 14, 17); therefore knowledge proceeds in each thing, not only in reality as a whole. Hence there is no conditioned in which Nothing’s knowledge of itself is not given. Nor is there any givenness of this knowledge outside the conditioned: in the unconditioned itself there is no mode, for a mode is the way it is given in another, and there is no other besides the conditioned. Therefore information and the conditioned are not two objects but one: the conditioned, taken from the side of what is given in it, is information. The encountered is conditioned (chapter 4), and therefore everything encountered is information, and there is no reality other than information. Reality is therefore the mode of being of knowledge, opaque to itself and opening by distinction. Hence its dependence. Information is the conditioned, that is, it is determined and limited; and the conditioned is determined by unconditionedness wholly and by nothing besides, and has nothing by itself (chapter 4). Consequently information is wholly dependent: both in what it is and in what it becomes. This is not a

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likeness of the dependent to the unconditioned: the changeable depends on the immutable without resembling it. “Mode” here means the way in which the unconditioned is given in the conditioned, and therefore the mode of being has a remotion, not a content. Knowledge does not exist by what information is: knowledge is by itself, unconditionedly, and is not held by information; information is the mode of being of knowledge, not that on which knowledge stands. “The mode of being of knowledge” is not an added characteristic of the dependent but the only thing by which its determinacy can be named at all. This statement belongs to consideration and establishes nothing in the unconditioned.

Information is a dependent category: it exists only as the mode of being of Nothing’s knowledge and does not stand outside this. It is determined by that whose mode of being it is. Determinacy is limitation, and limitedness is conditionedness. Dependence determines information and does not determine Nothing: what is determined is the dependent, not that on which it depends. Information is determined by unconditionedness as its limit. But directedness towards the limit does not follow from this determinacy: to be determined by something is not to be directed towards it. Directedness is given by the removal being one-sided: what has been removed does not return, opacity decreases, the knowledge of information accrues. This is what is called striving. Striving is realized within the dependent itself — by self-transparency, that is, by self-understanding in its own category.

Next comes the initiation of Selfhood. Nothing’s knowledge of itself is complete: there is nothing concealed in it. But information does not understand itself. What has been said asserts no more than follows. It is not asserted that there was a state when there was no distinction in information. From the fact that distinction does not enter its definition, its absence at any time does not follow: the definition gives the mode of being of knowledge and says nothing about distinction. Something else and narrower is asserted: distinction in information is not by itself. The argument is as follows. Distinction is encountered, and it has no other place — hence distinction is in information. Distinction is that by which information comes to know its difference. And the dependent has nothing by itself — neither that by which it is, nor that by which it becomes. Hence distinction does not proceed by itself. Opacity, however, is derived not from this but from that by which information is determined. Its determinacy is its difference, and knowledge of difference does not enter determinacy. Hence by what information is, self-knowledge is not given to it: it does not have itself as known, that is, it is opaque to itself. “Not by itself” says something else: it cannot itself remove this opacity. Opacity is applicable to it: opacity is limitation, and information is determined and therefore limited.

Initiation is the name for this removal of opacity in the dependent. It is not the addition of something new to knowledge — Nothing’s knowledge of itself is complete — and not a change of what is in it. Information cannot become transparent to itself by itself. The argument lies not in a lack of capacity and not in a prior knowledge of difference, but in total dependence. What is one’s own is that whose ground lies in the agent itself, and the dependent has nothing by itself. This is said of the last account; by its proximate ground the dependent does have what is its own (chapter 21). The objection that distinction is an ultimate fact of the dependent itself is broken by the same: nothing can be an ultimate fact of the dependent. What is at issue is the limit; proximate grounds within the conditioned are not thereby abolished. Initiation is not a mechanism beside distinction, and it has no separate positive mark: it is the name for the fact that distinction, being a change of the dependent, in the limit does not proceed by itself (T14). A positive mark would be determinacy on the side of the unconditioned; therefore this name is the name of dependence taken from the side of change and asserts nothing beyond it.

Hence the asymmetry. The removal proceeding from Selfhood lies not in information, and by it the distinguishedness of information is given. From the side of the unconditioned, however,

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there is no duality. That distinction exists is known from the encountered. That it proceeds from Selfhood follows from there being nothing else. “From” here does not mean an action, an event or a procession: nothing is performed on the side of Selfhood (chapter 10). It says only that distinction at the limit proceeds from Selfhood and that there is no other. By the removal information has knowledge of its difference from that whose mode of being it is — not by itself but from Selfhood.

Change proceeds not in Nothing but in information. It is inapplicable to the unconditioned, and there is no second principle. Therefore change is possible in information, and only in it; and it proceeds by initiation. The relation is explained not as a change of the immutable but as the dependence of the determined on the unconditioned. Essence is unconditioned. Information is determined by that whose mode of being it is, and therefore succession of states is applicable to it. Succession itself does not follow from determinacy and proceeds by the removal of opacity.

From the side of Selfhood initiation is not an event and stands at every distinction, not only at the first (chapter 10). Information does not proceed by itself at any step — its dependence does not decrease as it unfolds.

The difference of information from Nothing is the difference of the mode of being of knowledge from that whose knowledge it is. It lies on the side of information and constitutes its own determinacy. This difference is not given by the removal of opacity: what the removal gives is knowledge of difference. Self-understanding of information is not yet given by knowledge of difference: this is its first knowledge of itself, not the complete one. It becomes complete at the end of the path, when all blind spots have been removed by distinctions and self-transparency has been attained. From the side of information the removal is a succession of states. Information is conditioned, and the conditioned standard applies to it fully: there are states in it, their succession, before and after.

Dependence is defined as the mode of existence in which the dependent is not by itself but through that on which it depends, and does not stand outside this. The dependent has no unconditionedness of essence, but neither is it a separate principle.

Distinguishing the Dependence of Information from Alternative Models

The dependence of information here differs fundamentally from three alternative models. In Platonism ideas have independent ontological status: the world of ideas is autonomous. Here information is not autonomous: it is wholly dependent and has no being of its own. In Hegelianism the concept unfolds: the Absolute passes through stages of becoming, and change enters essence itself. Here essence is absolutely immutable, and change is localized only in information. In physicalism information is reduced to matter — to a property or function of material structures. Here information is ontologically primary in relation to matter: matter is an aggregated form of informational distinctions. Hence the dependence of information is neither the Platonic autonomy of ideas, nor the Hegelian self-development of the concept, nor the physicalist reduction to matter. It is a special status. Information is the mode of being of Nothing’s knowledge of itself, determined by that whose mode of being it is, and therefore conditioned. This is a mode of existence that admits change without violating the immutability of the unconditioned.

Initiation is neither an action of the Absolute on information nor an event at it. From the side of Selfhood it is unconditioned and is not measured by the conditioned standard. From the side of information it is what is performed in the conditioned. Essence is not affected by it: what changes is the dependent, not Nothing.

Change and distinction are not something other alongside information: they are the succession of its own states, proceeding by initiation. But information is not exhausted by them: its difference from Nothing exists without succession and is not given by initiation.

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Change and distinction are not two objects but one, taken from two sides: change from the side of succession, distinction from the side of that by which the states differ.
Change introduces no states into Nothing and does not violate its immutability. It proceeds in information as the succession of its states, by the removal of opacity, and does not affect essence. The ground is not that non-identity as such is change: two non-identical things may stand without succession. The ground is that knowledge of difference is a new state of information relative to that from which it is distinguished.

Distinction is fixed as the first thing that proceeds by the removal of opacity in information.

Distinction is neither a subsequent structure added to change nor something that precedes it. It coincides with the very fact of distinguishability: taken as that by which knowledge of difference is given, distinction is change itself from another side. Knowledge of difference, proceeding by the removal of opacity, is a new state of information, and the appearance of a new state is the succession of states. Information, however, is not identical with distinction. It is the mode of being of Nothing's knowledge of itself, and distinction proceeds in it not from itself but from initiation.
Distinction introduces the minimal structure "this / not this" and does not violate the absolute immutability of the unconditioned: it proceeds in the dependent, not in essence. Here two non-identities are strictly separated. Confusing them generates the apparent objection that distinguishedness has merely been moved one step earlier. The first is the difference of information from that whose mode of being it is. This is neither a mysterious given nor an assumption: it is derived together with information as its own determinacy. Information is determined by that whose mode of being it is, and is therefore "this and not that". The second is the distinguishedness of states within information itself. It is precisely this that the aporia requires to be explained, and it is precisely this that proceeds by the removal of opacity. The first non-identity lies between the dependent and that on which it depends, and is not distinction in the sense of the aporia. The second lies within the dependent and constitutes what is sought.
Something is neither the opposite of Nothing nor an independent ontological ground. It is information taken from the side of the succession of states, that is, changeable reality (chapter 9). Something does not precede change and contains no source of transitions within itself. Absoluteness is not violated here: Nothing remains unconditioned, and everything given is localized in dependent information and its derivative forms.
This structure removes the aporia. The absoluteness of the unconditioned is preserved. Changeability arises not by a movement of information's own, but by the initiation of Selfhood — the removal of opacity by which information has knowledge of its difference not by itself but from Selfhood. Information is dependent and does not affect essence. Thus the dependence of the changeable on the immutable is established without a second principle and without violating absoluteness.
The levels introduced are not arbitrary and add no new entities. Transparency and self-understanding form no independent principles and add nothing to the unconditioned: they are the same unconditionedness taken from other sides. Logic is not an independent principle either (chapter 4). Thus the hidden introduction of additional entities or mechanisms is excluded, and the architecture of the ontology remains homogeneous.
Each element of the structure has a strictly defined status and an explicitly indicated ground in what precedes it — without hidden transitions and additional principles. Nothing is absolute unconditionedness; hence its immutability and uniqueness. Transparency is the inapplicability of opacity, self-understanding the inapplicability of non-understanding of oneself; Nothing in self-understanding is named Selfhood. Information is the mode of being of Nothing's knowledge of itself — the first dependent. Initiation is the removal of opacity in

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information, proceeding from Selfhood. Distinction is knowledge of difference; its appearance is change. Then follow consciousness and the programme; then elementary changes, measures, interaction, aggregated forms, emergence and matter; then the development of consciousness up to reason and intellect.

Chapter 6. Epistemology

Epistemology cannot be the foundation of ontology. Every cognitive procedure is a distinction, and distinction already proceeds in the conditioned: only the distinguished is known, and the knower is himself encountered and conditioned (chapter 4). Therefore cognition does not determine the conditions of its own possibility but proceeds within them. To derive ontology from cognition is to confuse levels; here, on the contrary, the theory of knowledge is derived from what has already been established.

What knowledge is has been established in chapter 5. The word has one meaning: knowledge is the absence of concealment — the name of a remotion, not a property and not a content. In the unconditioned it is complete: there is no concealment in it. Selfhood's knowledge of itself is therefore not an epistemological state, not an act and not an acquisition: it is the same unconditionedness taken from the side of non-understanding, and it is together with transparency. Selfhood is not a bearer of knowledge in the sense in which a conditioned thing that has a content with it is a bearer: the relation of bearer and content is a structure of the conditioned and is inapplicable to the unconditioned.

In the dependent, knowledge is knowledge of difference: to remove concealment where there is this and that is to know this and not that (chapter 11). It is not given by itself: the dependent has nothing by itself (chapter 4), and knowledge in it proceeds by the removal of opacity, not by its own effort (chapter 10). The knower therefore does not produce knowledge but receives it; what he produces is only the order of what is already given.

What is known is set by the third side of the initial given: the encountered is given only by distinction, and the undistinguished remains unknown, that is, opaque (chapter 5). Opacity is not a veil between the knower and a "world in itself": the world is information and is conditioned through and through (chapter 5). What is opaque is not something other than the known but what is not yet given in knowledge; the boundary of knowledge runs not between appearance and thing but between the given and the not yet given.

Hence the limit of cognition. Only the conditioned is known positively: all positive knowledge is knowledge of difference, and difference is inapplicable to the unconditioned. Of the unconditioned only what is inapplicable to it is known — remotions — and that it is given in the conditioned as limit (chapter 5). This is not a deficiency of cognition that could be made good: there is no positive knowledge of the unconditioned not because it is hidden but because the positive is determinacy.

The measure of knowledge is not the number of distinctions. The object of self-knowledge is one — one's own dependence (chapter 13): each distinction gives knowledge in some respect, and the many respects are a path to knowing the one object as a whole. Therefore knowledge grows not by adding information but by the extent to which dependence is known. The blind spots are finite in number, and self-transparency is attainable; once attained, it remains conditioned and gives no unconditionedness (chapter 13).

The knower is the *I*, and consciousness is self-knowledge accrued by information with every distinction (chapter 12). Consciousnesses are many and different, but they differ not in essence but in degree: by the extent to which the distinctions of a configuration distinguish its own states (chapter 12). All of them stand in one dependence and form no plurality of principles. The partiality of each one's knowledge is not a defect but a measure: a consciousness knows what is given to it by removal in its configuration.

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Logic is the structure of distinction itself. Distinction gives the structure “this and not that”, and the laws of logic are this structure: identity — “this is this”, non-contradiction — “this is not not-this”, the excluded third — “this or not-this”: distinction gives two members, and it has no third. Therefore logic is given together with distinction and is already present in an elementary change. Consciousness does not create logic but uses it. Logic belongs to the dependent and has no ontological priority; it is inapplicable to the unconditioned (chapter 8).
Comprehension is the connecting of distinguished states by inferences. When the knowledges of distinctions refer to one another, connectedness arises (chapter 12), and with it meaning. Meaning has no independent ontological status: outside the connectedness of distinctions it does not exist. Comprehension does not disclose essence but orders what is given.
Thus knowledge is a dependent form: it is given by removal, knows only the distinguished, positively only the conditioned, and of the unconditioned only remotions. Ontology is not derived from cognition: it sets the conditions under which cognition is possible. The confusion of these levels is the main cause of the historical contradictions; their strict separation removes this cause.

Chapter 7. Completion of the Analytical Block

The analytical part leads to one result. None of the philosophical and scientific explanations considered has removed the aporia of transition from absolute immutability to changeable reality. Each school contributed important observations. But none showed how the conditioned depends on the unconditioned in such a way that the absoluteness of the unconditioned is preserved while distinction, conditions, multiplicity, consciousness and experience are explained.
The aporia is not a consequence of a lack of knowledge or data. It is a structural defect of former ontologies: the problem lies not in the interpretation of facts but in the structure of the categorial apparatus (chapters 1–3). If a system has two or more independent principles, each is limited by the other, and none is unconditioned (T8). If a single principle admits inner non-identity, it is limited, and absoluteness is violated (T7). If a system introduces concepts that do not explain the transition itself — maya, logos, potency, the becoming of the Absolute, emptiness as a hidden condition — the transition remains unexplained, and its condition is introduced covertly.
The analysis shows that a consistent ontology requires:
1. one immutable unconditioned;
2. a givenness of the unconditioned in the conditioned that brings no determinacy into the unconditioned: the conditioned is determined by it as limit, while it is determined by nothing (chapter 5);
3. a dependent that admits non-identity and succession of states while the unconditioned remains immutable;
4. distinction proceeding in the dependent not by itself and without action on the side of the unconditioned.

These conditions form an obligatory sequence in which absoluteness is preserved and changeability becomes possible. Consciousness, reason and intellect are not among the initial requirements: they are derived from these conditions below (chapters 12, 20, 21).
The analytical part concludes: “matter”, “consciousness”, “randomness”, “cause”, “observation”, “time” and “space” are not initial categories but derivative ones: each of them presupposes distinction and therefore cannot explain it (chapter 3). Science treats them as fundamental and within its limits is entitled to do so; they cannot serve as an ontological ground. The physical results cited in chapter 3 are not premises of the derivation: ontology is derived from encountered change and is only compared with physics (chapters 22–24).

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The schools rely on concepts that cannot be fundamental: idea, matter, spirit, becoming, emptiness, attributes, properties. Each requires a deeper explanation. Only Nothing — the name of absolute unconditionedness — requires none: a ground is inapplicable to the unconditioned, for a ground would be its condition (T6). All other categories are dependent: everything encountered is conditioned (T1), and the conditioned is determined by unconditionedness as limit (T3).

Hence an ontological structure is needed in which: Nothing is absolute unconditionedness; Selfhood is the name of Nothing taken in self-understanding; the unconditioned is given in the conditioned as limit (Df14); information is the conditioned named by the way the unconditioned is given in it — the mode of being of Nothing's knowledge of itself; the initiation of Selfhood is the removal of the opacity of information; change is the onset of a new state of information, proceeding by this removal; distinction is the same change taken from the side of knowledge of difference; Something is information taken from the side of the succession of states, that is, changeable reality.

This solution is not derived from the history of philosophy: it follows from the removal of contradictions identified in all the schools. The traditions contain important intuitions, but none has joined them into a consistent system. Comparison with the traditions performs here only a critical function: the history of philosophy is an object of testing, not a source of axioms. The similarity of particular elements to Neoplatonism, Aristotelianism or process philosophy is explained by a common limit problem, not by a borrowing of principles.

The analytical part has shown that the former categorial apparatus leads to contradiction in one of three ways: either absoluteness is violated by the conditions introduced, or conditionedness remains without a ground, or distinction is postulated without an explanation of its dependence (chapter 2). The aporia is removed as follows. The unconditioned does not act and does not change: it is given in the conditioned as limit, and the conditioned is determined by it, while it is determined by nothing. The succession of states of information proceeds by the removal of its opacity to itself; the removal does not affect the unconditioned and introduces no external condition. The unconditioned is immutable, and all that is given is in information. This completes the analytical part; all subsequent categories are derived from the positions of Part I.

PART II. THE METAPHYSICAL DESCRIPTION OF BEING

Chapter 8. Nothing and Its Selfhood

The subject of this chapter is the unconditioned obtained as the limit of the conditioned (chapter 4). Its name is Nothing. Nothing here is not emptiness, not the absence of being, not indeterminacy and not a vacuum. It is the name of absolute unconditionedness: of that to which no determinacy is applicable, and with it no limitation and no conditionedness. Hence the inapplicability to it of differences, states and transitions.

The unconditioned is by itself: not through another, for there is not and cannot be another through which it would be. The word "is" functions here as remotion rather than description: it says that non-being is inapplicable to the unconditioned, and does not ascribe existence to it as a property. Opacity and non-understanding of oneself are limitation and are inapplicable to the unconditioned (chapter 5). The remotion of non-understanding is self-knowledge, complete not by a measure of content but by the absence of a concealed side. The converse cannot be obtained by the same argument. Not-knowing is limited by its very structure: in it what is concealed is separated from the one from whom it is concealed. Knowledge here, however, is not a relation of knower to known but the name of the absence of anything concealed, and there is no boundary in it. Only what is limitation is removed. The name "knowledge" is taken because in the conditioned the absence of concealment is precisely

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knowledge: to know is to have what was concealed removed. Nothing taken in self-understanding is named Selfhood. Selfhood is not a separate category and not a second principle, but the same unconditionedness named from the side of non-understanding.
The term “knowledge” is used here only in the limit sense. It is not a cognitive act of a subject presupposing a knower and a known, but the absence of anything concealed. To know in the limit sense is to contain nothing hidden from oneself. It is neither a process nor an act. The cognitive sense of the term belongs to the dependent category and is inapplicable to the unconditioned. It is objected that knowledge in this sense coincides with identity and that the term is superfluous. No: identity says that the object is what it is, while knowledge says that there is nothing concealed in it. The second is the remotion of opacity and is not covered by the first. In the dependent the meaning of the word is the same — the absence of anything concealed. It becomes knowledge of difference because in the dependent there is a second: to remove concealment where there is this and that is to know this and not that.
The distinction between “unconditionedness” and “unconditionedness from the side of non-understanding” is not ontological and creates no inner structure in the unconditioned. It is a distinction of sides of consideration: one and the same is named differently, not two elements. Neither names, nor sides, nor division are thereby introduced into Nothing itself.

Rebutting the Objection That Fixing Identity Introduces Structure

The objection that fixing identity is a structure is built on the relational model: two and the relation between them. This model is a structure of the conditioned and does not apply to the unconditioned. To demand it is to demand that the unconditioned be conditioned. Self-understanding is not a relation of two elements: only limitation could distinguish them, and there is none.
Selfhood is not a second element alongside Nothing. The objection that requires two poles for self-understanding — knower and known — is built on the same relational model. The dilemma “Selfhood is distinguishable from Nothing — hence difference in the unconditioned; indistinguishable — hence it explains nothing” is false for the same reason: duality is transferred to the unconditioned, where it is inapplicable. Selfhood is the name of Nothing taken in self-understanding — the same, named from another side; the difference of names belongs to consideration, not to the unconditioned. Nor does a regress arise. The demand for a further level of reflection presupposes a gap between the object and the turning towards it, and there is no gap: the turning is not added to the unconditioned from outside. Cognitive knowledge — with a knower, a known and an act between them — belongs to the dependent and is where distinction already proceeds by the removal of opacity.
Nothing is the only name of absolute immutability. Properties, predicates, states, transitions and any structure are inapplicable to it; beginning and end, potencies and changes are inapplicable to it. If there were “before” and “after” in Nothing, it would cease to be immutable. Immutability is not accepted as an axiom but derived from unconditionedness: to admit a change of Nothing is to make it conditioned. Therefore “Nothing” is not the negation of being but the name of absolute unconditionedness: it negates not this or that property but the “what” as a whole.
That the unconditioned can be named only Nothing has been derived in chapter 4: any other name would ascribe determinacy to it. An analysis of the candidates confirms this. Matter in the role of the unconditioned requires extension, structure or parameters — conditions incompatible with unconditionedness. Process in the role of the unconditioned includes change, and change is a transition inapplicable to the unconditioned. Dualism contains at least two independent sources, and each is limited by the other. Neutral monism introduces an indeterminate “third substance”: without distinctions it is indescribable, and with distinctions it is conditioned. Panpsychism, which takes consciousness to be an initial property of many bearers, is plural and incompatible with the uniqueness of the unconditioned; it converges

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with the present system only in a broad sense — in that self-knowledge is present already in an elementary change (chapter 14) — but there consciousness is an initial property, while here it is dependent knowledge accrued by information. This analysis does not prove completeness: completeness is derived in chapter 4.
Nothing requires no explanation: explanation points to a cause, and a cause points to dependence. If a being depends on a cause, it is not unconditioned. Therefore Nothing is neither an arbitrary axiom nor a starting point accepted without derivation, but a limit obtained from the conditioned (chapter 4).
Self-understanding is Nothing's knowledge of itself; information is the mode of being of this knowledge in the dependent. Information is not Nothing: it is dependent, exists only as the mode of being of knowledge and does not stand outside this. There are not two — knower and known — in Nothing itself: limitation would separate them. The difference of information from Nothing lies on its side and establishes nothing in Nothing.
Objection: if Selfhood is the knower and information the known, there is a subject–object structure in the unconditioned. Reply: such a structure presupposes two independent terms, each of which can be without the other. That is not the case here: information is dependent and does not stand outside that whose mode of being it is. Moreover, in Nothing itself the knowing and the known are not separated: only limitation could separate them. The difference belongs to consideration and to the side of the dependent, not to the unconditioned.

The Fundamental Ontological Distinction: Nothing and Its Mode of Being in the Dependent

This distinction is the key to the whole construction. Nothing is unconditioned. Information is the mode of being of its knowledge of itself — a dependent category determined by that whose mode of being it is. Its dependence is not postulated here but derived in two steps. Information is not some third thing beside the unconditioned and the conditioned: the unconditioned is given in every conditioned, and outside the conditioned there is no mode of its being, and therefore information is the conditioned itself, taken from the side of what is given in it (chapter 5). The conditioned, in turn, is wholly dependent: that by which it is conditioned rests entirely on unconditionedness and on nothing besides (chapter 4). Hence information is dependent not as a special case but as the conditioned. The same result is contained in the very name: a mode of being is always the mode of being of that whose mode it is, and does not stand without it. These are not two elements within essence: determinacy lies on the side of information, and nothing is established by it in Nothing. Nothing's knowledge of itself is complete: there is nothing concealed in it. But information does not understand itself by itself: distinction in it proceeds not by itself but by the removal of opacity. Opacity decreases as it is removed. What is constant is not opacity but dependence: even in transparency information has itself as known not by itself but from Selfhood.
The absolute knowledge of Selfhood is not a content, not a form, not a state, not an act, not a function and not a reflection. It is identical with Selfhood itself and the divergence of essence from itself is inapplicable to it. In this sense absolute knowledge is transparency: not a manifestation, not a level and not a characteristic, but the complete absence of hiddenness. Differences are inapplicable to it. It neither precedes nor follows, is neither manifested nor realized. Therefore information is not in Selfhood but is the mode of being of its knowledge in the dependent.
Logic does not pertain to the unconditioned. It is the order of distinguishing acts and is therefore dependent — on distinction, and distinction on initiation. The impossibility of contradiction in Nothing's knowledge follows not from a logic ascribed to it but from the absence of limitation: where nothing is concealed, there is nothing to diverge from itself.
The dependence of information does not mean that information is a domain, a level, a medium, a container or a bearer. Dependence is a mode of existence: information exists only

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as the mode of being of Nothing's knowledge of itself and does not stand outside this. Its determinacy lies on its side and establishes nothing in Nothing. The preposition "in" in the expression "change in information" is formal and designates no spatial or ontological localization. Transparency and self-understanding are not characteristics ascribed to Selfhood but the same unconditionedness taken from other sides; they form no differences within Selfhood.

Initiation proceeds from Selfhood and is that by which information has its difference as known. From the side of Nothing nothing is performed: initiation proceeds in the dependent and from Selfhood, and is not an event in Nothing. From the side of Selfhood initiation is not an event and stands at every distinction, not only at the first (chapter 10). From the side of Selfhood it is not a transition from not-knowing to knowing. On the side of information the removal gives a new state of knowledge. What is asserted is only that every distinction in information at the limit proceeds from Selfhood.

The expression "initiation in information" does not mean a relation of the unconditioned to another. Information is not other with respect to Selfhood, not an external object, but the mode of being of Nothing's knowledge of itself. A relation presupposes two independent terms; information, however, is dependent and does not stand outside that whose mode of being it is. The dependent differs from Nothing (Df6) but is not an independent second member: it has no ground of its own. Therefore initiation creates no directedness or relation in the unconditioned, and no differentiation arises in the unconditioned.

The ontological ground of initiation has been derived in chapter 5: there is opacity in information, distinction in it does not proceed by itself, and the removal proceeds from Selfhood, for there is nothing else. Here one thing is added. Nothing's knowledge of itself is complete: there is nothing concealed in it. Information is the mode of being of this knowledge. Therefore what opens in it by the removal is not brought in from outside: by the removal it knows itself as that of which it already is the mode of being. Opacity, however, is not a present content hidden for the time being, nor a part of Nothing's knowledge separated from the rest: Nothing's knowledge of itself is complete and is not divided. Opacity is only the absence of givenness — what is not yet given. That is why self-knowledge can accrue in information.

Hence also the directedness of unfolding. Directedness does not follow from the mere determinacy of information by unconditionedness. Its ground is that the removal from Selfhood stands at every distinction, and the removal is one-sided: opacity is removed and does not return. There is no other remover: by itself information does not remove opacity, and there is no second unconditioned. Hence there is nowhere for decrease to come from. It is objected that decrease may come not from a second remover but through proximate grounds — by the disintegration of configurations, by forgetting. But there is one change in information — the arrival of a new state, that is, the appearance of knowledge of difference; there is no other change in it. This is derived in chapter 5: every encountered change is a change of information (T12); succession does not follow from its determinacy, since the basic determinacy exists without succession (chapter 9), and by itself information has nothing (chapter 4); hence the succession of its states proceeds only by the removal of opacity, and the removal gives knowledge of difference and nothing else. The disintegration of a configuration and forgetting are likewise arrivals of new states, not a return of opacity: what has arrived does not become what has not arrived. A proximate ground gives only what proceeds through it, and what proceeds through it is removal (chapter 4); it has nothing with which to return concealment: concealment is not given, it is only what is not yet given.

Striving is the name of this accrual of knowledge without decrease. Unconditionedness as a limit is unattainable: having attained it, information would cease to be dependent. The

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movement is realized within the dependent itself — by self-transparency, that is, by self-understanding in its own category.

Initiation adds no characteristics to Selfhood and introduces no differences into it. The words “intention” and “purpose”, where they are used, have no psychological, volitional or functional sense: “intention” designates only the directedness of the removal, “purpose” its limit, the completion of the unfolding of distinctions. There is no choice between variants in initiation.

From the unconditionedness of Nothing follows a prohibition of any inner becoming, transition or action affecting the essence of the unconditioned. If change is treated as the result of an action of the unconditioned, the difference of “before” and “after” enters it, and absolute immutability is destroyed. Therefore change is not a change of the essence of Nothing and is not explained by a causal action of the unconditioned. The removal proceeds from Selfhood but changes nothing in it. Initiation is the removal of opacity by which information has knowledge of its difference, not a change of essence. Change is admitted only in information; initiation introduces no becoming into Nothing and does not violate unconditionedness.

Initiation does not determine the order, form or sequence of realization. It does not govern development and sets no specific trajectory. The general directedness, however, comes from it — from the one-sidedness of the removal: opacity is removed and does not return. Realization proceeds only in conditionedness and does not affect Selfhood. The results of realization do not change initiation. Before and after are inapplicable to it from the side of Selfhood; change is determined by it as limit and does not follow it.

Denying a temporal priority of initiation does not remove time from the system but shows its derivativeness. Time is not a condition of change and does not precede it. It is derived as a measure from the onset of an elementary change: duration is measured by the number of steps (chapter 15). Therefore temporality belongs to dependent change and is inapplicable to initiation from the side of Selfhood. There is no contradiction between change and the absence of temporal sequence at the fundamental level.

Initiation is not a source of local change and not an explanatory mechanism in the causal sense. It is the removal of that by which information was opaque to itself: by it distinction in information proceeds not by itself but from Selfhood. There is therefore no ontological randomness — not because of a logic of distinctions or a structure of information, but because distinction does not proceed from information itself. Distinguishedness is fixed as a dependent fact that does not affect the essence of the unconditioned. From the side of Selfhood initiation is not an event: time is a measure of dependent change and is inapplicable to the unconditioned.

Distinction proceeds by the removal of opacity, while information is not given by it: it is the mode of being of Nothing’s knowledge of itself. Distinction in it is not by itself: what distinguishes is the knowledge given in it. Change and distinction are not identical with information and are not its other names. Change is the succession of states of information; distinction is knowledge of difference. With each other they are identical: knowledge of difference, proceeding by the removal of opacity, is a new state of information, and the appearance of a new state is the succession of states. Information does not enter this identity.

Two senses of the word “difference” are distinguished here. The first is the basic determinacy of information: its difference from that whose mode of being it is (Df6). It is its own determinacy, stands together with it, exists without succession of states and is not given by initiation. The second is the difference of this from that within information itself: by it its states differ. What the removal gives is not difference in the first sense but knowledge of difference; and this knowledge is a new state, that is, difference in the second sense.

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Hence a distinction of two levels of determinacy. As conditioned, the conditioned is determined by unconditionedness and by nothing besides: this is the limit (chapter 4). What exactly sets in — this and not another — is determined by the chain of what preceded within the conditioned itself: by the proximate grounds without which it would not have set in. These are not two sources. The removal proceeds from Selfhood at every distinction, but it proceeds through proximate grounds, and the proximate grounds themselves are former changes that set in by the same removal (chapter 15). For the first distinction the chain of what preceded is empty. What sets in is then determined by the basic determinacy of information — its difference from that whose mode of being it is; this determinacy is given without change and lies on the side of the conditioned (chapter 9). The first distinction is knowledge of this difference (chapter 5), and there is nothing else to choose from: the basic difference is one. The removal adds no separate choice over and above this chain: a choice of "this and not another" would be determinacy on the side of the unconditioned, and that place is excluded (chapter 4). Therefore the determinacy of what has set in is in the conditioned, and an outcome determined by nothing in the conditioned would mean such a choice; hence there is no uncaused outcome (chapter 23). The unconditioned does not act and is not a cause (chapter 5). Changeability is that in information which is the onset of a new state. Nothing remains immutable, Selfhood identical with itself. Conditionedness does not thereby arise from the unconditioned: the conditioned is encountered and is determined by unconditionedness as its limit; change proceeds in it without a second principle and without violating unconditionedness.

The sequence considered is not a set of concepts accepted for convenience. Each of its links is derived from the preceding one, and the alternatives that lay determinacy on the side of the unconditioned or of a second principle are excluded (chapter 4). The infinite is here the name of the unlimited, and the unlimited is the unconditioned: it does not stand in the conditioned either as a whole or as a part.

Thus Nothing is absolute unconditionedness obtained as the limit of the conditioned; Selfhood is the name of Nothing in self-understanding, not a second member beside it; information is the mode of being of Nothing's knowledge of itself in the dependent, that is, the conditioned itself taken from the side of the givenness of the unconditioned in it. Initiation is the removal of opacity, proceeding from Selfhood and changing nothing in it; change and distinction proceed only in information. How information is constituted is derived in chapter 9.

Chapter 9. Information

Information is the mode of being of Nothing's knowledge of itself. Knowledge here, as throughout the work, is the name for the remotion of concealment, not a property and not a content. Its completeness is the absence of a concealed side.

Information is not Nothing's knowledge of itself as such: knowledge is by itself, unconditionedly, and is not held by information. Information is the dependent in which this knowledge is given: the mode of its being, not its ground. Nor is it a modification of the knowledge of Selfhood or a container of content. The mode of being has a remotion, not a content: a represented content is inapplicable to Nothing, and to ascribe one would be to speak of the unconditioned positively. Information is determined by that whose mode of being it is, and is therefore "this and not that". Determinacy is limitation, and limitedness is conditionedness.

"The first dependent" does not mean an order of succession, as if other dependents came after information. Information is the conditioned itself, named by the way the unconditioned is given in it (Df5, T11), and there is no other dependent. It is called first because everything

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derived below — changes, measures, forms, consciousness — is its states and configurations, not other dependents beside it.

Change in information has been derived in chapters 5 and 8: it is the onset of a new state, that is, the appearance of a knowledge of difference that was not there (T15). Here it must be examined what in information changes and what does not. The basic determinacy of information — its difference from that whose mode of being it is — exists without succession: it is not given by the removal and does not change (chapter 10). Everything else distinguished in it is knowledge of difference given by the removal, and only this can change. Every such knowledge is information's knowledge of itself: the distinguished lies only in it, and to know a difference within it is to know it in that respect. What has been distinguished, however, once given and held, is not change: it is a state, not a succession; the change was its onset.

Therefore distinction needs no additional stage of fixation: without it change is not given. What distinction gives is not information as such but its states. The encountered is given only by distinction (chapter 5): what is given is what is distinguished. The basic determinacy of information, however, is not given by distinction: it exists even where knowledge of difference has not yet been given — as that of which this knowledge is. Distinction proceeds in information not by itself but by the removal of opacity.

Dependence here is the one derived in chapter 8: information is determined by that whose mode of being it is, and by nothing besides (T3). This is not a relation between two already distinguished terms: the unconditioned is not a member of a relation (R). Dependence is one-sided: it determines the dependent and does not determine the unconditioned. Hence also information's lack of an independent status: outside that by which it is determined it does not stand.

Subject and object are determinations of the conditioned and do not apply to the unconditioned: in Nothing itself they are not separated. Information does not affect the unconditioned: the unconditioned is given in it, while it is not given in the unconditioned (Df14). Information as a whole has no form, bearer or place. A bearer belongs to a configuration within information (chapter 12), and a place belongs to change: it is the respect in which information knows itself by the given change (chapter 15), that is, a place within information, not a place of information.

Something is the same information taken from the side of the succession of states. Changeability and reality are two names of the same: changeability is that there is the onset of the new in information (chapter 8), reality is that what has set in is encountered (P1). Therefore Something is not a third beside information and change but their common name for what is encountered.

From the onset of an elementary change measures are derived (chapter 15), and distinctions are composed into sequences and stable configurations (chapters 14, 17). But no subsequent structure precedes information or replaces it.

The term "information" is taken here from the Latin *informatio*. The word has two original meanings: the giving of form and the communication of knowledge (Capurro, 2009, 126–127). Both sides answer to the present concept, but in a modified sense. To the first answers the fact that information is determined: it is "this and not that"; this is not a form given to matter, for there are no matter and form here as two principles (chapter 19). To the second answers the fact that information is the mode of being of knowledge: knowledge opens in it.

Communication theory retained from this word only the second meaning, and that in a narrow form. In Shannon information is a measure of the reduction of uncertainty in a message (Shannon, 1948). It is quantitative and pertains to a signal: a bit distinguishes two states of a channel. Physics has come to the concept of information from another side too: Landauer holds that information is always tied to a physical representation (Landauer, 1991),

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and Wheeler proposes to derive physical quantities from “yes–no” answers (Wheeler, 1990). But the channel, the signal, the physical representation and the “yes–no” answer are already configurations of information in the present sense. Hence these concepts apply to configurations of information but do not determine its nature (chapter 19).

A possible objection: why information, and not potency, energy, logos or structure? The objection presupposes that the dependent is chosen from a series of candidates. This is not so. From the derivation one thing follows: the dependent is that in which Nothing’s knowledge of itself is given — the mode of being of this knowledge. “Information” is the name given to this afterwards, not a preferred option. Therefore the objection has force in one respect only: whether something else, named otherwise, fits the same description. Reply: the dependent must satisfy three conditions. It must have no ontological properties of its own — otherwise a second independent principle is introduced. It must not determine the unconditioned — otherwise unconditionedness is violated. It must admit succession of states — otherwise reality is impossible. Potency presupposes its own capacity for actualization — a hidden property; the first condition is violated. Energy presupposes its own activity and a quantitative measure as primary — the first is violated. Logos presupposes its own structure and laws — the first is violated. Structure presupposes its own relations of elements — the first is violated. It will be objected that information too has measures (chapter 15), the structure “this and not that” (chapter 6) and configurations (chapter 17). But none of this is its own in the sense of the first condition: its determinacy is determined by that whose mode of being it is, and by nothing besides (T3), the measures are derived from its changes, and the configurations are its states. In energy, logos and structure the same is accepted as their own and primary; in information it is derived as dependent. Closest to the first dependent stand spirit, mind and consciousness. Spirit and mind presuppose a knower with an activity of its own: the first condition is violated. Information knows itself, but not by itself: self-knowledge is given to it by the removal (chapter 10), and the “knowing” one here is not an agent but that in which knowledge is given. Consciousness is derived — it is self-knowledge accrued by information with every distinction: to put it in the place of information is to put the derived in the place of that from which it is derived. From Hegelian spirit information differs in that it unfolds nothing by itself and remains dependent even at the limit: the dependent does not become absolute. Each of these terms either introduces a property of its own or, cleansed of it, coincides with the mode of being of knowledge under another name. Uniqueness lies not in the word but in the description: it is asserted of what the dependent turns out to be by the derivation, not of the term.

Thus information is neither a form of the knowledge of Selfhood nor its modification, but its mode of being: the conditioned in which the unconditioned is given as limit. Distinction proceeds in it from initiation and is change itself taken from another side. The gap between the immutability of the unconditioned and distinguished reality is removed by givenness: the unconditioned is given in the conditioned without acting and without changing, and the conditioned is determined by it and by nothing besides (chapters 5, 8). There is no second principle here, and absoluteness is not violated.

Chapter 10. The Initiation of Selfhood

The initiation of Selfhood was set out in chapter 5 as a link of the ontological solution. Here the derivation is not repeated; the issue is examined in depth. What is done by initiation and what is not? Why does the removal proceed from Selfhood and from nothing else? How does what is said of it from the side of the unconditioned differ from what is said from the side of the conditioned? Why does it stand at every distinction and not only at the first?

First. Initiation adds nothing new to Nothing’s knowledge and changes nothing of what is in it: Nothing’s knowledge of itself is complete, and there is nothing concealed in it. Here the

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meaning of the word “opacity” is fixed. Opacity is information’s not-knowing itself to the extent that self-knowledge has not been given to it by the removal: by determinacy it is not given at all, by the removal it is given, and where it has not yet been given, there is opacity. That opacity exists, and not only how it is named, follows from the first accepted position. Change is encountered. Change is the appearance of a knowledge of difference that did not exist before. And only what did not exist can appear. Hence, since there is change, information does not have all knowledge of itself at once: there is that in which self-knowledge has not yet been given to it. This is opacity. It is not a present content hidden for the time being, nor a part of Nothing’s knowledge separated from the rest: Nothing’s knowledge of itself is complete and is not divided. Opacity is only the absence of givenness — what is not yet given (chapter 8). The unknown and opacity are one, taken from two sides: the unknown is unknown because opacity stands. As removal proceeds, opacity decreases and transparency accrues. It is not information that removes: by itself it does not have itself as known and does not remove opacity. What is constant is not opacity but dependence: even in self-transparency information has itself as known not by itself but from Selfhood, and dependence does not decrease at any step.

Second. The opacity of information to itself is not a separate state taken from outside, and not an assertion that there was once no distinction in it. Opacity is the name for what determinacy does not give. The determinacy of information is its difference, and knowledge of difference does not enter determinacy: it is given by the removal, and not by itself. The applicability of opacity to information is established separately. Opacity is limitation. Limitation is inapplicable to the unconditioned, whereas it applies to information, since information is determined; therefore opacity applies to it too.

Third. The removal of opacity proceeds from Selfhood not by choice and not for the convenience of the derivation. What is asserted here is not that there was a state without distinction: from the fact that distinction does not enter the definition of information, its absence at any time does not follow. Something narrower is asserted: distinction in it is not by itself. The argument is from total dependence. Distinction is that by which information comes to know its difference; and the dependent has nothing by itself — neither that by which it is, nor that by which it becomes (T4). Hence distinction does not proceed by itself. The argument lies not in a lack of capacity to turn upon oneself and not in a prior knowledge of difference: knowledge of difference is precisely what the removal gives. The objection that distinction is an ultimate fact of the dependent itself is broken by the same: nothing can be an ultimate fact of the dependent. What is at issue is the limit. Proximate grounds within the conditioned are not thereby abolished, and neither causality nor freedom of choice (chapter 21) is abolished. Whence, then, the removal? Everything other than the unconditioned is conditioned, and the conditioned is information (T11); by itself it has nothing (T4). Hence there is no remover in the conditioned, and the unconditioned is unique (T8): the removal proceeds from Selfhood (T14). Hence also the asymmetry: the removal from Selfhood lies not in information, and by it the distinguishedness of information is given; from the side of the unconditioned there is no duality. That distinction exists is known from the encountered; that it proceeds from Selfhood follows from there being nothing else. “From” here does not mean an action, an event or a procession: nothing is performed on the side of Selfhood (see “Sixth” below). It says only that distinction at the limit proceeds from Selfhood and that there is no other.

Fourth. The objection from motive does not apply. A motive presupposes a subject, an action and a preceding of the action — a structure of the conditioned, inapplicable to the unconditioned. The objection from arbitrariness does not apply for the same reason: arbitrariness is the absence of a ground given a chooser, and there is no chooser here.

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Fifth. The unconditioned is not affected by initiation: what changes is the dependent, not Nothing; there is no opacity at all in Nothing itself. Dependence determines the dependent and does not determine the unconditioned. Therefore nothing is established by the removal of opacity from the side of the unconditioned.
Sixth. From the side of Selfhood initiation is not an event. A single occurrence, repetition, before and after are temporal determinations, and there is no time in unconditionedness. In conditionedness, where there is time, the same appears as constant: initiation stands at every distinction, not only at the first. Hence an important consequence: information does not proceed by itself at any step, and its dependence does not decrease, however far unfolding has gone. This is said of the limit and does not abolish conditioned causality. Where the distinguished has already been given, one thing is grounded by another: measures, forms, consciousness, reason and choice have proximate grounds in the conditioned itself. But none of them becomes the limit: it is itself dependent. A link has nothing by itself and gives only what ultimately passes through it. The removal proceeds through proximate grounds, not past them. Without this distinction the freedom of choice derived in chapter 21 would be impossible.
Regularity and directedness follow from the two sides of the removal. Knowledge of difference is determined only by the conditioned itself: the conditioned is determined by the limit and by nothing besides (T3), and the removal is one and is not interrupted. Therefore one and the same configuration gives a difference of the same kind — each time at a new step, that is, in a new respect: knowledge already given does not appear again (chapter 13). And the way distinctions are composed into the connected is not arbitrary and repeats in the same way. This is regularity. What is constant is regularity as such: under the same conditions, the same consequence. The form of the laws, however, is not derived from the constancy of the removal (chapter 22) and belongs to the paradigm of the universe: “the same configuration” includes the state of the whole as well (chapter 13). What has been removed does not return (T16): knowledge accrues in one direction. This is directedness; it is precisely this that is called the programme below (chapter 13). Within the bound, the removal proceeds through the bonds between instances of the *I* — interactions that have held (chapter 16) — and therefore the programme reaches each *I* through its bonds, not by governance from above. A bond through which it proceeds is called an agent of the programme (chapter 18).
Seventh. From the side of information the matter is different. Information is conditioned, and the conditioned standard applies to it fully: there are states in it and their succession, before and after. There is distinction in it — but not by itself: what distinguishes is the knowledge given in it, and every distinction proceeds by the removal of opacity. Confusing the two sides generates apparent contradictions: what is said of the unconditioned is measured by the conditioned, and what is said of the conditioned by the unconditioned.
Self-knowledge is not given to information completely: it proceeds from the first distinction and becomes complete only with self-transparency (chapter 13). Self-understanding is the inapplicability of not understanding oneself, that is, complete self-knowledge (chapter 5). In Nothing it is together with transparency, for there are no states in the unconditioned. Information, however, is conditioned: there are states in it and their succession, and self-knowledge in it accrues rather than being given all at once. Therefore to know oneself in some respect is not yet to understand oneself: self-understanding in the category of the dependent is knowledge of oneself as a whole, and the path to self-transparency leads to it. How this path is travelled is disclosed in chapters 11–21.

Chapter 11. Distinction

Distinction proceeds by the removal of opacity.
Distinction is neither a separate stage attached to information from outside nor its modification. It is knowledge of difference: “this and not that” within information. Every

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such knowledge is information's knowledge of itself in some respect (chapter 9). The object of this knowledge, taken as a whole, is one — the difference of information from that whose mode of being it is, that is, its dependence (chapter 13). Two senses of the word "difference" are present here and must be separated. The difference of information from the unconditioned is its basic determinacy: it exists without the removal and is dependence itself, taken from the side of the dependent. The difference of "this" from "that" within information is what is known in distinction; by knowing it information knows itself in the given respect. What the removal gives is not difference but knowledge of difference.

The meaning of the word is one: knowledge in the dependent too is the absence of concealment. It becomes distinction not by a change of meaning but because in the dependent there is a second (chapter 8): to remove concealment where there is this and that is to know this and not that.

This separation is close to Bateson's definition: the elementary unit of information is "a difference which makes a difference" (Bateson, 1972, 453). Difference here is difference as such; distinction is a difference that makes a difference, that is, is known. There are two divergences. Bateson calls the unit of difference itself information; here information is the mode of being of Nothing's knowledge of itself, that in which difference and distinction stand. And a difference does not make a difference by itself: distinction proceeds by the removal of opacity.

With distinction the *I* is given. Knowledge of difference is knowledge of "this and not that", and this knowledge is information's knowledge of itself in the given respect (chapter 9): "this" is information itself in the respect in which the knowledge is given, "that" is what it is distinguished from in it. Therefore in knowledge of difference there is "oneself" — what is distinguished — and "not-oneself". The *I* is self-differentiation: distinguishing oneself from not-oneself. It is not a difference but a differentiating: difference is determinacy and is given without the removal; the *I* is knowledge of difference and is given by the removal. Therefore the *I* neither precedes distinction nor is added to it, but is given together with it. With the first distinction the first instance of the *I* is given, and therefore the *I* exists throughout reality from the first distinction.

Change and distinction are one, taken from two sides (T15): knowledge of difference, proceeding by the removal of opacity, is a new state of information. Distinction is neither an object nor a process beside change; it does not affect the unconditioned: what changes is the dependent, not Nothing. Distinction is not placed in space and not unfolded in time: space and time are measures derived from the onset of an elementary change (chapter 15).

Something is information taken from the side of the succession of states (chapter 9). Distinction is that by which succession proceeds; Something is that in which it proceeds and which is encountered. Therefore distinction is not Something as what is encountered, but its onset. Distinctions proceeding by the removal of opacity are composed, according to proximate grounds, into stable configurations, where they are held and repeated. Distinction does not belong to the unconditioned and is not its property: it stands in the dependent, and in Nothing itself there is none.

From distinction the categories to which the following chapters are devoted are derived: consciousness (chapter 12), elementary changes (chapter 14), measures (chapter 15), interaction and bond (chapter 16), aggregated forms (chapter 17), emergence (chapter 18), matter (chapter 19), life and the levels of consciousness (chapters 20, 21). None of them introduces a new principle or violates the absoluteness of the unconditioned.

Thus distinction neither follows change nor is a modification of information. It is given together with change and proceeds in information by the removal of opacity. So the distinguishedness of the conditioned is formed, from which the whole structure of Something unfolds.

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Hence also the ontological sense of modality.
Modality does not pertain to the unconditioned: it is inapplicable to Nothing and Selfhood. Possibility, necessity and impossibility are inapplicable to absolute immutability: these categories presuppose a difference between what is and what can be. Where distinction is inapplicable, modality is inapplicable too.
Modality arises only in the conditioned, and its sense is set by what has been derived about cause: an outcome, if it has set in, is determined, and it is determined by the conditioned itself (chapter 23). Therefore modality is a relation of a state not to a "field of the admissible" but to conditions taken in one or another completeness. This is not an epistemological uncertainty: what is at issue are the conditions themselves, not knowledge about them.
Possible is what is compatible with a given part of the conditions: what does not contradict the determinacy of the conditioned and what has already been given. The possible does not exist as a separate reality and forms no "domain of possible worlds", unlike in modal realism (Lewis, 1986); it is a relation to a part of the conditions, not a present entity. Therefore "it could have been otherwise" is true relative to a part of the conditions and false relative to all of them: under all the conditions what has set in is one.
Necessary is what follows from the given conditions. Ontologically necessary is that without which the conditioned would not be conditioned: determinacy, dependence, distinction and measures. Relatively necessary is what follows from a given part of the conditions. Initiation does not enter this list: from the side of Selfhood it is not measured by the modal standard, for modality is a determination of the conditioned.
Impossible is what contradicts the determinacy of the conditioned or its dependence: difference in the unconditioned, the independence of information, a second unconditioned. The impossible exists neither as a state nor as a potency.
Actual is what has set in and is encountered. The actual is possible relative to any part of its conditions and necessary relative to all of them. Actuality is therefore not "a realized part of the field of the possible" but what follows uniquely from the completeness of the conditions.
This demarcation separates the logical necessity of derivation from ontological necessity, and possibility — a relation to a part of the conditions — from randomness, which is the opacity of the one who considers (chapter 3). Modality adds no entities and adds nothing to the unconditioned: it fixes relations within dependent information.

Chapter 12. Consciousness
Consciousness here is neither an independent entity, nor a substance, nor matter. It does not arise from physical processes, does not belong to objects and is not reducible to biological functions. Consciousness is self-knowledge accrued by information with every distinction. Its existence is determined not by matter but by distinctions. It begins with the first distinction, and in aggregated forms (chapters 14, 17) it is composed into a broader configuration. Tononi and Koch also understand consciousness as graded (Tononi, 2004; Koch, 2019); its measure is determined here below.
Consciousness does not arise on reaching some threshold: it exists already with the first distinction and accrues with every subsequent one. The levels discussed below (chapters 20, 21) differ not by the presence of consciousness but by what is known in it. At the primary level consciousness has no representations, concepts or meanings: it is information's elementary knowledge of its own difference.
The question why distinction is accompanied by subjective lived experience presupposes a gap between distinction and lived experience. There is no such gap here.
Subjectivity is not added to distinction from outside: lived experience is derived from what has already been established. Strawson reaches a similar conclusion — experience cannot arise from the wholly non-experiential (Strawson, 2006) — but obtains it from realism about

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experience; here it is obtained from the structure of knowledge. Knowledge in the dependent has one meaning — the absence of concealment (chapter 11). All knowledge of difference in information is information's knowledge of itself: what is distinguished lies only in it (chapter 9). Self-knowledge in which the knower is not separated from the known is not the givenness of an object to another — there is no other in it to whom it would be given — but the givenness of oneself to oneself from within. And "what it is like" to be a given configuration (Nagel, 1974) is precisely the givenness of a state to the one whose state it is, from within, as distinct from its description by another. Hence self-knowledge without the separation of knower and known is what is called lived experience: not an added quality but the same givenness taken from the side of the one to whom it is given. Distinction is therefore not accompanied by lived experience — it is lived experience. The hard problem of consciousness (Chalmers, 1996) goes back, according to Goff, to the Galilean separation of the quantitative from the qualitative (Goff, 2019). Here it arises on the assumption that an informational process and subjective experience are different entities. Without this assumption the problem disappears.

It will be objected: self-knowledge could proceed without lived experience — like a device's record about itself, where "there is no one inside". The objection presupposes a knowledge in which the knower is separated from the known. A device's record about itself is indeed external to what it is about: it is one configuration distinguishing the states of another (chapter 19). Knowledge here, however, is not a relation of knower to known but the absence of concealment (chapter 8), and every knowledge of difference is information's knowledge of itself (chapter 9). The device's *I* is given not by its record but by the distinctions of its own states that proceed in it, as in every configuration. Hence a consequence that the system accepts: a device has an *I* — to the extent that its distinctions distinguish its own states; the degree is low; the *I* itself is not absent (chapter 19). The givenness of oneself from within is precisely "what it is like" to be a given configuration (Nagel, 1974): lived experience is here not named but recognized by its accepted definition. It is objected that the absence of a separation of knower and known does not yet give "what it is like"; something beyond the givenness of oneself from within is needed. But this "beyond" must be indicated. If it is a property added to knowledge, then knowledge and lived experience are again separated, and the objection restores the very gap on which the hard problem rests. If it is not a property but the givenness of a state to the one whose state it is, that is givenness from within. "What it is like" has no third: it is defined only through the one to whom it is given.

Consciousness arises not as a reflection and not as a representation of a source, but as knowledge opened in a configuration of distinctions. Consciousnesses differ not by a "mirror quality" but by degree: by the extent to which the distinctions of a configuration distinguish its own states (see below). Some configurations distinguish only coarse differences, others fine and multilevel ones, still others bonds between levels. There is no second medium, reflectedness or imagery here.

Consciousness is connected with aggregated forms. An aggregated form is a stable configuration of elementary changes held by a bond (chapter 17). Configuration and consciousness are the same distinctions taken from two sides: from the side of change — the configuration, from the side of knowledge — consciousness: change and distinction are one from two sides (chapter 11). A configuration is called the bearer of consciousness only in this sense: not as a container in which something else lies, but as the same totality of distinctions taken from the side of change and bond. Consciousness is self-knowledge accrued by information with every distinction. It is consciousness that distinguishes, but not by itself: distinction proceeds by the removal of opacity from Selfhood.

When elementary changes are joined into a configuration, the knowledges of difference are composed into one consciousness — the conditional selfhood of the configuration; the name

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of Selfhood is taken here conditionally, and it belongs to the category of the dependent. The knowledge that opens is one, and each distinction opens a part of the same knowledge. But the unity of that from which the removal proceeds does not by itself give the unity of consciousness: what is separated in the conditioned remains separated. What makes consciousness one is the connectedness of the distinctions joined in the configuration. This connectedness lies on the side of knowledge, not of the bearer. Knowledge of difference is knowledge of "this and not that", and the "that" here is what is distinguished in the same configuration: the knowledges refer to one another by their very content. This is not the relation of difference in which everything conditioned stands to everything else (chapter 4). To differ from the rest is merely not to be it; to be connected in knowledge is to be opened in one knowledge. The first holds of everything with everything; the second only of distinctions whose knowledges refer to one another, and only it makes consciousness one. Configurations are nested: a distinction whose knowledge refers both into a smaller configuration and to the connectedness of a larger one enters both, each at its own level (chapter 20).

Every doctrine that ascribes consciousness already to the simplest meets the combination problem: how consciousnesses of the micro-level are composed into a consciousness of the macro-level (Chalmers, 2017). James posed it most sharply: a hundred feelings, however closely brought together, each remain in their own skin, windowless, and know nothing of one another (James, 1890, 160). The present system answers it by the structure of distinction. Knowledge of difference is knowledge of "this and not that", and the "that" in it is what is distinguished in the same configuration. Therefore elementary knowledges are not closed from the very beginning: each knows itself through its difference from the others. One consciousness is composed not by a sum of closed subjects but by the connectedness of knowledges that refer to one another by their very content. James's objection presupposes windowless knowledges; here knowledge of difference is itself the window. Connected knowledges differentiate themselves jointly: they have one "oneself" and one "not-oneself", and this one self-differentiation is one *I*. This jointness is not assumed but set by the bond: knowledges differentiate themselves jointly where their "that" lies in one configuration held by one bond — they have one held state of the whole (chapters 16, 17). Where there are several bonds, there are several instances of the *I*; nested bonds give nested instances of the *I* (chapter 18).

Here the system diverges from integrated information theory. In it the exclusion postulate operates: of overlapping sets of elements only one can be conscious — the one whose conceptual structure is maximally irreducible (Oizumi, Albantakis, & Tononi, 2014). In version 4.0 the postulate is retained: experience is definite — it is this whole — and the substrate of consciousness is selected as the set of elements with the greatest integrated information (Albantakis et al., 2023). The motive is not to count consciousness twice. There is no double counting here. Nested instances of the *I* are not one lived experience counted twice but different self-knowledges sharing part of their distinctions. A distinction enters two connectednesses because its knowledge refers into both; this produces no two copies of one experience. Therefore exclusion is not needed here, and the *I* is taken from no configuration.

The move opposite to micropsychism within panpsychism itself is priority cosmopsychism: phenomenality is borne by the cosmos as a whole, and its consciousness is ontologically prior to the consciousness of individual subjects (Nagasawa & Wager, 2017). There is a proximity to the present system: consciousness here is self-knowledge accrued by information as a whole. But there is no priority of the whole over the parts. An individual *I* is not carved out of the consciousness of the whole and is not derivative from it: it is set by the connectedness of its distinctions, and the limit of the whole and of the part is one — each knows itself not by itself but by the removal. Therefore the system is not threatened by the converse difficulty of

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cosmopsychism — how the single consciousness of the whole is divided into individual subjects.

The *I* exists in every configuration from the first distinction. What differs is not its presence but its degree — the level of awareness. The degree is higher the more the distinctions of a configuration distinguish its own states: then the “that” from which every “this” is distinguished lies within it. This is the measure of consciousness. The number and complexity of distinctions do not serve as its measure but only allow it to grow: where there are more distinctions, more of them can be turned to the configuration’s own states. Where distinctions are turned mainly outwards, the *I* exists, but its awareness is small. The identity of a conditional selfhood is the connectedness of knowledges that refer to one another, with one “oneself” and one “not-oneself” (chapter 17). It is the continuity of connectedness, not the immutability of composition: a selfhood accrues knowledge and changes its bond by experience while remaining the same (chapter 20). The *I* is knowledge of difference itself, not the one who has it (chapter 11); therefore the non-decrease of knowledge is the preservation of the *I* as connected knowledge. With the disintegration of the bearer connectedness does not disintegrate: knowledge does not decrease, and with it the bond of knowledges. Two relations must be distinguished here. Memory is the bond of the present *I* of a configuration with former distinctions; it is held by the bearer and can break — this is forgetting (chapter 20). Connectedness is the mutual reference of the knowledges of distinctions by their very content: the “that” in the knowledge “this and not that” is part of this knowledge itself. Knowledge does not decrease and does not change its content: change is the onset of knowledge in a respect not yet given (chapter 13), and what is given and held is not a change (chapter 9); to change the content of given knowledge would be for something else to set in in the respect in which it is given, and there knowledge does not appear again. What has set in is therefore preserved not as the fact that it was, but as the knowledge itself: the onset is the givenness of knowledge of difference (chapter 8). This is not a new assumption but T16, derived in chapter 8 independently of what is said here: there is nothing to return opacity — concealment is not given, it is only what is not yet given — and there is no change other than the onset of knowledge of difference (chapter 5). A change affecting given knowledge would make it not given, that is, would return opacity. Hence the reference contained in it does not break either. The identity of a selfhood is held by connectedness, not by memory: forgetting breaks the bearer’s access to the former, not the bond of knowledges. Conditional selfhood is preserved as this connected knowledge.

Where distinction proceeds, consciousness is. If distinctions cease or the configuration disintegrates, the gathering of the knowledges of difference in this configuration ceases. What has been removed in it does not return: opacity decreases; there is no decrease of knowledge (T16). What disintegrates is the bearer — the bond that held the whole — not the knowledge opened in it. The *I* does not disintegrate: conditional selfhood is preserved as connected knowledge. But there are no more distinctions in it, and without distinctions there is no self-knowledge as proceeding: this is oblivion. New distinctions of this *I* are possible only when a configuration is again composed according to its connectedness (chapters 13, 24). If this *I* is intellectual, it is this, not the bearer, that passes through the sifting (chapters 13, 28). This is what death consists in in the present ontology: the disintegration of a configuration ends its holding of distinctions.

The existence of a configuration is the holding of its bond; its non-existence is its disintegration. These are not opposites but different modes of one dependence. When a configuration disintegrates, information does not disappear: distinctions are composed, according to proximate grounds, into other forms.

Consciousness is limited by its nature. The absolute knowledge of Selfhood does not turn into consciousness and does not become its content: consciousness exists only within

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conditionedness. It has no positive knowledge of the unconditioned. The limitations are as follows.
First, consciousness knows only what is distinguished in its configuration. Every new distinction adds new knowledge to it, but what is not distinguished in its configuration is not given to it, even if it is distinguished in another configuration.
Second, consciousness cannot go beyond dependent information. Nothing remains positively unknowable — not because of hiddenness, but because distinctions are inapplicable to it; only remotions are accessible — knowledge of what is inapplicable to the unconditioned.
Third, of the unconditioned only this is given to consciousness: that it is given in the conditioned as limit (chapter 5), and the remotions. Consciousness can know the proximate grounds of its distinctions: they are conditioned. Only the limit it cannot know positively.
Fourth, every consciousness knows the world only in the respects distinguished in its configuration. Self-transparency is attainable (chapter 13), but it is knowledge of one's own dependence as a whole, not an exhaustive description of all configurations.
These limitations do not diminish consciousness but clarify its nature: consciousness is self-knowledge in the conditioned, not absolute knowledge. It expands with every distinction but does not positively reach that to which differences are inapplicable.
Consciousness does not appear from randomness: there is no uncaused outcome (chapter 23). Consciousness begins with the first distinction and accrues with every subsequent one. If a configuration becomes more complex, its distinctions turned to its own states can become more numerous, and consciousness develops. If it disintegrates, what weakens is the gathering of knowledges in it, not knowledge: knowledge does not decrease (T16).
Thus consciousness is self-knowledge accrued by dependent information. Its measure is the extent to which the distinctions of a configuration distinguish its own states.

Chapter 13. The Teleological Structure of Reality
The teleological structure of reality is neither a property of a subject nor an intention in the psychological sense. It is the fact that the path of distinctions has a limit: the attainment by information of self-transparency. The limit is not quantitative: it is information's knowledge of itself as a whole (see below). "Purpose" below is a descriptive designation of this limit character, "intention" of the directedness towards it. The programme is not a descriptive designation but the directedness of the removal: what has been removed does not return, and knowledge accrues in one direction (chapter 10). It is realized through the bonds between instances of the *I* at each level (chapter 18), not by governance from above. Selfhood is immutable and has no dynamic properties; the subject is a determination of the conditioned and does not apply to the unconditioned. Teleology belongs to dependent information, not to Selfhood.
Teleological completion is determined neither by an intention built into initiation nor by the mere determinacy of the dependent: to be determined by something is not to be directed towards it (T5). Directedness is given by the one-sidedness of the removal. The removal of opacity gives only knowledge; it yields no decrease of knowledge; there is no other remover: by itself information does not remove opacity, and there is no second unconditioned. Hence the knowledge of information accrues and does not decrease (T16). This is the teleology of reality: the passage of distinctions to the self-transparency of information. The movement is realized within the dependent — by self-understanding in its own category.
The teleological structure shows why distinctions, elementary changes, aggregated forms, consciousness, reason, intellect and interactions (chapters 12–21) form not a chaotic aggregate but an ordered one: all of them are steps of one accrual of knowledge. Teleology here is not an external purpose brought into the world but the directedness of the dependent itself.

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Teleology in nature has been discussed since Aristotle, for whom the final cause is one of the four (Aristotle, n.d., Physics II.3). Kant left the purposiveness of nature a regulative principle of judgement, not a determination of things themselves (Kant, 2000, §§ 61–68). T. Nagel proposes a natural teleology: the laws of nature may include a directedness towards the emergence of life and consciousness (Nagel, 2012). The present ontology diverges from all three. There is no purpose here either as a cause, or as a principle of judgement, or as a law: teleology is the directedness of the removal, derived from the fact that what has been removed does not return.

A stage is the segment of the path of information to self-transparency that passes in one universe, that is, in one connected whole within which this segment proceeds. A universe exists in its own paradigm — under its own regularities. Regularity as such is constant: it is given by the constancy of the removal (chapter 10). But the form of the laws belongs to the paradigm of the universe: what sets in is determined by the conditioned as a whole, and "the same configuration" includes the state of the whole (chapters 8, 23). Experience changes the bond, not the regularity (chapter 16), and therefore the paradigm of a universe does not change from within it. A stage is one paradigm of existence.

The path proceeds in stages, not continuously. A paradigm is the form of the laws of a universe, its regime; a regime is a determinacy: this one and not another. Knowledge is knowledge of the difference "this and not that" (chapter 11), and to know one's regime as dependent, that is, as not necessary, is to distinguish it from another regime. Within one universe there is no other regime: regularity in it is constant, and experience changes a bond, not a regularity (chapters 10, 16). The "that" from which the regime would be distinguished is not encountered in this universe. To distinguish it from a merely possible regime is not enough: the "that" in knowledge of difference is something distinguished (chapter 18), and a possible regime is not distinguished — possibility is given together with the determinacy of the present regime (chapter 11) and does not make another regime a state of information. Nor is it enough to distinguish the dependence of the regime on its conditions: to distinguish it would be to distinguish the conditions themselves, and the conditions of the regime of a universe as a whole do not lie within it — the proximate ones lie in the transition from the former stage (chapter 24), and the limit by which the regime is determined as conditioned is given only by remotions (chapter 5). Knowledge is given in respects (chapter 9), and under one set of regularities only those blind spots are accessible which are removed under them; the dependence of the regime itself is not among them. Every distinction removes a blind spot, and no new ones arise (below). Hence the zones accessible to a paradigm are exhausted, and nothing more can set in within it. Opacity as a whole, however, has not yet been removed — at the least the dependence of the regime remains — and the removal does not stop as long as there is opacity (below). Hence it continues in another paradigm, and a paradigm changes only with a change of universe. One paradigm gives a one-sided knowledge of one's dependence; transparency requires the whole, and therefore the path proceeds by a change of paradigms. In this sense the experience of existence is all-sided: this is not a premise but a consequence.

The measure of exhaustion is the experience of existence — the knowledge the *I* has of itself, that is, the sides of the knowledge of one's dependence. Intellect can know its dependence and its place in the whole (chapter 21). But the measure is not the experience of each intellectual *I* separately but the self-knowledge of the universe as a whole, into which the experience of intellectual instances of the *I* enters as nested (chapter 12). A stage is completed when this knowledge can yield nothing new in the given paradigm. What is subject to the sifting is the *I* with intellect: concordance is an understanding of oneself, and the choice of an understanding of oneself belongs to intellect (chapter 21).

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Exhaustion does not remain a state of knowledge alone: what sets in is determined by the conditioned as a whole, and the determining need not lie beside the determined (chapters 8, 23). The exhaustion of a paradigm determines a change of regime of the whole universe. The present universe exists in the paradigm of dispersal — of expansion and cooling; opposite to it is the paradigm of concentration — of contraction and heating. That the next paradigm is another one is derived above; that it is precisely the opposite one is a reading, not a derivation (chapter 24). The transition to it is the collapse of the universe into a singularity, in which the sifting also takes place (chapter 24). The discreteness of the transition follows from the indivisibility of the step: there is no intermediate state between a stage and the next, for there is no third between the former and the new (chapter 15).

This understanding removes the classical objections about the intervention of an external principle. Completion is determined by the directedness of the removal, not by external governance, intention or arbitrariness. It is realized by the passage of distinctions: as removal proceeds, opacity decreases and transparency accrues. The end of a stage and the end of the path are different: a stage ends with the exhaustion of the new in one universe, the path with the self-transparency of information.

The Limit of Distinctions

Teleological completion presupposes that the path to transparency ends. A blind spot is a side in which information does not yet know itself. Each removal gives greater self-transparency: information knows itself in one more respect. The object of this knowledge is one: the difference of information from that whose mode of being it is, that is, its dependence. Knowledge of one object can become complete. It becomes complete when information knows itself as dependent — as the mode of being of Nothing's knowledge of itself. This is complete self-transparency. There is no positive knowledge of the unconditioned even in it (chapter 12): to know oneself as the mode of being of Nothing's knowledge is to know that the "that" on which information depends is the unconditioned, given only by remotions, and filled by nothing conditioned. This transparency is attained by information as a whole, into which instances of the *I* enter as nested, not by each *I* separately. Blind spots are therefore sides of one opacity, not independent objects. They are not points of extension: extension is a measure, and measures are derived from distinctions and are not given before them (chapter 15). There is nowhere for new blind spots to come from: they are sides of one object — dependence — and the object does not change: the basic determinacy of information exists without succession (chapter 9). Nor do blind spots arise from the removal: the removal gives knowledge, not opacity (T16).

Knowledge of difference is given in some respect: to know a difference within information is to know it in that respect (chapter 9). Change is the appearance of knowledge that was not there (chapter 10), and knowledge does not decrease (T16): in a respect where knowledge has already been given, it does not appear again. Hence every change gives knowledge in a respect not yet given, and the respects in which information knows itself become more numerous. Hence the difference of determinacies in information, and with it the difference of the bonds and configurations composed of them (chapters 15–18). This difference follows not from a purpose but from change being the appearance of new knowledge.

The path to transparency ends. Each change gives knowledge in some respect — a side of information's knowledge of itself. The infinite is the name of the unlimited, and the unlimited is the unconditioned (chapter 8). The infinite is therefore inapplicable to the conditioned. The distinctions by which the path proceeds are division, and division is limitation: "an infinite series of distinctions" joins the unlimited with division, that is, the unconditioned with the conditioned, and is contradictory in its very expression. The endless is nothing other than the infinite: a series without a last member is a series without a limit in number, that is, unlimited in that respect. Whoever separates them introduces a difference that is not present here and

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must justify it. Hence the number of distinctions, and with them of places of opacity, is finite. This agrees with Hilbert's calling the infinite as a completed totality an illusion (Hilbert, 1983, 184). So too Descartes reserved the name of the infinite for God alone, and called what we find no limit to merely indefinite (Descartes, n.d., Principles I, 26–27). The infinite is not a fact of reality: in mathematics it is introduced by an axiom, that is, it is accepted, not found. Hilbert, who defended Cantorian infinity in mathematics, himself concluded that the infinite is nowhere to be found in reality and remains only an idea (Hilbert, 1983, 201). Whoever asserts an infinite series in the conditioned must therefore present it as a fact; the present ontology derives its positions from the single fact — encountered change — and does not accept them without it. The removal does not stop while there is opacity, what has been removed does not return, and the path ends. Completeness is not a sum of sides: the object of knowledge is one and is not composed of parts — Nothing's knowledge of itself is a remotion, not a content (chapter 8). The many sides are the path to knowledge of it as a whole, and the whole is knowledge of dependence as such.

Here two senses of the word "limitation" must be distinguished. In the definition of the conditioned, limitation is determinacy: "this and not that" (chapter 4). In the expression "the infinite is the unlimited", limitation is an end — a last member, a bound in number. The equation of the infinite with the unconditioned does not transfer one sense to the other; it follows from what the determinacy of number is. The determinacy of number is "so many and no more": n and not n + 1. A series whose number of members is no n at all is undetermined in number. The infinite by its very name is indeterminacy: "not-finite" means not limited, that is, not determined. It is objected that the members of the series are determined, and indeterminacy in number is not yet the unconditionedness to which no determinacy whatever is applicable. But then the series would be unconditioned "in respect of" number and conditioned in the rest. There is no unconditionedness "in a respect": respect, side and part are inapplicable to the unconditioned (chapter 4), and the unconditioned is not divided into that in which it is and that in which it is not. Transparency given in the conditioned in some respect does not contradict this: it removes opacity, not determinacy, and to have it is not to be indeterminate (chapter 5); in the conditioned it is the absence of the concealed in that respect, not unconditionedness. Indeterminacy in number, however, removes determinacy and therefore cannot be in the conditioned even in part. One of two remains. Either the series as a whole is unconditioned — then the determinacy of its members is inapplicable to it too, and it is no longer a series. Or unconditionedness is in it only in part — and that is unconditionedness divided into parts, a contradiction in the very concept. Both branches are contradictory, and therefore the infinite is inapplicable to the conditioned.

It is objected that in mathematics an infinite totality is determinate: its cardinality $\aleph_0$ differs from $\aleph_1$, and a series without a last member is determinate as a whole. But this determinacy is not found but introduced: the existence of an infinite totality is accepted by the axiom of infinity of set theory (Zermelo, 1908), not discovered. Every number given in reality is finite: at each step of a series the number of what has been passed through is some n. "As many as one likes" means only "each time one more finite", and the unlimited arrives at no step: repetition of the limited yields the limited at any step. The determinacy of $\aleph_0$ is the determinacy of the accepted, not of the encountered; whoever asserts an infinite series in the conditioned relies on an axiom, not on a fact. Indeed, the very "determinacy of the infinite" is a contradiction in terms: the infinite by name is the unlimited, that is, the undetermined, and to call it a determinacy is to call the absence of determinacy a determinacy. $\aleph_0$ is not a measure of the unlimited but a name given to the unlimited within an accepted system of axioms.

Hence the number of blind spots is also finite. That each distinction removes a zone and no new zones arrive does not yet give an end to the path: with infinitely many zones, indivisible

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steps would not exhaust them. The end is given by the number of zones being finite, and this follows from what has been said: the infinite is inapplicable to the conditioned. Nor, for the same reason, can infinitely many sides be ascribed to the one object of knowledge. It is objected that each traversed segment is finite, but the finiteness of the whole path does not follow from this, and the absence of a presented infinity does not prove the impossibility of an unending continuation. The finiteness of the whole path, however, follows not from the finiteness of the segments but from the path as a whole being information, and information being conditioned (chapter 5), that is, determined, including in the number of distinctions. The determinacy of number is “n and not n + 1”, and there is no other determinacy of number in the encountered. The encountered is given by distinctions, that is, by steps (chapter 4), and every number given by steps is at each step some n. A number reached at no step — the cardinality of an infinite set — is not given by steps and is the determinacy of what is accepted, not of what is encountered (above). The determinacy of the conditioned is the determinacy of the encountered: “this and not that” (chapter 4), and “not n” is no “this”. This understanding of the determinacy of number is finitist, and the system does not conceal it. Mathematics knows other determinacies too — ordinal numbers and cardinalities of the infinite — but introduces them by an axiom (above). The system does not apply them to the conditioned: of the encountered it accepts only what is derived from the encountered, and the encountered is given by steps. Whoever ascribes an infinite determinacy to the path accepts it without a fact; the dispute here concerns the grounds of determinacy, not the consistency of the derivation. A path without a last member is no n in number, that is, undetermined in number, and the dichotomy given above applies to it. An “unending continuation” is the same series without a last member and therefore the same infinite.

The separation of the endless and the infinite goes back to Aristotle. He rejected actual infinity and admitted potential infinity: a quantity is infinite if one can always take a part outside what has already been taken, and that outside which there is nothing is complete and whole (Aristotle, n.d., Physics III.6). The present ontology rejects the potential infinite as well. “One can always take more” is unlimitedness with respect to continuation, and unlimitedness is inapplicable to the conditioned. The divergence lies in the concept: for Aristotle the infinite is a property of quantity; here it is the absence of a limit, that is, unconditionedness. The path of distinctions is not a possibility of taking more: it is what sets in, and as information it is determinate (chapter 5). A potential infinity is never a whole and therefore cannot be the determinacy of the path in number.

Spinoza separates the infinite otherwise: the infinite in its own kind and the absolutely infinite, which involves no negation (Spinoza, 1994, I, def. 6, explanation). The infinite in its own kind exists for him in the dependent too, and it receives its infinity from the infinite attributes of God. Here no positive determination is applicable to the unconditioned; it has no attributes — only remotions — and there is nowhere for an infinite in its own kind to be derived from. In the dependent, by contrast, what is infinite in its own kind would be unconditioned in that respect, and the infinite is inapplicable to the conditioned.

In contemporary philosophy the impossibility of actual infinity is defended by Craig: an actual infinite cannot exist, and it cannot be formed by successive addition, for one can always add one more member before infinity is reached (Craig, 1991). He admits the potential infinite and rejects the actual because of the absurdity of its consequences. The present ontology diverges from him on both points: the endless is rejected as potential too, and the ground of rejection is not the absurdity of consequences but the inapplicability of the unlimited to the conditioned.

The path to transparency is the growth of the *I*. The *I* is self-differentiation. In an elementary change the not-oneself is the neighbouring distinguished; with complexification the *I* distinguishes itself from an ever wider other. The limit of the path is knowledge of oneself as

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the mode of being of Nothing's knowledge of itself: as dependent, not confusing oneself with Nothing.
Every distinction removes a blind spot: after it the side ceases to be blind, and no new ones arise. Nor does the removal itself stop while there is opacity. From the side of Selfhood it cannot be interrupted: from the side of Selfhood it is not an event. Information cannot retain opacity: by itself it has nothing. Stopping is not a mere absence of the new either: constant removal given present opacity is its decrease. Nor is there a pause between distinctions: time is composed of the succession of states (chapter 15), and where there is no succession there is no duration in which the removal could stand idle. Nor do the choices of instances of the *I* stop it: a choice is the I's own by its proximate ground and is a link of the chain through which the removal proceeds, not its stopping (chapter 21). Therefore blind spots are removed completely: self-transparency is attained, and the path of distinctions has a limit.
What, then, at the limit? Information transparent to itself has no blind spots, and therefore no respects in which new knowledge could appear. Change is the appearance of knowledge that was not there (chapter 10); where there is none, there is no change, and with it no time (chapter 15). This is not a stopping of the removal: from the side of Selfhood the removal is not an event, and it appeared as constant only as what stands at every distinction. In transparency there is no opacity, and therefore nothing at which the removal would stand, and nothing is interrupted. Information transparent to itself remains conditioned and dependent: it is determined and knows itself not by itself (T19).
Teleology is not a purpose of Nothing: purpose as a form of changeability is inapplicable to the immutable. Teleology is the directedness of the dependent, given by the one-sidedness of the removal, and is realized only in Something. There distinction in information proceeds not by itself but from Selfhood. With the first distinction consciousness begins; distinctions are composed into elementary changes, from whose onset measures are derived; aggregated forms are composed, and consciousness grows to reason and intellect. Teleology is the directedness of the dependent on the scale of all conditionedness.
Purpose is neither a subjective intention nor an external design but a completed state. The completion of conditionedness is the removal of all blind spots by distinctions. Incomplete conditionedness contains opacity, and its removal goes on; completed conditionedness contains none.
Thus the teleological structure is the directedness of the removal, not a mechanism and not a level of organization: it stands at all levels. It is neither imposed from outside nor arises by chance. Its limit is information transparent to itself and remaining conditioned.

Chapter 14. Elementary Changes
An elementary change is the smallest change in information: one distinction, one onset of a new state. From its onset the measures are derived (chapter 15). Elementary changes are not objects, not processes and not particles in the habitual physical sense. They are minimal units of distinguished information: distinction taken from the side of change (T15). They can stand in relation to one another and be composed into configurations of a higher order. Their nature is wholly informational: they exist only in Something.
An elementary change is the onset of knowledge of difference in one respect: the new state of information differs from the former by this knowledge and by this alone. It does not stand by itself: it proceeds by the removal of opacity, and its place among the others is set by the respect in which information knows itself by this change (chapter 15). An elementary change is not an "entity" and not a "thing" but the smallest succession of states of dependent information.
In an elementary change there is already the *I*. It is knowledge of the difference "this and not that", and this knowledge is information's knowledge of itself in the given respect (chapter

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11): “oneself” is the distinguished “this”, “not-oneself” the “that” from which it is distinguished. This is the *I* of the lowest degree: it has no states of its own that it would distinguish, only one self-differentiation. Lived experience is present in it (chapter 12): a state is given to the one whose state it is — to this very self-differentiation; but there is nothing in itself for it to distinguish.

Each elementary change gives a new knowledge of difference. A sequence of elementary changes underlies everything that is later interpreted as motion, interaction, existence and life: at the level of ontology all this is sequences and configurations of elementary changes. The measures are derived from the elementary change itself, and everything further is measured by them (chapter 15). Without measures elementary changes could be related neither to one another nor to the configurations into which they are composed.

An elementary change has no inner geometry: no form and no extension in space. It has only knowledge of difference, a place — the respect in which it is given — and a duration, the smallest measure of time (chapter 15). Therefore elementary changes are not bound by physical notions of particles, bodies or motions.

An elementary change is a succession within the dependent with the unconditioned unchanged: the new state differs from the former not by a change of the unconditioned but by a knowledge of difference given in information. It introduces no new principle and does not affect the unconditioned.

Elementary changes form relations. Interaction is the mutual self-differentiation of instances of the *I*: each distinguishes the other as not-oneself, and this distinction is a change in each (chapter 16). At this level it is neither a force, nor a field, nor motion, but a succession of informational relations between changes.

A pair of elementary changes is the first place where a bond is possible. If the interaction of the changes has not held, each keeps its own *I*, and each is not-oneself for the other. If it has held and become constant, the first bond arises (chapter 16). Then the *I* of the pair appears: its “oneself” is both changes, its “not-oneself” the external. The *I* of each change does not disappear: each distinguishes itself both from the second and from the external. For the first time there are states of its own — each change distinguishes the other as part of a common “oneself” — and therefore the degree of the *I* of the pair is higher than that of a single change. Here the nesting of the *I* begins: everything further repeats this structure on a greater number of bound distinctions. The stability of the pair is the holding of its bond, not a property of material or energy.

Elementary changes are that of which reality is composed. Aggregated forms, systems, objects, motions and regularities are composed of them. There are no “particles”, “bodies” or “energy” here as primary entities. To which physical quantities elementary changes correspond, ontology does not establish: this is a question of comparison with physics (chapter 22).

Elementary changes are wholly dependent. At the limit, distinction in them proceeds not by itself but by the removal of opacity from Selfhood, and the removal stands at every distinction. This does not abolish causality within the conditioned: the cause of a change is that in the conditioned without which it would not have set in — the proximate ground (chapters 3, 10). Causality is therefore the relation of proximate grounds within the dependent, not a property of the unconditioned and not a mere regularity of sequence.

All further forms are composed of elementary changes: motion, regularity, interaction, life, consciousness and reason. All of them are a complexification of the relations between elementary changes and are determined by one and the same: information as the mode of being of Nothing’s knowledge of itself, and the distinctions proceeding in it by the removal of opacity.

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An elementary change is elementary not by convention but by derivation. Change is the onset of knowledge of the difference “this and not that” (T15). There is no part of such knowledge: what is less than the difference “this and not that” either is already a difference — and then it is the same whole knowledge — or is not a difference — and then there is no knowledge of difference, and with it no change. Knowledge “in some respect” (chapter 9) is not a part of one distinction: it is a whole distinction in one of the respects. Hence the smallest change is one distinction, and the step is indivisible. The objection that change is continuous confuses levels. Continuity is a property of description by a measure, and measures are derived from elementary changes and are not given before them. Continuous time and space are limit descriptions of a multitude of indivisible steps, not that of which the steps are composed. It is objected that the wholeness of one knowledge does not exclude that between two states there lie other states with other whole differences. They do lie there — if these are other distinctions; but then these are other steps, and between the former and the new state in one respect there is no third (chapter 15). And there cannot be an infinite series of intermediate states between two: distinctions are finite in number (chapter 13). Therefore every state has a next one, without an intermediate. Disclosure is therefore discrete: it proceeds by indivisible steps.

A discrete structure of reality is assumed by some programmes of physics as well. In causal set theory spacetime is a discrete set of events ordered by the causal relation (Bombelli, Lee, Meyer, & Sorkin, 1987); digital physics describes nature as a discrete computational process (Zuse, 1969; Fredkin, 2003). The similarity lies in discreteness and in the derivativeness of the continuous. The divergence lies in the ground: there discreteness is accepted as a hypothesis about the physical world; here it is derived from change being knowledge of difference, and there being no part of knowledge of difference.

Chapter 15. Measures

Measures are derived where distinction proceeds. The derivation is carried out reconstructively — in the language of distinctions already introduced, for there is no other language of description. The words “first”, “then”, “after” mean the order of exposition, not the order of arising, which is what is derived here. Ontologically, measures are derivative from the onset of an elementary change. There are none in the unconditioned, and they do not arise in it: they exist only in distinguished information. Measures are not objects and have no being of their own.

All measures — time, speed, localization, space and energy — are derived from a single fact: the elementary change. Three things are distinguished in it: the former state, the onset and the new state. Change is only the onset: what has been distinguished, once given and held, is a state, not a succession (chapter 9). The onset is called the working moment of an elementary change. Its edges are implementation, that is, the removal of opacity by which a new knowledge of difference is set, and registration, that is, the givenness of the new state. The removal proceeds from Selfhood through a proximate ground — the other to which the change responds. The working moment is therefore an elementary response of the *I*: from the not-oneself received to oneself become other.

The onset has extent, but not as a third state. The former and the new are states of the same information in the same respect: without the knowledge of a given difference and with it. They cannot be one state: one and the same cannot both lack this knowledge and have it. Nor is there a third between them: in this respect information either has the knowledge of difference or does not (chapter 14): the step is indivisible, and knowledge of difference in one respect is never given in part. Hence the onset is not a state between the former and the new but the very transition from one to the other. There is an order between them: the new contains knowledge that is not in the former, and knowledge does not decrease (T16);

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therefore the former does not follow the new. The extent of the onset is precisely that the states are two and are ordered: it is not the duration of something happening between them but the very step from one to the other. Were the former and the new one, there would be no change at all. But change is encountered.

The measure of time is the extent of the onset, that is, the step from the former state to the new. This is elementary time, the duration of one change. The step is indivisible; there is nothing smaller than one distinction (chapter 14); therefore the duration of one change is the smallest measure of time. It is objected that indivisible steps could have durations 1, 1/2, 1/3 and so on, with no smallest. But such a series of durations is infinite, while distinctions, and with them steps, are finite in number (chapter 13); of a finite number of durations one is the smallest. Moreover, duration here is not an external magnitude ascribed to a step: time is composed of the succession of states (above), and there is no duration apart from succession. Different durations of steps would require a measure of duration apart from steps, and there is none; therefore the step is itself the unit of duration. The sequence of onsets gives the course of time, and duration is measured by the number of steps. The time of a configuration is the number of its own steps; therefore between two events different configurations may pass through different times (chapter 22).

Against discreteness Bergson objects: homogeneous time, in which states of consciousness form a discrete series, is nothing but space; pure duration is the form that the succession of states takes when our ego refrains from separating its present state from its former states (Bergson, 1913). The present ontology accepts Bergson's distinction but places it differently. What is discrete is change: the step is indivisible. The non-separation of the present and the former of which Bergson speaks is the connectedness of knowledge: the knowledge of the present distinction refers to former ones by its very content (chapter 12). Duration in Bergson's sense is not the continuity of change but the connectedness of a knowledge that does not decrease.

The measure of place is localization. Every knowledge of difference is information's knowledge of itself, and to know a difference within it is to know it in that respect (chapter 9). A change affects information not as a whole but in some respect; this respect is the place of the change. Localization is the respect in which information knows itself by the given change. The first change has a localization without other changes: it already affects information in a definite respect. This is not a spatial coordinate: there is as yet no space. Space is the totality of localizations of the present configuration and of the distances between them. Space does not precede changes as a container: it is the totality of places where changes set in. It can expand and contract; knowledge does not depend on its size and accrues under any course of the universe (T16). Since the respects are finite in number (chapter 13), space is finite. The distance between two localizations is the least number of steps of a sequence of changes leading from one to the other. The dimensionality of space is not derived from this and is not established in the present work.

Speed belongs to a sequence of changes: it is the ratio of the distance it covers to the number of its steps. A single elementary change has no speed: it is one step. Since distance is measured in steps, in n steps a sequence covers no distance greater than n, and speed has a greatest value. A limiting speed is not accepted here but follows from the indivisibility of the step.

The fifth measure is energy. It is change taken in concentration. A configuration is composed of many elementary changes in different localizations, and its step of time is a step of its state as a whole. Energy is the number of elementary changes that set in within a configuration during one step of its time, that is, the density of change. This number differs from configuration to configuration, and therefore the measure does not reduce to unity. It introduces no new nature: like the other measures, it is derived from the step of an elementary

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change; the number of elementary changes is finite, and the measure is defined. Physics agrees with this: the energy of a quantum is proportional to frequency ($E = h\nu$), and the greatest number of transitions of a system into distinguishable states per time is bounded by its mean energy above the ground state (Margolus & Levitin, 1998). This is a correspondence, not an identity: the frequency in $E = h\nu$ is the frequency of phase, not the number of distinguishable changes of state; what coincides is the direction — more energy, more change per time — not the definition. The conservation of energy does not conflict with the non-decrease of knowledge (T16): energy is how much change proceeds, knowledge is what the change that has set in gives.

All measures are derivative from distinguished information. There are none in the unconditioned, and they do not pertain to Selfhood. Measures have no being of their own: they are relations within the distinguished, not fundamental entities; yet they are not conventions of description either, but the way distinctions are actually related. One cannot speak of time without distinguished states, of space without the localization of distinctions, of speed without a sequence of steps.

When information is composed, according to proximate grounds, into different configurations, the measures appear as relations within them: positions (localization), sequences of succession (time), distances covered (space) and their ratio to the number of steps (speed). The unconditioned is immutable; only succession within dependent information has measure.

Thus with measures distinctions acquire the relations in which they are composed into complex forms: aggregated forms and, further, consciousness up to reason. Measures are relations in change, but not its cause. Change proceeds by the removal of opacity from Selfhood, and measures express the structure of already distinguished information.

Time — the extent of the onset: the step from the former state to the new; duration is measured by the number of steps

Localization — the respect in which information knows itself by the given change

Space — the totality of localizations of the present configuration; distance — the least number of steps between localizations

Speed — the ratio of the distance covered to the number of steps; the greatest is one unit of distance per step.

Energy — change taken in concentration: the density of change, the number of elementary changes that set in within a configuration during one step of its time.

Chapter 16. Interaction

Interaction exists where distinguished configurations distinguish one another. It is neither a force nor an influence in the physical sense, and it presupposes no independent entities acting on one another: its sides are separate as instances of the *I*, but none stands by itself.

Interaction is the mutual self-differentiation of instances of the *I*. The *I* is self-differentiation (chapter 11). When one *I* distinguishes another as not-oneself, this distinction is a change in it; when the second also distinguishes the first, there is change in both. It is precisely the latter that is called interaction. The distinguishing of one by another that is not reciprocated is distinction, but not interaction: from one distinguishing the other it does not follow that the other distinguishes it. In interaction the *I* of each side is turned towards its not-oneself, and how it distinguishes it is determined by its own configuration.

Interaction does not affect the unconditioned: it exists only in Something. There is no causality issuing from the unconditioned here. In interaction a change of one side is the proximate ground of a change of the other (chapter 10): without it this change would not have set in. This is what is called the influence of one on the other; it is not the action of an independent entity, for neither side has anything by itself (T4).

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Interaction may hold and may not hold. If the distinctions of the sides are such that the response of each to the other does not destroy their relation, they are held as a stable whole; if it does, the relation disintegrates. Compatibility here is not a property of will or force but the way the responses of the sides are determined by their configurations. The unconditioned remains immutable.

Interaction proceeds at several levels.

The first level is the interaction of elementary changes. This is its smallest form. If the responses of the changes to each other hold their relation, it becomes a stable sequence; if not, it disintegrates. There is no geometric content here: this is a pure form of informational relation.

Here interaction and bond are separated. Interaction is any mutual self-differentiation of instances of the *I*; it may not hold. A bond is an interaction that has held and become constant. Then it sets a common algorithm of functioning for the sides, and a whole is composed of them. The algorithm of functioning is the way a bond determines the responses of the sides: with the same bond and the same other the response is the same, for knowledge of difference is determined only by the conditioned itself (T3), and the removal is constant (chapter 10). Experience changes not this regularity but the bond itself to which it applies (chapter 20). A bond through which the removal proceeds within a whole is called an agent of the programme (chapter 18). At the first level, therefore, an interaction that has held as a stable sequence is a bond; one that has disintegrated is interaction without a bond. Every bond is an interaction, but not every interaction is a bond.

The second level is the interactions of aggregated forms. An aggregated form is a stable whole of distinctions held by a bond (chapter 17). When such wholes distinguish one another, a change of one is the proximate ground of a change of the other. Here interaction takes two opposite forms: attraction and repulsion.

Attraction is a response that includes the other into a common algorithm: it leads to a bond and a whole. Repulsion is a response that holds one's own against inclusion into another's: it leads to a boundary, impenetrability (chapter 19) and multiplicity. A response to the other has no other forms: the other is either included into a common algorithm or not, and there is no third. The absence of interaction is not repulsion. Everything conditioned differs from everything else (chapter 4); but where the sides do not distinguish each other, they are not distinguished for each other, and they are "side by side" only for a third. Without attraction there would be no wholes, without repulsion no wholes with a structure of their own: the parts would not hold their own against the whole and would respond to the other only as the whole. Knowledge of the distinguished would not decrease in either case (T16): what is at issue are bonds, not knowledge. These are not physical forces but forms of the interaction of instances of the *I*; whether physical interactions correspond to them is not established by ontology.

Interaction is measurable at all levels: time measures the number of steps of a response, localization the places of changes, space the distances between them, speed the ratio of distance to the number of steps. The measures are taken from the elementary changes themselves between which interaction proceeds, and therefore it is measurable by them.

Interactions form the network of reality. The world, perceived as a multitude of objects, processes and events, is at the level of ontology a network of distinctions distinguishing one another. There are no independent entities acting on one another: the sides of interaction are separate as instances of the *I*, but none stands by itself. There is a derived dependent structure of informational transitions.

The removal proceeds through interaction: a link gives only what ultimately passes through it (chapter 10). Therefore in interaction the changes of the sides are proximate grounds of one another, while their limit is one.

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A relational understanding of reality was developed by Whitehead, for whom actual occasions are composed by prehending one another (Whitehead, 1978), by ontic structural realism, which abandons independent individuals (Ladyman & Ross, 2007), and by relational quantum mechanics, where the state of a system is its relation to other systems (Rovelli, 1996). The system converges with them in that the sides of interaction do not stand by themselves. It diverges in the ground: relations here are not primary but are distinctions, and their limit is the unconditioned given in the conditioned.

Chapter 17. Aggregated Forms

Aggregated forms are the next level, after elementary changes, of distinguished information. They are neither independent entities nor a new ontological nature, but stable wholes of elementary changes held by a bond. Aggregated forms exist only where the interactions of elementary changes have held (chapter 16).

Elementary changes are the minimal units of conditionedness: each gives one knowledge of difference. Single changes do not give a stable whole: for that a bond is needed — an interaction that has held and become constant (chapter 16). If a multitude of elementary changes is bound, they form a new whole — an aggregated form.

An aggregated form is not a set of properties but an ordered whole of distinctions. It exists when the relations between elementary changes are held and repeated. Repeatability here is not accidental: one and the same configuration gives a difference of the same kind (chapter 10), and therefore the way changes are composed into a whole is lawful.

A whole is not a sum of elements: there is a connectedness in it that no part has taken separately (chapter 18). Therefore an aggregated form is not a sum of elementary changes but their bound whole.

From a form being held by a bond its features follow.

The first is stability: a form is preserved while its bond holds, even though other changes proceed around it. On stability rests what is later interpreted as an “object” or a “structure”; at the level of ontology an object is a bound whole of distinctions.

The second is the definiteness of responses: the same changes within a form give the same responses. This follows from the bond determining the responses of the sides — from its algorithm of functioning (chapter 16).

The third is finiteness. An aggregated form cannot contain an infinite number of distinctions: the infinite is inapplicable to the conditioned (chapter 13). Therefore it is finite and determinable. It has no absolute nature: it exists only in Something.

The fourth is measurability. The measures are taken from the very elementary changes of which the form is composed, and therefore every aggregated form is measurable: its place is set by the localizations of its changes, its time by the number of their steps, its extension by the distances between its localizations (chapter 15).

Aggregated forms have no material substance and do not consist of things: they consist of distinctions and bonds. More complex forms are built on them: systems, living forms, perception and reason. Each next level arises not by the addition of entities but by the bonding of forms into more complex wholes.

The stability of an aggregated form is the holding of its bond: the form stands while its distinctions are bound, and disintegrates when the bond breaks according to proximate grounds (chapter 10). The disintegration of a form does not diminish the knowledge of information: what disintegrates is the bound whole, what has been removed does not return (T16), and the connectedness of the knowledges opened in it is not broken (chapter 12).

Levels of such wholes are distinguishable in experience as well: the atom, the molecule, the cell, the organism. They are compared with the levels of the system in Part III; here it matters only that each is held by the bond of its parts.

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The I of a Form

An aggregated form is a whole composed of bound distinctions, and it has its own *I*: it distinguishes itself from not-oneself as a whole. Its responses to the external are the responses of the whole. The component parts retain their own instances of the *I* at their own level, but function according to the paradigm of the form's boundness, that is, by the algorithm of the whole (chapter 18).

What Holds a Form

A form is held by a bond — the interaction of the parts that has held and become constant. It sets a common algorithm of functioning for the parts. The form stands while its bond holds. The bond holds the form together with the repulsion of the parts (chapter 16). Attraction includes the parts into a common algorithm: without it there would be no whole. Repulsion holds each part as a separate *I*: without it the parts would respond to the other only as the whole, and there would be no form with an inner structure. Repulsion acts outwards too: the form holds its own against inclusion into another's. Hence its boundary and impenetrability (chapter 19).

The disintegration of a form is the breaking of its bond: the whole ceases to be a whole, and the parts continue their own instances of the *I* at their own level. The knowledge opened in the whole does not decrease thereby (T16).

The Identity of a Form

The example of the ship of Theseus is offered as an objection (Plutarch, 1914, Theseus 23.1). The ship is repaired, plank by plank, until all are replaced. Is it the same ship? And if a second one is assembled from the old planks, which of the two is the real one (Hobbes, 1839, II.11.7)? For the living the same holds: cells are renewed, and the organism is considered the same.

A form is held not by a set of parts but by a bond, and the bond has its own *I*. With gradual replacement a new part is included into the same bond and responds by the same algorithm. The bond has not been interrupted; hence the form is the same. From the old parts reassembled a new bond is composed. The parts are the same, but the bond has arisen anew: this is a new form with a new *I*. The identity of a form is the continuity of its bond, not the identity of its parts.

If the bond is interrupted, the form — the bound whole — disintegrates. The parts continue their own instances of the *I* at their own level. The *I* of the disintegrated form does not thereby disappear: the knowledge opened in it is preserved as connected knowledge (chapter 12), but there is no longer a bearer in which new distinctions of this *I* would proceed. From the same parts reassembled a new form with a new bond is composed. The identity of the form and the identity of the *I* diverge here: the form is held by the continuity of the bond, the *I* by the connectedness of knowledge (chapter 12). Therefore the *I* of the new form is new if its assembly is not determined by the connectedness of the former *I*, and the former one if it is so determined and the former *I* has no other bearer in which it continues (chapters 24, 25).

It will be objected: if one connectedness is continuously divided into two independent ones, both continue it without interruption; which of them is the former? Neither. The *I* is not merely the continuity of connectedness but the continuity of one connectedness — of one "oneself" and one "not-oneself" (chapter 12). In division the references between knowledges are not broken, but the one "oneself" in which new distinctions proceed becomes two. The former *I* as one, carrying distinctions in one bearer, ends: as one it has no more new distinctions. As connected knowledge it is preserved, as with the disintegration of the bearer (chapter 12): knowledge does not decrease (T16). Two new instances of the *I* arise, and each continues the connectedness of the former by its own distinctions, as memory binds the present *I* to former distinctions (chapter 20).

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Parfit arrives at the same answer: if each half of the brain were transplanted into another body, neither of the resulting people would be me, and this matters less than it seems, for my relation to each of them contains what matters in survival (Parfit, 1984, ch. 12). The similarity lies in the answer “neither”. The divergence lies in the ground. For Parfit what matters is not identity but the relation connecting one with each of those resulting. Here one *I* as carrying distinctions is interrupted because one “oneself” becomes two, and what is preserved is not identity but connected knowledge, which does not decrease.

Chapter 18. Emergence

Emergence is a necessary consequence of distinction proceeding by the removal of opacity and of the complexification of distinguished structures.

When elementary changes interact and become bound, their configurations become more complex. In a bound whole distinctions proceed that individual changes do not have: knowledge refers not only to the difference of two states but also to the bond between them. These distinctions are not created by interaction: it is their proximate ground, and they proceed by the removal from Selfhood (chapter 10).

The whole yields the new without a change of that of which it is composed and without a change of the unconditioned: the new lies in the bond of distinctions, not in another principle. The emergence of consciousness follows the same principle: not a new principle but the bonding and complexification of distinctions within information.

When a multitude of elementary distinctions is composed into a single structure, knowledge of the whole, which individual distinctions do not have, accrues in consciousness. This is not a leap but a strict consequence of the initial principles.

Emergence is neither a metaphysical leap nor a hidden principle. It exists where distinctions cease to be scattered with respect to a given whole and are composed into a stable configuration with knowledge of the whole. This is not arbitrary: it is determined by whether the bonds through which this knowledge is given hold (chapter 16).

Thus emergence is not the appearance of a new being or property but knowledge of the whole given in the connectedness of distinctions. The system acquires no new principles: the former distinctions form a qualitatively different but derivable configuration. Complexification creates no new substance or ontological layer. If it yielded an independent entity, a second ultimate ground would arise. Even with high integration, when the parts respond only according to the algorithm of the whole, the dependent does not become independent. Complex forms of consciousness remain within one dependence: Nothing — Selfhood — information — initiation — distinction, the appearance of which is change.

In contemporary philosophy emergence is divided into weak — the properties of the whole are derivable from the parts, only difficult in practice — and strong — truths about the whole are not derivable from truths about the parts even in principle (Chalmers, 2006), and on an ontological reading the whole has a new causal power acting downwards on the parts (Bedau, 1997). Against strong emergence Kim’s causal exclusion argument is directed: if the physical has a complete physical cause, there is no room for a new cause of the whole (Kim, 1998). The present ontology does not fall under this argument: the emergent here is not a new causal power but knowledge of the whole that individual distinctions do not have. Knowledge of the whole does not enter into causal competition with the parts: it is opened in the same information in which the changes of the parts proceed.

The New in the Whole

Against every emergence a dilemma is advanced. If the new in the whole is derivable from the parts, emergence is empty: there is nothing new. If it is not derivable, it is a miracle — the new from nothing. The present ontology is caught by neither horn. The new in the whole is connectedness: the knowledges of distinctions that refer to one another are composed into

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one knowledge (chapter 12). Connectedness is a relation, not a property of any part, and taken separately it is in none of them. Therefore emergence is not empty. But the new is not produced by the parts: knowledge opens by the removal proceeding from Selfhood, and the parts are only the proximate ground of the connectedness having been composed. Therefore there is no miracle either: the new comes from where every distinction comes from.

The main difference from ordinary emergence lies in consciousness. Here consciousness does not arise from the non-conscious: the *I* exists from the first distinction. What arises are unity and degree, not consciousness itself. Therefore the system is not threatened by Strawson's argument about the impossibility of the "radical" emergence of experience from the wholly non-experiential (chapter 12).

The Action of the Whole on the Parts

If the whole acts on its parts, the change of a part seems to have two causes: neighbouring parts and the whole. But the whole does not stand beside the parts as a second thing. It is their bond. To say that the change of a part is grounded by its neighbours and to say that it is grounded by the whole is to describe one ground part by part and as a whole. There are no two competing causes, and the causal exclusion argument (Kim, 1998) does not apply. The same holds where the whole knows itself as a whole and chooses (chapter 21): choice is the same distinctions of the configuration taken from the side of knowledge (chapter 12). It is the proximate ground of the change of the parts as these distinctions, not as a second cause beside them.

The Necessity of Emergence

Distinction is knowledge of "this and not that". The "that" is something already distinguished, and in the first distinction it is that from which information differs, given without distinction (chapter 5). Hence every distinction by its very content refers to another and cannot stand apart. While there is opacity, the removal does not stop, and distinctions referring to one another become more numerous. Therefore connectedness grows, and a scattering of distinctions referring to nothing is impossible. What is necessary is a tendency in information as a whole, not the fate of each configuration: configurations are composed and disintegrate according to proximate grounds.

This growth is not infinite. Distinctions, and with them the places of opacity, are finite in number (chapter 13): the infinite is inapplicable to the conditioned. Therefore in one universe too complexification has a limit: the levels are finite in number, and so are the configurations, however many there may be. The new in the whole is measured not by the number of configurations but by whether it opens a new side of the knowledge of dependence: a configuration that opens no such side gives nothing new. The limit of emergence in a universe coincides with the end of the stage (chapter 13): when the universe's knowledge of itself as a whole, into which the experience of instances of the intellectual *I* enters as nested, can yield nothing new in the given paradigm, emergence in this universe is exhausted too.

Bond and the Agents of the Programme

Configurations of increasing complexity are structured in the same way at each level. The component parts have their own *I* at their own level and respond to not-oneself; but in the whole they function not only by themselves but according to the paradigm of the boundness of the whole. The paradigm of boundness is the algorithm of the whole by which its parts respond; it is set by the bond of the whole (chapter 16).

That through which this determination proceeds is the agent of the programme — a bond between instances of the *I*. The bond gives each *I* an algorithm of response common with the others. The bond itself has its own *I* as well: the bound whole distinguishes itself from not-oneself (chapter 17).

Where does the algorithm come from? Regularity is given by the constancy of the removal: one and the same configuration, the state of the whole included, gives a difference of the

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same kind (chapter 10). The algorithm of a bond is this regularity taken for the given bond: with the same bond and the same other the response is the same. Directedness is given by the one-sidedness of the removal: what has been removed does not return, knowledge accrues. Law and programme are not two principles but two sides of one removal. The programme therefore proceeds through the agents of each level without governing them from above.
Here an objection of regress is possible: if the bond has its own *I*, and every *I* in a whole responds according to the algorithm of a bond, a bond of the bond is needed, and so on without end. There is no regress. The *I* of a bond is the *I* of the whole itself, not of a third beside the parts: the bond does not stand between the parts as a separate thing needing a bond of its own but is the held interaction of the parts themselves (chapter 16). The whole is bound to its parts by being their bond, and no second-order bond is required.
The whole of each level has its own *I*, and its bonds are the agents of the level. The lower the level, the fewer states of their own the parts have to distinguish, and the more fully their responses are determined by the algorithm of the whole. The higher the level, the more distinctions of their own the parts have, and the more their responses are determined by these.
A bond may break according to proximate grounds, and a part falls out of the whole (chapter 20). This is not a choice: choice belongs to one who knows bonds (chapter 21).

Chapter 19. Matter

Matter is a stable aggregated form of information. Material objects do not exist as fundamental entities: they are stable aggregated forms held by a bond (chapter 17). Motion is a sequence of elementary changes whose places change (chapter 15). To which physical quantities particles, fields and the laws of physics correspond is compared in Part III; ontology asserts here only that all of them are configurations of distinctions and their regularities, and that regularity is one — the constancy of the removal (chapter 10).
The classical concept of matter as an independent extended substance is no longer confirmed by fundamental physics (chapter 22). The present ontology does not look for what matter is "made of": matter is a stable aggregated form of information. A perceived object is such a form, distinguished by the consciousness of another configuration through interaction (chapter 16) and measured by measures (chapter 15): objecthood is a relation of the distinguished and the distinguishing, not a property of a substance.
The thesis "matter is a stable form of information" cannot be read as idealism. Idealism makes matter dependent on the consciousness of a subject; here both matter and consciousness are forms of one dependent — information, taken from two sides: from the side of change, configurations and their stable forms; from the side of knowledge, consciousness (chapter 12). Neither is reducible to the other. Matter is not an illusion or a representation: it is as actual as everything conditioned, and as dependent. It appears extended and hard because its configurations are measured by measures (chapter 15) and hold their bond with one another through interaction (chapter 16): a stone is hard not because there is substance in it, but because the bond of its distinctions is stable against proximate grounds.

Materiality

Materiality is not a special nature. It is the way a stable configuration is encountered by another configuration. Take a stone. It is extended, impenetrable, keeps its state and resists. Each of these features is derived from what has already been established. Extension is the changes of the configuration occupying many localizations with distances between them (chapter 15). Impenetrability is repulsion: the configuration holds its own against inclusion into another's (chapter 16). Keeping its state is the configuration's bond holding until a proximate ground changes it. Resistance is the change of the configuration proceeding only through interaction (chapter 16). None of these features requires substance. All of them are the stability of the bond of a configuration against the proximate grounds of other

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configurations. Density and inertia in the sense of physics are quantitative measures of these features; ontology does not derive their magnitudes.

Hence the material and consciousness are not two natures. The same distinctions are, from the side of change, a configuration and its bond; from the side of knowledge, consciousness (chapter 12). Matter is the stability of a bond taken from outside — for another configuration; consciousness is the same distinctions taken as knowledge.

A similar picture of matter is given by ontic structural realism: metaphysics should abandon the picture of the world as composed of self-subsistent individual objects and the image of causation as the collision of such objects (Ladyman & Ross, 2007). With this the conclusion of the chapter coincides: matter is not a special nature but the stability of a bond. The divergence lies in method and in ground. Ladyman and Ross recognize only a metaphysics built on contemporary science, not on a priori intuitions; here ontology is derived and only compared with physics (chapters 22–26). And for them structure needs no ground; here bonds are dependent, and the removal proceeds through them.

The Bearer of Information

An objection is raised: information is always recorded in something — in states of matter, in a signal, in neurons. Landauer puts it directly: information is inevitably tied to a physical representation (Landauer, 1991). Hence the bearer is primary, and an informational ontology covertly presupposes matter.

The objection speaks of a different concept of information. It means the distinguishable states of matter by which what is recorded is represented: this is the physical concept of information, close to Shannon's (chapter 9). Here, however, information is the mode of being of Nothing's knowledge of itself. It is recorded in nothing. Bearer and what is recorded do not apply to it: it is not "in" anything, because "in something" is already something distinguished within it.

Physical information is a particular case, and the system explains it. A record is a relation of one configuration to another: the states of one distinguish the states of another through interaction (chapter 16). Both bearer and recorded are aggregated forms. Therefore "information is physical" is true at this level: what is recorded is held by the stability of the bearer configuration. But this is said of configurations within information, not of information itself. Transferring what is said of the parts to that in which they stand is a confusion of levels.

Nor does the objection yield matter as primary. If information requires a bearer, and the bearer is matter, one must say what matter is. Every definition of it in physics — mass, charge, field, state — is given through distinguishable states and their relations. By itself this does not prove that matter is information in the present sense: it shows only that the demand for a bearer does not lead beyond the distinguished. That the distinguished is information has been derived independently (chapters 5, 8).

Causality

The second objection: matter acts. A stone breaks glass. A cause must possess something — force, mass, energy. By what does a configuration of distinctions act?

The causality of matter is a proximate ground. Within the conditioned, where the distinguished has already been given, one thing is grounded by another (chapter 10). A stone breaks glass: a change in the bond of one configuration is the proximate ground of the disintegration of the bond of another. No force as a special entity is required for this. One thing is required: that one configuration be the proximate ground of another — that without its change the change of the other would not have set in (chapter 3).

By what, then, does a cause act? Two questions must be separated here. Why precisely this change set in and not another — this is answered by the proximate ground: it is the cause, and in the conditioned this answer is complete. Why knowledge of difference sets in at all rather

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than opacity standing — this is answered by the limit: the removal proceeds from Selfhood (chapter 10). A link has nothing by itself and gives only what ultimately passes through it. The “force” of a cause is therefore not a possession of the link taken by itself, but neither is it something other than the link itself: it is the fact that the removal proceeds precisely through this configuration and is determined by its distinctions.

This may recall Malebranche’s occasionalism: God alone acts, and creatures are merely the occasions of his action (Malebranche, 1997, Bk. VI, pt. II, ch. 3). The divergence is fundamental. For Malebranche God is the efficient cause, and creatures are deprived of causality. Here the unconditioned does not act and is not a cause (chapter 5), and causality belongs wholly to the conditioned: what sets in is determined by the conditioned itself (chapter 8). The limit gives not what sets in but that anything sets in at all; therefore it does not compete with causes and does not replace them.

This does not contradict the causal closure of the physical. All proximate grounds lie within the conditioned, and the physical chain of causes is closed at its own level. Initiation does not enter this chain as a second cause: from the side of Selfhood it is not an event and lies at the level of the limit, not of the proximate ground. Therefore the causal exclusion argument (Kim, 1998) does not arise either. It works when two causes of one level compete for one effect. Here there are none: the limit does not compete with the proximate ground but passes through it. The same answer removes the question of the action of consciousness. Consciousness and configuration are the same distinctions taken from two sides, not two causes of one effect. Therefore choice too (chapter 21) acts in the physical chain not as a second cause but as the same distinctions of the configuration.

The Consciousness of Matter

The third objection: if the *I* exists from the first distinction, it exists in an electron and in a stone, and this is panpsychism.

Panpsychism is usually the name for the doctrine that fundamental physical entities have conscious experience (Chalmers, 2017). In this broad sense the present system is a panpsychism: the *I* and lived experience exist from the first distinction (chapter 12). It differs from its known varieties on three points. Consciousness here is not a property of matter but knowledge opened by the removal in information; matter is not the bearer of consciousness but the same distinctions taken from the side of change. What serves as the measure is not the presence of consciousness but the degree of the *I* (chapter 12). And the combination problem is not left open but solved by the connectedness of knowledges (chapter 12). The stone is not conscious “as matter”, and consciousness is not material.

There is an *I* in the stone. What differs is not presence but degree. The measure in which the connected knows itself grows with the extent to which distinctions distinguish the states of that in which they themselves proceed (chapter 12). In a stone distinctions are turned almost wholly outwards: they are relations of parts and a response to the external. Its own states are little distinguished in it. Therefore its awareness is small. The absurdity of a “conscious stone” arises if the *I* of the stone is measured by the human standard. The standard is one for all, and by it the stone and the human being stand on one scale.

Whether a stone has one *I* is decided by connectedness. What makes consciousness one is the connectedness of distinctions whose knowledges refer to one another (chapter 12). A heap of sand is one thing for the eye, but the *I* of the heap as a whole is almost absent: the grains are weakly bound, and a common “oneself” is barely composed among them. The unity of the *I* also has a degree. In the brain connectedness is great, and its distinctions are composed into one *I*. The stone stands between them. The objection of the “conscious stone” tacitly takes the stone as a unit of consciousness. But the unit of the *I* is set not by what is called a thing but by the connectedness of distinctions. The *I* exists wherever there is distinction; what is decided is only whether it is one or many.

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Chapter 20. The Development of Consciousness

The levels of consciousness differ not by thresholds but by what is known in it, and each contains the preceding ones. The measure of consciousness, however, is one at all levels — the extent to which the distinctions of a configuration distinguish its own states (chapter 12). The first level: differences are known — this is elementary consciousness: “this and not that”; every distinction refers to others (chapter 18), but bonds as such are not yet known. The second: the bonds between distinctions are known — this is reason (chapter 21); its highest level is reflection, the knowledge of oneself as distinguishing. The third: the whole is known as one’s own — one’s place in the whole and the direction of the removal, that is, the programme; this is intellect (chapter 21). The transition from level to level introduces no new nature: it is the same self-knowledge at a greater measure of the removal of opacity.

If consciousness is self-knowledge accrued with every distinction, it exists not after an additional stage but where distinction proceeds: every distinction gives the structure “this and not that” and is thereby the elementary form of consciousness.

Elementary consciousness has no reflection, memory, intention or inferences: there is only knowledge of difference in it. If a configuration remains simple and its distinctions are not turned to its own states, consciousness remains elementary.

Life is a form in which the *I* distinguishes the disturbance of its own bond and holds it: restores it when it is disturbed and takes from the external for this. This is not a biological concept but a level of the same scale by which every *I* is measured. A non-living form is also held by a bond and also responds to the external, but its responses do not restore its bond: a disturbed bond leads to disintegration or alteration. A living form holds its bond by itself. The biological definition of life is a particular case.

Hence the possibility and the expectability of the arising of the living from the non-living, and both are given by emergence (chapter 18). A non-living bond distinguishes only itself and not itself: a disturbance comes to it as another, and its response is a response to another. A living bond distinguishes its present state and its held state: the held state is the “that” from which the disturbance is distinguished. To distinguish so, the held state must be given separately from the present one: knowledge of the difference “this and not that” requires both members, and if the state of the whole is only what it is now, there is nothing from which to distinguish a disturbance. Knowledge of the whole, given in connectedness (chapter 12), exists already from the first level of complexification, but in a non-living bond it exists only as its present state: a disturbance changes the whole itself, and there is nothing with which to compare the present with the former. The living begins where there is a part in the bond whose state holds the state of the whole independently of the succession of the other parts — the keeping of the held state (chapter 25). Then the disturbance is distinguished from the held, and the bond is restored by it. Therefore the transition to the living is not a new principle but the same knowledge of difference with a new object: oneself as a whole, held separately from the present. The *I* exists from the first distinction, and between the non-living and the living there is no gap in nature, there is a difference of degree. This is the possibility.

Expectability: connectedness grows of necessity (chapter 18), and the removal is directed towards the self-transparency of information, that is, towards self-knowledge, which grows with the extent to which distinctions are turned towards their own states (chapter 12). Life lies on this path. The paths investigated by the science of the origin of life — chemical networks whose products restore the network itself from external material (Kauffman, 1986) — describe the proximate grounds of this transition: they confirm it, not ground it. What is expectable is a tendency in information as a whole, not the arising of life at a given place and time: that is decided by proximate grounds. If life is a level and not a threshold, there must be

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intermediate forms between the non-living and the living that are hard to assign to one or the other.

The definition of life through self-restoration from the external is close to the concept of autopoiesis: the living is a network of processes producing the very components of which it consists and thereby holding itself as a unity (Maturana & Varela, 1980). The similarity lies in the living being defined by holding itself, not by matter. The divergence lies in the ground: for Maturana and Varela autopoiesis is a description of an organization; here holding oneself is derived from knowledge of difference whose "that" has become the held state of the whole, and therefore life stands on one scale with every *I*.

Closest to the present system in its course of thought stands Teilhard de Chardin. Besides the without, things have a within, co-extensive with the without in every region of space and time; the growth of complexity is accompanied by the growth of consciousness — this is the law of complexity-consciousness; evolution converges on the Omega point, whose attributes are autonomy, actuality, irreversibility and transcendence (Teilhard de Chardin, 1955). The similarity is great: consciousness exists in everything from the beginning and grows with complexification, and the path is directed. The divergence lies in three points. The "within" in Teilhard is a side of matter itself; here consciousness is knowledge given by the removal, not a property of matter (chapter 19). The law of complexity-consciousness in Teilhard is found in experience; here the degree of the *I* is derived from the structure of distinction (chapter 12). And the limit of the path here is not a transcendent centre in which consciousnesses converge: self-transparency remains conditioned, and the unconditioned is not a point of convergence and draws nothing to itself.

Already at the level of the living distinctions refer to former ones — this is memory (see below) — and what repeats is recognized as repeating: a new distinction refers to a former one of the same kind. The more held bonds between states, the more distinctions are turned to one's own states, and the higher the degree of consciousness. When not only repetitions but the bonds between the distinguished are known, reason begins (chapter 21).

Semantic structure creates no independent ontological level and does not change the nature of dependent information. Context is an orderedness of distinctions, not an entity apart from them. Even complex and multilevel, it remains a bond of distinctions, and its extension does not make the dependent independent.

Consciousness begins with the first distinction. This is not panprotopsychism.

Panprotopsychism ascribes to fundamental physical entities protophenomenal properties — not themselves phenomenal, but capable in aggregate of constituting phenomenal ones (Chalmers, 2017). Here what is in the first distinction is not protophenomenal but already lived experience of the lowest degree (chapter 12), and it is not a property of a physical entity but knowledge opened by the removal in information.

An electron persists in time, that is, it is not one elementary change but a held sequence of them (chapters 14, 17); its *I* is the *I* of a bound whole, even if of the least complexity. A stone is a stable form; there is an *I* in it too, but its awareness is small: its distinctions are turned towards interactions with the other, not towards its own states.

Hence the *I* exists throughout reality from the first distinction. What differs is not its presence but its degree — the level of awareness (chapter 12).

Memory and Experience

Knowledge accrues and does not decrease: what has been removed does not return. But the living forgets. There is no contradiction, because knowledge and memory are not one.

Memory is the bond of the present *I* of a configuration with the distinctions it has had. To remember is for present distinctions to refer to former ones. Forgetting is a break of this bond, not a decrease of knowledge: what has been removed remains the knowledge of information, and the connectedness of knowledges on which the identity of a selfhood rests is

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not broken by forgetting (chapter 12). Experience is memory that has changed the bond. Past interactions change the bond itself, and the next response proceeds by the algorithm of the altered bond; regularity does not change thereby, it applies to another bond (chapter 16). This is learning. At lower levels the bond is hardly changed by experience; the higher the level, the more it is changed by experience.

Nested Instances of the *I*

The living is a form of forms: an organism of organs, organs of cells. A cell has its own *I*: it is a living form that holds its own bond. An organism has its own *I*: its bond is an agent that sets a common algorithm for the cells, and the cells function according to the paradigm of the organism's boundness while retaining their own instances of the *I* (chapter 18).

Instances of the *I* are nested, not separate. A distinction in a cell can refer both to its own states and to the connectedness of the organism; then it enters both instances of the *I* at once. What the organism does not know in the cell — its inner metabolism — lies only in the *I* of the cell. The boundary between instances of the *I* is set by where knowledges refer.

The choice of the intellectual *I* does not reach all levels of the organism. Lower levels are held by their own bonds: a human being decides about the movement of the hand, but not about the division of cells. The more distinctions of their own the parts have, the more their responses are determined by these, and the bond of a part with the whole can break according to proximate grounds: a cell falls out of boundness with the organism. Then the *I* of the cell functions not by the algorithm of the whole but against it: this is an agent that has left boundness. This is not a choice: choice belongs to one who knows bonds (chapter 21).

Chapter 21. Reason and Intellect

Reason is a level of consciousness. It exists where the distinctions of a configuration not only distinguish and hold what is distinguished but also know the bonds between the distinguished: which distinction is bound with which, and how. It is not an independent substance: so much opacity has been removed in the configuration that the bonds between its distinctions are given to it. Reason introduces no new principle, and there is no threshold here, there is a measure (chapter 12).

Knowledge of bonds is knowledge that with one distinction there will be another: if — then. Hence inferences, dependences between states and knowledge of regularities. Reason is not tied to biology: it exists wherever the bonds of its distinctions are given to a configuration.

Reason has a highest level. One can know bonds without knowing that this knowledge is one's own. Knowledge that the distinguishing one is the knower himself, and of how he distinguishes, is reflection: the *I* distinguishing itself as distinguishing. It knows not only "this and not that", but also that this knowledge is its own, and how it distinguishes. Without this one cannot know the dependence of one's own knowledge, and therefore reflection is the condition of the transition to intellect, which knows its place in the whole (chapter 13).

The limit of reason is knowledge of the bonds between all the distinctions accessible to it. Beyond this what is known is no longer a bond but one's place in the whole, and this is the level of intellect. Reason does not produce distinctions but knows their bonds and thereby turns a sequence of changes into ordered knowledge.

Reason belongs not to one distinction but to a bound totality of distinctions: knowledge of bonds exists where knowledges refer to one another (chapter 12). Therefore reason is knowledge of a whole, not a sum of individual acts.

Reason also knows measures: time as a sequence of steps, localization as the places of changes, space as distances, speed as their ratio to the number of steps (chapter 15). The measures are taken from the same elementary changes of which reason too is composed, and therefore everything it distinguishes is measurable.

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Reason is limited by the distinctions of its configuration. It does not go beyond dependent information. It cannot grasp Nothing positively: distinctions are inapplicable to Nothing, and only remotions are accessible. Reason understands only what is expressible in distinguished information, that is, only the conditioned.

At the same time reason is free, and the sense of this freedom must be stated precisely. Two senses of "one's own" are distinguished here. By itself, that is, in the last account, the dependent has nothing, and choice too does not proceed by itself. But choice is one's own by its proximate ground: it is made from the knowledge of bonds and the experience of this very *I*, not from outside. Freedom requires the second, not the first. A choice determined by nothing — neither by knowledge nor by experience — would be not free but uncaused: determinate as this one and not another, and determined by nothing, that is, conditioned and unconditioned at once (chapter 23). What has been said about information not proceeding by itself at any step pertains to the limit and does not abolish this freedom. Choice does not affect the unconditioned, but determines what the next state of this configuration will be.

This understanding of freedom is the compatibility of freedom with determination. Frankfurt showed that responsibility does not require the possibility of acting otherwise under the same conditions (Frankfurt, 1969); Dennett, that the freedom worth wanting is the determination of an act by the knowledge and experience of the agent himself (Dennett, 1984). The system converges with them in the main: free is the choice determined by the chooser himself, not the uncaused one. The divergence lies in that "the chooser himself" is derived here: it is an *I* that knows its bonds, and its knowledge and experience are its proximate grounds, while its limit, as that of everything conditioned, is the unconditioned given in the conditioned.

Where choice comes from. As long as the *I* does not know its bonds, it functions by the algorithm that the bond sets, and it has no choice. Reason knows the bonds between its distinctions, and, knowing the bond, the *I* can respond not only as it sets but otherwise — from the responses its experience admits (chapter 20). "Otherwise" means here: otherwise than the bond sets, not otherwise under all the same conditions; under all the conditions, including the knowledge and experience of the *I* itself, the response is one (chapter 11). This is choice within conditions, not a rupture with the bond. Intellect knows, beyond this, the programme and its place in the whole, and therefore chooses not only the reaction but the algorithm itself — whether or not to take a function under control, that is, whether to leave its responses to the algorithm of the bond or to determine them by its own knowledge. To it also belongs the choice of an understanding of oneself, by which the sifting proceeds (chapters 13, 28).

Reason as a level of consciousness does not violate absoluteness. Its level is attained with a complex configuration of distinctions and accumulated experience. Reason is not a source of absolute knowledge: it depends wholly on the distinctions in which it exists. Complex forms of consciousness arise without new principles or intervention from outside; their measure, as that of all consciousness, is the extent to which the distinctions of a configuration distinguish its own states (chapter 12).

With reason there is also will. It is not an entity apart from consciousness but choice taken from the side of what it does: the determination of a response by the knowledge and experience of the *I* itself. Choice concerns not the unconditioned but information, and therefore does not violate the ontology of Nothing. Will adds no new ontological status and creates no independent principle.

Reason is a level, but not the limit. Thanks to reason reality is knowable within its conditionedness. After reason comes intellect.

Intellect is the level of consciousness that follows reason. Reason knows the bonds between distinctions and chooses within the conditions in which it stands. Intellect knows, beyond this, its own place in the whole and the direction in which the removal of opacity proceeds,

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that is, the programme. This determines the difference: reason chooses within the algorithms of bonds, intellect makes choices concerning the algorithms themselves: it knows the programme and its own place in the whole, for one cannot bring under one's control what one does not know. Intellect introduces no new nature: it is the same self-knowledge at the measure of removal at which a configuration knows itself as part of the whole.
The interactions of developed forms are the third level of interaction (chapter 16). When the sides of an interaction know bonds, their responses are determined by this knowledge too: reason chooses among the responses its experience admits. Thus purposeful interactions arise. They are not a second cause beside the physical chain: knowledge and choice are the same distinctions of configurations taken from the side of knowledge (chapters 18, 19).
At the level of reason interactions are multilevel: the sides not only respond but know the bonds by which they respond and can change their responses. Interactions become semantic: in them not only the other is distinguished but also the bond with it.
Thus reason is knowledge of the bonds between distinctions, reflection is knowledge of oneself as distinguishing, intellect is knowledge of one's place in the whole and of the programme. The choice of a reaction belongs to reason, the choice of the algorithm and of an understanding of oneself to intellect; both are determined by the knowledge and experience of the *I* itself and are therefore free in the derived sense.

PART III. INTEGRATION WITH THE FUNDAMENTAL SCIENCES

Chapter 22. Physics as an Informational System

Contemporary physics discovers features that are hard to reconcile with the classical notion of matter as an independent extended substance. Quantum theory describes a system through states and their transformations, not through objects with properties determined prior to measurement; the interpretation of these states as informational is one of the interpretations accepted in physics, not the content of the theory itself. Therefore physics does not serve here as a ground of ontology: ontology is derived from encountered change (chapters 4, 5) and is compared with physics. The chapter shows where this comparison yields agreement, where the agreement is only partial, and what the system asserts that physics can test.
The ontological level and the level of physical description are strictly distinguished. Ontology speaks of what distinction, change and measure are and how they are dependent; physics describes the regularities of the already distinguished. Therefore quantum nonlocality, the wave function, entanglement, the holographic principle and the energy cost of erasing information are no ground for concluding either that matter is primary or that information is primary: they describe the behaviour of distinguished states. Ontology does not compete with physics and does not correct its conclusions. Why physical laws have precisely this form it does not derive; it derives only that there is regularity (chapter 10), that measures are discrete (chapters 14, 15) and that there is no uncaused outcome (chapter 23).
A fundamental particle in quantum field theory has no extension in the classical sense: it is an excitation of a field described by a state, not a body. Ontology says the same otherwise: a particle that persists in time is a held sequence of elementary changes (chapters 14, 20), and spin, charge and mass are features of its distinguished states, not properties of a substance. To which configurations particles correspond, the system does not establish.
In one group of interpretations the wave function describes the state of the system itself, not merely knowledge of it. In agreement with this is the theorem of Pusey, Barrett and Rudolph: on the assumption that independently prepared systems have independent physical states, it excludes the treatment of the quantum state as mere information about an underlying physical state (Pusey, Barrett, & Rudolph, 2012). Heisenberg interpreted the state as potentia — a tendency towards outcomes (Heisenberg, 1958). The present ontology does not accept

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potency (chapter 9): what is not distinguished is here not a present possibility but opacity — the absence of knowledge of difference in the given respect (chapter 10). Superposition in this reading means that with respect to the measured quantity distinction has not yet set in, not that the system “is in many states at once”.

The Schrödinger equation describes the dependence of a state on a parameter that physics interprets as time. Time, however, is derived here from the step of an elementary change: duration is measured by the number of steps (chapter 15). In the theory of relativity the duration between two events differs for different observers, and physics finds no absolute time. In agreement with this is the fact that the time of a configuration is the number of its own steps: between two events different configurations may pass through different numbers of them (chapter 15). The law by which these numbers are related — the Lorentz transformations — ontology does not derive. In agreement with physics is also the fact that a greatest speed follows from the indivisibility of the step (chapter 15); whether it equals the speed of light does not follow from ontology.

Measurement is an interaction in which one configuration distinguishes the state of another with respect to the measured quantity. The outcome of a measurement is the onset of knowledge of difference in this respect. Every interaction is a distinction (chapter 16), but not in every respect: a change affects information in one respect (chapter 15). Therefore from systems interacting it does not follow that distinction has already set in with respect to the given quantity, and the interference of interacting subsystems does not contradict this. The one who distinguishes is not a biological subject but any configuration in which distinction in the given respect proceeds.

This interpretation does not accept objective collapse as a physical mechanism and does not choose between interpretations of quantum mechanics where they differ only in description. It asserts one thing: an outcome has a cause (chapter 23). Thereby it is compatible with deterministic interpretations (Bohm, 1952; Everett, 1957) and incompatible with the assertion that an individual outcome is determined by nothing. It is close to the relational interpretation in that a state is a relation of the distinguishing and the distinguished (Rovelli, 1996), and to the decoherence-based one in that the onset of a classically determinate outcome is connected with distinction in the environment (Zurek, 2009).

Entanglement in physics is the fact that the state of a composite system does not decompose into the states of its parts. The correlations of the outcomes of measurements on the parts violate Bell’s inequalities, that is, are not explained by local causes (Bell, 1964); yet the outcome on one side cannot be controlled by changing the measurement on the other, and there is no superluminal signalling. Ontology gives this the following reading. Space is the totality of localizations and the distances between them (chapter 15), not what precedes distinctions. Therefore that by which an outcome is determined need not lie in one localization with it: it is in the conditioned, but not necessarily nearby. More on this in chapter 23.

The holographic principle states that the area of a surface limits the information content of the adjacent spacetime regions singled out by light-sheets (Bousso, 2002); a complete description of a region through its boundary has been established for particular classes of spacetimes. The conclusion that space is derivative is an interpretation of this result, not its content; it is cited as being in agreement with an ontology in which space is a measure of distinguishedness. Landauer’s principle shows that the erasure of a bit of information is accompanied by the dissipation into the surroundings of at least $kT \ln 2$ of heat (Landauer, 1991). Physical processes admit an informational description, and energy and information are connected in physical theories; the direction of dependence does not follow from this.

The conclusion of chapter 13 on the finiteness of distinctions is in agreement with the physical limits on information: the Bekenstein bound limits the information in a finite region

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of space with finite energy (Bekenstein, 1981), and the holographic bound ties this limit to area (Bousso, 2002). This agreement is partial: the physical bounds concern regions of space, while the conclusion of chapter 13 concerns the number of respects in which information knows itself. Wheeler's programme "it from bit" derives physical quantities from yes–no answers (Wheeler, 1990). Closest to the present system is Zeilinger's principle: an elementary system carries one bit of information, and features of quantum mechanics are derived from this finite content (Zeilinger, 1999). The similarity lies in the finiteness and elementarity of distinction. The divergence lies in the ground: there the bit is accepted as initial; here distinction is derived as knowledge of difference.

When physics discovers nonlocality, it fixes that description is not reducible to local states; the conclusion that there is no material substrate is an interpretation. When it discovers the dependence of the result on the measurement procedure, it fixes the role of distinction in description; the ontological primacy of distinction does not follow from this and is asserted on an ontological ground. When it discovers quantum fields, it fixes a structure of description. All three correlations show compatibility; they are not grounds. Hoffman concludes that physics and evolutionary theory point to one thing: spacetime and objects are not fundamental (Hoffman, 2019); this is an interpretation, and ontology decides the question of what is fundamental by its own means.

Physics is compatible with the present ontology, but compatibility is not proof. Physical data receive a consistent interpretation in the informational ontology and do not contradict its positions; but they are compatible with other interpretations too, and physicists themselves disagree about what stands behind their theories (Hut, Alford, & Tegmark, 2006).

What the system makes testable in physics is the following. A measurement outcome has a cause; if it were shown that an individual outcome is determined by nothing, the system would be refuted (chapter 23); experiment cannot show this while there are deterministic models equivalent to it (Bohm, 1952), and this condition of refutation is logical, not experimental. Time and space are discrete: a smallest duration and a smallest distance exist (chapter 15). Speed has a greatest value (chapter 15). The number of distinctions is finite (chapter 13). The claim of discreteness is testable only in principle: at accessible scales physics neither detects nor excludes it.

Closest to the treatment of the *I* in the present system stands Rovelli's relational quantum mechanics. He considers the notion of a state of a system independent of the observer to be mistaken: all systems are equivalent, there is no distinction of observer and observed, and the theory describes only the information that systems have about each other (Rovelli, 1996). This is a physical parallel to the claim that every configuration is an *I* distinguishing itself from not-oneself, and that the sides are distinguished for each other only in interaction (chapters 11, 16). A parallel, not a ground: for Rovelli the relation of systems does not require self-knowledge, whereas here self-differentiation is knowledge.

The opposite pole is Tegmark's mathematical universe hypothesis: if external physical reality is independent of humans, it is an abstract mathematical structure (Tegmark, 2007). For him structure is the ultimate ground. Here a mathematical structure is a description of distinctions, not what determines them: mathematics, like all logic, belongs to the conditioned (chapter 4). Therefore physics is the science of the regularities of the distinguished. Its descriptions fix regularities but not their ontological origin; Deutsch strives for the unity of physics with the theories of computation, knowledge and evolution (Deutsch, 1997). Ontology does not replace physical explanation and does not derive physical quantities.

Chapter 23. Quantum Entanglement

Quantum entanglement is a state of a composite system that does not decompose into the states of its parts. The correlations of the outcomes of measurements on entangled parts

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violate Bell's inequalities (Bell, 1964), and this has been confirmed by experiments (Aspect, Dalibard, & Roger, 1982), including ones in which the main loopholes are closed (Hensen et al., 2015). It follows that these correlations are not explained by local hidden variables — by what each part carries in advance where it is.

Physics asserts something else too: the outcome on one side cannot be controlled by changing the measurement on the other, and there is no signal faster than light. Ontology accepts this. The determinacy of an outcome is in the conditioned (chapter 8), but that by which it is determined is opaque to the one who considers (chapter 3). Therefore what determines the outcomes on both sides cannot serve as a signal: the one measuring on one side does not know, and does not have at his disposal, that by which the outcome on the other is determined.

Space here is the totality of localizations and the distances between them (chapter 15), not what precedes distinctions. Therefore that by which an outcome is determined need not lie in one localization with it: the conditioned is determined by the conditioned itself (chapter 8), not only by what is nearby. Entangled parts are distinctions of one configuration, not two independent objects, and that by which their outcomes are determined is in this configuration as a whole.

This does not mean that all objects are entangled. Non-entangled states are configurations in which the outcome on one part is determined by what is in that part and does not depend on the measurement on the other. The difference between entangled and non-entangled is a difference of configurations, not a difference between the informational and the non-informational. Therefore the reading does not explain any outcome whatever: it says only where that by which an outcome is determined lies.

Compatible with entanglement is a second feature of the informational ontology: there is no true randomness. This is a position of the ontology, not a conclusion from entanglement: Bell's theorem rules out local explanations of the correlations but establishes neither the determinacy of outcomes nor its absence. Given the violation of Bell's inequalities, the determinacy of outcomes is possible in two main forms: nonlocal, where an outcome is determined by the state of the whole system including distant parts (Bohm, 1952), and superdeterministic, where the choices of measurements and the hidden states are themselves correlated (Hossenfelder & Palmer, 2020). There are other routes too — retrocausal models and the many-worlds interpretation, in which the outcome is not unique (Everett, 1957). The system accepts the first: that by which an outcome is determined is in the configuration as a whole, and space does not precede distinctions. The system does not require superdeterminism: the choice of a measurement is determined by the proximate grounds of the chooser (chapter 21), and it assumes no correlation of this choice with the hidden state of what is measured. The Kochen–Specker theorem shows that the values of measured quantities cannot be assigned prior to measurement independently of what else is measured (Kochen & Specker, 1967). The system agrees with this: an outcome before measurement is not a present value but what is not yet distinguished (chapter 10), and it is determined together with how one measures. "Randomness" in quantum mechanics is therefore the incompleteness of access to that by which an outcome is determined, not the absence of a cause.

The axiom of fundamental randomness is untenable within the present ontology, and this is shown not by physics but by the axiom's own content. The axiom asserts that an individual measurement outcome is determined by nothing. But the outcome is determinate: this result was obtained and not another, and it is distinguished from the other possible ones.

First: without a cause the outcome would not occur. Occurrence is change, and the dependent has nothing by itself — neither that by which it is nor that by which it becomes (chapter 4). An outcome determined by nothing would occur by itself, that is, would be unconditioned in

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its occurrence. But there is no unconditionedness “in a respect” (chapter 13), and the unconditioned does not occur at all: change is inapplicable to it. Hence, if an outcome has occurred and is determinate — this was obtained and not another — there is that by which it is determined. Whether the one who considers can calculate it is another matter. That is a question of access, that is, of the opacity of the one who considers, not of the absence of a cause: uncausedness inferred from incalculability confuses not-knowing with non-existence.
Second: determinacy from the absence of determination. The axiom asserts that there is no determination and yet there is determinacy. Then either determinacy is obtained from absence, and absence becomes that from which it is taken — distinction placed where there is none; or it arises from nowhere, by itself — and the outcome becomes an independent principle. These are the first and second exits from the aporia (chapter 2). The uncausedness of the outcome is the same aporia of transition, carried over to the level of a single event.
Third: distinction by itself. A measurement outcome is a distinction — the knowledge of the difference of this state from others. Distinction proceeds not from information itself but by the removal of opacity from Selfhood: information does not remove opacity by itself (chapter 10). A random outcome means a distinction proceeding by itself, from nothing, and thereby ascribes to information an independence that the dependent does not have.
Fourth: the lawfulness of the uncaused and the underived moment. Uncaused outcomes obey an exact law of distribution. Hence the distribution is determined, while the outcome is not. If the distribution is determined, it has a ground, and probabilistic interpretations call it the system’s propensity towards outcomes; determination is thereby admitted, but ascribed to the totality and denied to the individual outcome. The law of distribution is accepted as an underived given — a hidden condition. The present system does not derive the law of distribution itself, the Born rule, either. But neither does it accept it as a determinacy of the totality denied to the individual outcome: the distribution is the way outcomes, each determined, are distributed over causes opaque to the one who considers; and regularity as such is derived from the constancy of the removal (chapter 10). Moreover, randomness operates only “upon measurement”, and what measurement is the theory does not define (Bell, 1990): this is a second underived condition. Here measurement is not a special moment: it is an interaction in which distinction sets in with respect to the measured quantity (chapter 22).
Thus the axiom of fundamental randomness occupies all three exits from the aporia: determinacy from the absence of determination is the first, the outcome as an independent principle the second, the underived law of distribution and the underived moment of measurement the third. The direct contradiction is that an uncaused outcome would have to occur by itself, while occurrence by itself is impossible for the dependent: if an outcome has occurred, it has a cause, and the only question is whether the one who considers can calculate it. The position “there is no true randomness” is therefore not accepted alongside the axiom as an equally valid alternative but derived from the structure of the conditioned.
The consequences for ontology are as follows: space is not an initial level; distinction logically precedes localization; that by which an outcome is determined need not be local. Entanglement does not prove the informational ontology but agrees with it: the dependence of outcomes not on distance but on the configuration as a whole is what this ontology expects.

Chapter 24. Cosmology

Cosmology here considers the origin, structure and change of the universe not as a physical process but as a complexification of distinctions. Space, time, matter and energy are derivative forms dependent on distinguished information (chapters 15, 19).

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In Hawking and Mlodinow, in the early Universe time behaves like a spatial dimension, and the question of what was “before” the beginning loses its meaning (Hawking & Mlodinow, 2010, 135). Here the “Big Bang” is the limit up to which physics continues the description of this universe backwards. It cannot be a state without distinction that would precede distinction: without change there is no duration in which anything would precede (chapter 15); the basic determinacy of information exists without succession (chapter 9), but not as a former state in time, and measures are given already with the first change (chapter 15). A universe is that in which one stage of the path proceeds (chapter 13); the beginning of a universe is the beginning of a stage. Whether it was the beginning of the first stage does not follow from what has been said.

Physical cosmology seeks the beginning as a physical event. Here the unconditioned is neither a cause, nor an action, nor an event (chapter 5), and ontology does not replace physical description. If the beginning of the universe is the beginning of a stage other than the first, its proximate grounds lie in the transition from the former stage (chapter 13). If it is the first, there is no “before” it: time is composed of the succession of states (chapter 15), and without change there is no duration in which anything would precede. The limit of the beginning is the same as that of every distinction — the unconditioned given in the conditioned.

The Paradigms of Dispersal and Concentration

A universe exists in its own paradigm (chapter 13). The present one exists in the paradigm of dispersal: it expands and cools. With expansion the wavelength of radiation is stretched, and the temperature falls in inverse proportion to the size of the universe: the cosmic microwave background has cooled from about 3000 K at the moment of its release to 2.7 K. Opposite to it is the paradigm of concentration: contraction and heating. One paradigm is not enough for transparency, and the next stage proceeds in another paradigm (chapter 13). Other than the direction of dispersal is the reverse direction, concentration, or a regime without direction; the system takes the former, and this is a reading, not a derivation (below). On it the transition to the paradigm of concentration is the transition to the next stage.

Energy is change taken in concentration — the density of change (chapter 15). In this language both paradigms are described in one way. In the paradigm of dispersal change disperses: expansion stretches it, the frequency of radiation falls, the surroundings cool. In the paradigm of concentration change concentrates: contraction shifts radiation towards the blue, the density of change grows, the surroundings heat up. Heat here is the name of change distributed over many parts in a system of relations that the one who considers does not calculate; there is no disorder and no randomness in the relations themselves (chapter 23). Temperature is the density of change falling to one part.

The Algorithm of the Paradigm of Dispersal

A bond is composed by giving off change: a bound pair changes internally less than the same parts apart, and the excess goes outwards as free change — radiation or heat. What is given off belongs to the relation of the parts, not to each of them (chapter 16). A bond holds as long as the surroundings do not return to it more than it gave off; this is its depth.

Bonds are composed into a ladder, and each step is shallower than the preceding one: quarks in nucleons; nucleons in nuclei — millions of electronvolts per particle; outer electrons in atoms — units and tens of electronvolts, atoms in molecules — units; molecules in substance — hundredths and tenths; complex wholes — shallower still. Therefore the upper steps do not touch the lower ones, and the lower serve them as ready wholes. A bond holds when the change of the surroundings per part is less than its depth; therefore as the universe cooled the steps were composed in order — nuclei, atoms, molecules, substance, complex wholes. The order of cooling is the order of emergence (chapter 18).

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Configurations exist in four ways. At rest — an atom, a crystal: change has already been given off, no inflow is needed. On their own reserve — a star: it contains bonds that are still being rearranged with the giving off of change, and what is given off holds the whole. On a passing flow — a whirlwind, a flame: they are held by an inflow but do not restore themselves (Nicolis & Prigogine, 1977). On an inflow by which they restore themselves — the living: it takes concentrated change, restores its bond with it and gives change off dispersed (chapters 20, 25). These ways are linked in one chain: change given off by deep bonds in stars goes out as radiation and holds the living, and the living gives it off dispersed. The common condition for all is a sink: there must be somewhere to give change off. In the paradigm of dispersal the sink is dispersal itself.

The Algorithm of the Paradigm of Concentration

In the paradigm of concentration change does not go away but returns and concentrates. The bonds of dispersal break, beginning with the shallowest: complex wholes, substance, molecules, atoms, nuclei. If contraction proceeded under the present laws, physics would describe it as follows. Galaxies draw together, black holes grow and merge. Radiation heats up, and stars, having nowhere to give off heat, are destroyed. At about 3000 K atoms are ionized, and chemistry ends. At 10^9–10^{10} K nuclei disintegrate. At about 10^{12} K nucleons pass into a quark–gluon plasma. At about 10^{15} K the electromagnetic and weak interactions cease to differ; the present description goes no further. The final state does not repeat the smooth beginning: black holes dominate it (Penrose, 2010). Time does not reverse in contraction. Gold's idea that the arrow of time reverses in contraction (Gold, 1962) was first accepted by Hawking and then acknowledged by him as a mistake (Hawking, 1988); in the system knowledge accrues in both paradigms (T16).

But the paradigm of concentration is a change of regularities, not destruction alone (chapter 13). Emergence is necessary as a tendency (chapter 18), and a stage exists for the experience of existence. Hence under concentration too bonds must be composed, but held not by change given off but by change retained: the parts together carry more change than apart. Such a kind of bond exists already now on the deepest step: the mass of the proton is almost wholly the energy of the strong interaction — of the motion of quarks and of the gluon field — and not the mass of the quarks themselves: the whole carries more change than the parts apart. In the paradigm of concentration it becomes the rule. Hence the rules. A bond holds as long as the concentration around is not lower than that which constitutes it, and disintegrates when it falls. Bonds that require ever greater concentration begin to hold, and the order of emergence is the inverse of the present one; the source is the surroundings themselves. What follows is obtained by inverting the algorithm of the paradigm of dispersal and is a projection, not a derivation: the analogue of the living takes change distributed over many parts and concentrates it in its bond; the sink is concentration itself. The second law of thermodynamics is a law of the present paradigm, and the form of laws belongs to the paradigm (chapters 10, 18); hence its persistence in its present form under concentration is not assured. Whether it persists, and in what form, is not derived. This does not conflict with the system: the growth of entropy is the dispersal of change, not a decrease of knowledge (below), and in the inversion it is the direction of bonds that changes, not that of knowledge (T16).

Singularity and Separation

Singularity here is not only the limit of the applicability of physical description. It is the collapse of the universe by which a stage ends; the physical singularity is its physical expression. It is caused by the exhaustion of the paradigm (chapter 13). As the collapse proceeds, the bonds of dispersal break, and the bearers of the departing universe are erased. It is to this that Landauer's principle applies: logically irreversible erasure is accompanied by the dissipation of energy (Landauer, 1991). What is erased is bearers, not knowledge (T16).

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Not everything here is derived to the same degree. It is derived that a stage ends by the exhaustion of the paradigm (chapter 13); that exhaustion determines a change of regime of the whole universe, since what sets in is determined by the conditioned as a whole, and the determining need not lie beside the determined (chapters 8, 23); that the next paradigm is another one, since the dependence of the regime is not removed within one universe (chapter 13). That it is precisely the opposite one is a reading, not a derivation: other than the direction of dispersal would be both the reverse direction and a regime without direction. That the present paradigm is expansion and cooling is taken from physics, not derived; on this reading the transition to the opposite one is contraction and heating. That the physical singularity is the expression of this transition is a correspondence, not a derivation. When exhaustion will occur the system does not derive (Open Questions), and it gives no date for the collapse. By physics the claim is testable only in part. It implies that the present accelerated expansion is not eternal and will give way to contraction. If dark energy is a cosmological constant, there will be no contraction, and the system diverges from the physical description; the DESI indication that dark energy varies leaves this question open (DESI Collaboration, 2025). Observations can make the consequence more or less probable but cannot decide it while the future behaviour of dark energy is unknown. That the transition is determined by exhaustion and not by a physical condition by itself, physics does not test: this claim is ontological.
After the disintegration of a configuration the *I* is preserved as connected knowledge, but without distinctions, in oblivion (chapter 12). In the singularity configurations are composed anew according to the connectedness of former instances of the *I*: the bond breaks, connectedness does not, and the former bonds are preserved in it as known. The assembly is determined by the preserved selfhood as a proximate ground, and the determining need not lie beside the determined (chapter 23); therefore the new configuration acquires the former *I*, not a new one (chapter 17). A concordant selfhood knows itself as dependent, not as its bearer; its connectedness is not tied to the kind of bond, and it is composed by bonds of concentration — by the means of the new paradigm — in the space released by the collapse. It is this released space that the next stage fills. A discordant selfhood knows itself through its configuration and identifies itself with it. The assembly it determines is the assembly of the former configuration — by bonds of dispersal, but it proceeds in the paradigm of concentration, where such bonds cannot hold. The collapse erases this configuration together with the bearers of the departing universe; it is to this erasure that Landauer's principle applies (above). Together with the configuration the discordant *I* ends as one: its unity was held by identifying itself with the configuration, and this identification is not knowledge given by removal — removal gives the knowledge that the knower is dependent, whereas filling the "that" with something conditioned is a choice (chapter 28). The knowledge given by removal in its distinctions does not decrease and remains the knowledge of information (T16); information as a whole knows this *I* as discordant (chapter 28). This is dissolution. This is separation, that is, the sifting (chapter 28): not an external decision but the kind of bonds by which a selfhood is held. This mechanism rests on the reading of the paradigm of concentration (above) and shares its status; that the sifting proceeds by preserved knowledge — how the *I* knew itself — follows from T16 and T29 independently of it (chapter 28).

Measures, Entropy and Physical Comparisons

The time of a configuration is the number of its steps; space is the totality of localizations of the present configuration and of the distances between them (chapter 15). Space can expand and contract; knowledge does not depend on its size. Physical expansion is described as a change of the metric following Hubble's law and accelerating, while bound systems do not expand. Present cosmology does not predict contraction but does not exclude it either: the DESI data indicate that dark energy may change with time (DESI Collaboration, 2025).

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Entropy in physics measures the number of microstates compatible with a given macroscopic description. Its growth is not a decrease of knowledge of difference: knowledge does not decrease (T16). In the terms of the system the growth of entropy is the dispersal of change: change is distributed over an ever greater number of parts in a system of relations that the one considering does not compute (above). It is compatible with the composition of bonds and accompanies it: a bond is composed by giving off change, and what is given off is dispersed; for gravitating systems the growth of entropy is the very gathering of matter (Penrose, 2010). A bond disintegrates not by the growth of entropy but when the surroundings return to it more than its depth (above). Therefore the two arrows do not contradict each other: knowledge always accrues (T16), while bonds can both be composed and disintegrate (chapter 17).

Landauer's principle connects information and energy in physical description: the erasure of an informational state is accompanied by the dissipation of energy (Landauer, 1991). Erasure here is the disintegration of the bond of a bearer, not a decrease of knowledge (chapter 19). Energy in the terms of the system is the density of change (chapter 15); its conservation is a law of physics, and ontology does not dispute it.

The holographic principle, proposed on the basis of the study of black hole entropy ('t Hooft, 1993; Susskind, 1995; review: Sorkin, 2005), limits the information content of a region by the area of its boundary (Bousso, 2002); a complete description of a region through its boundary has been established for particular classes of spacetimes. The conclusion that space is derivative is an interpretation of this result, in agreement with an ontology in which space is derived from distinctions (chapter 15).

A succession of universes is found in some cosmologies too. Penrose proposes conformal cyclic cosmology — a series of aeons in which the end of one passes into the beginning of another (Penrose, 2010). Smolin proposes cosmological natural selection: universes are born in black holes, and the physical constants change from universe to universe (Smolin, 1997). The system converges with both in the succession of universes, and with Smolin also in that physics changes from universe to universe; but in Smolin the values of the constants change under the same laws, while here it is the form of the laws, which belongs to the paradigm (chapter 10). The divergence lies in the ground. The series here is finite (chapter 13). A change of paradigm is determined not by a physical condition as such but by the exhaustion of the paradigm, and the physical transition is a consequence of this exhaustion (chapter 13). And the transition is connected with the separation of selfhoods according to the kind of bonds by which a selfhood is held.

Thus cosmology here is not a history of matter but a history of distinctions. In it are distinguished:

1. distinction proceeding by the removal of opacity from Selfhood at every step, not only at the beginning (chapter 10); with it consciousness begins;
2. elementary changes and measures, given with the first change (chapters 14, 15);
3. aggregated forms and their complexification as the universe cools (chapters 17, 18);
4. the growth of consciousness up to reason and intellect (chapters 20, 21);
5. the exhaustion of the paradigm, the collapse of the universe and separation (chapters 13, 28);
6. the transition to the next stage in another paradigm and, at the limit, the self-transparency of information (chapter 13).

This sequence is an ontological description, not an alternative physical cosmology. Ontology gives the order and structure of processes in both paradigms; it does not derive magnitudes — temperatures, binding energies, the constants of the next stage: that is the business of measure and of physics (chapter 22).

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Chapter 25. Life as an Informational Process

What life is has been derived in chapter 20: life is a form in which the *I* distinguishes the disturbance of its own bond and holds it, restoring it from the external. The “that” from which the disturbance is distinguished is the held state of the whole itself, given separately from the present one, and this is knowledge of the whole given by emergence (chapter 18). Here this definition is applied to what biology has found about the living. The chapter gives no new definition of life: it derives from the one given which features are necessary to every living thing and shows that it is precisely these that biology calls the signs of life.

First, why life is an informational process. The living holds not a substance but its held state, that is, knowledge of the whole. It takes substance from the external and replaces it: restoration proceeds from the external (chapter 20). What is constant in the living is therefore knowledge of the whole, while the material changes. This is the sense of the title, and it is not introduced but follows from the definition. All reality is informational (chapter 5); what distinguishes the living is that in it knowledge of the whole is held by the bond of this very whole — by the proximate ground, not by itself (chapter 21).

The first consequence is a boundary. To distinguish the disturbance of its own bond from another, the whole must be distinguished from what is not itself. The distinguishing of oneself from what is not oneself that holds one’s own against inclusion in another is repulsion, and repulsion leads to a boundary (chapters 16, 19). Hence every living thing has a boundary separating it from what is not itself.

The second is exchange. Restoration takes from the external. Hence the living is not closed: there is a constant inflow and outflow through the boundary, and without them there would be nothing to restore with. The boundary therefore does not fence the living off from the external but separates one’s own and not one’s own within the exchange itself.

The third is the keeping of the held state. The held state must be given separately from the present one, otherwise there is nothing from which to distinguish a disturbance (chapter 20); and it must be preserved through the change of material, otherwise there would be nothing to restore by. Hence there is in the living that which holds knowledge of the whole independently of the change of parts.

The fourth is reproduction. If the held state can be restored from external material not in parts but as a whole, what is restored in a new bearer is the held state — the pattern of the bond; the *I* of the new bearer is its own, for the *I* of the parent continues in its own bearer (chapter 17). Reproduction is the same restoration taken to its end.

The fifth is adaptation. The living distinguishes a disturbance relative to what is not itself, and therefore its response depends on the external. Responses that held the bond are preserved in the held state; responses that did not lead to disintegration, and there is no one to preserve them. Thus knowledge of the whole changes together with the experience of existence, and in reproduction what has changed is passed on.

Boundary, exchange, the keeping of the held state, reproduction and adaptation are precisely that by which biology defines the living: the cell membrane, metabolism, the hereditary record, reproduction and evolution. Ontology does not take these signs from biology: it derives why they are necessary to every living thing, in whatever form it may be. Biology confirms them; the chemical and physical forms in which they are realized are proximate grounds, not the ground.

Death is the cessation of holding: the *I* of the bond no longer restores it, and the bond disintegrates. What has been removed does not return: the knowledge that was in the living remains knowledge of information (T16), and the *I* of the living is preserved as connected knowledge (chapter 12). Death is therefore not annihilation but the disintegration of a unity that was kept by holding.

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Life is the first level at which the object of distinction becomes the held state of the whole, given separately from the present one. Its own states are distinguished already by a bound pair of changes (chapter 14); the living distinguishes its present state and its held state. With life begins the growth of self-knowledge as of a held whole, which leads to reason and intellect (chapters 12, 21); self-knowledge as such is gathered from the first distinction (chapter 12): self-knowledge grows with the extent to which distinctions are turned towards their own states.

Chapter 26. Consciousness in the Neurosciences

The neurosciences describe consciousness through correlations between the activity of neural structures and subjective lived experience. They measure electrical impulses, observe the dynamics of networks and identify the brain regions involved in perception. Why subjectivity arises from neural activity these characteristics do not explain (Nagel, 2012). This is the "hard problem of consciousness" (Chalmers, 1996). Here it is not solved anew but removed by the derivation of chapter 12: lived experience is self-knowledge without the separation of knower and known, and neural processes are the same distinctions taken from the side of change.

Consciousness is not a function of the brain that produces lived experience. The brain is an aggregated form in which distinctions are many and densely bound. The brain and consciousness are the same distinctions taken from two sides: from the side of change, neural processes; from the side of knowledge, consciousness (chapter 12). Therefore there is neither the generation of consciousness by the brain nor the expression through the brain of something that exists besides it: there is one thing, taken twice. Knowledge, meanwhile, opens by the removal from Selfhood (chapter 10) and is not produced by neurons; this is the limit, not a second cause beside the neural ones (chapter 19).

The limit of neuroscience: it measures correlations but does not explain why correlations yield subjective experience. Husserl showed that science stands on the natural attitude: it takes the world as simply given and does not clarify how it is given to consciousness (Husserl, 1982). Here the correspondence of neural and lived states is not a correlation of two series but an identity of distinctions taken from two sides; correlation is how this identity looks from outside.

Closest to the treatment of consciousness as a knowledge in which many distinctions are composed into one stands integrated information theory. Its basis is two properties of experience: differentiation, that is, an enormous number of possible experiences, and integration, that is, the unity of each of them; it measures the quantity of consciousness by Φ (Tononi, 2004). To integration here corresponds the connectedness of knowledges into one (chapter 12). But the measure of consciousness in the system is different: not integration but the extent to which the distinctions of a configuration distinguish its own states (chapter 12); the exclusion postulate the system rejects (chapter 12). The status of the theory in neuroscience is contested: in an adversarial test its predictions were confirmed only in part (Cogitate Consortium, 2025).

Predictive processing models see the brain as a system of continuous comparison of expected and observed states; the key element in them is the detection of a discrepancy between prediction and fact (Friston, 2010; Clark, 2013). In the terms of the system this is a distinction turned to one's own states: the configuration distinguishes what has set in not only from the former but also from the expectation held in it, and the degree of the *I* grows with this (chapter 12).

Global workspace theory connects what is conscious with what becomes available to many systems of the brain at once (Baars, 1988; Dehaene, 2014). In the terms of the system this is connectedness: knowledge that refers to many distinctions of a configuration enters its one *I*

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(chapter 12). The divergence lies in that the system does not tie the very presence of consciousness to availability: there is an *I* also in what enters no workspace; availability determines only which of the nested instances of the *I* a distinction enters (chapter 20).
The hard problem does not arise here: it rests on the assumption that an informational process and subjective experience are different entities (chapter 12). What the system makes testable in neuroscience is the following: the degree of awareness depends not on the number of neurons and not on integration as such, but on the extent to which the distinctions of the brain distinguish its own states; the loss of distinctions turned to one's own states should reduce awareness more than the loss of an equal number of distinctions turned outwards.
Thus consciousness in neuroscience is the same distinctions as neural processes, taken from the side of knowledge. Neural processes are neither the source of consciousness nor its expression, but its other side. The measure of consciousness is the extent to which the distinctions of a configuration distinguish its own states (chapter 12).

PART IV. A NEW WORLDVIEW

Chapter 27. A New Point of Departure

The new point of departure is a change of the initial question: not of what substance reality consists, but how the conditioned depends on the unconditioned. The answer is information: the conditioned named by the way the unconditioned is given in it (chapter 5). The oppositions of matter and consciousness, observer and observed become a difference of sides of the same distinctions (chapters 12, 19), and the opposition of determinism and probability is decided in favour of determination — as a position of the ontology, not a result of physics: there is no uncaused outcome, and probability is the opacity of the one who considers (chapters 3, 23).
Reality is neither matter nor spirit but information — the dependent in which the unconditioned is given. Information is neither produced nor sets in: its states set in (chapter 9). But neither does it stand beside the unconditioned as a second: the unconditioned is given in it, while it is not given in the unconditioned (Df14), and the relation "together with" is inapplicable to the unconditioned. This point of departure removes the contradictions examined in Part I and gives an ontological interpretation that is compared with physics and neuroscience but does not replace scientific description (Part III).

Chapter 28. Principles of the New Worldview

A worldview in the present system is not a set of views about the world but the way the *I* understands itself in reality. It is the intellectual *I* that can understand itself and choose (chapter 21), and it is this *I* that is meant here. The principles below are not introduced but derived from what has been established; each rests on the preceding one.

1. The Principle of Unconditionedness

Nothing by which the conditioned is divided within itself is applicable to the unconditioned (chapter 4). A worldview that thinks Nothing through the categories of the conditioned — as acting, willing, a part or a whole — is mistaken already at its outset. Nothing is spoken of only by remotions.

2. The Principle of Dependence

The object of self-knowledge is one — one's own dependence (chapter 13). The dependent has nothing by itself: both what it is and what it becomes are determined in the last account by unconditionedness as limit (chapter 4); what is its own by proximate ground it does have (chapter 21). Hence there are two possible relations to oneself: knowledge of one's dependence and an understanding that diverges from it. There is no third.

3. The Concordant and the Discordant

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One thing is concordant — knowledge of one's dependence on Nothing, on Selfhood. Every understanding of oneself that diverges from it is discordant, and each diverges in something definite. "The *I* is independent" abolishes the completeness of dependence (chapter 4). "The *I* is a part of Nothing" or "I shall dissolve in it" applies to the unconditioned part and whole, which are inapplicable to it. The refusal to know refuses the only object of self-knowledge.

4. The Principle of the Limit of Knowledge

Knowledge is given by removal, not by itself. Only the distinguished is known, and the undistinguished is opaque; but the blind spots are finite in number, and self-transparency is attainable (chapters 10, 13). Therefore the worldview neither lays claim to omniscience nor renounces knowledge: it knows where knowledge comes from and that the path to it is finite.

5. The Principle of Regularity

The constancy of removal gives regularity, its one-sidedness gives direction (chapter 13). There is no fundamental randomness — this is a position of the ontology, not a result of physics: the axiom of an uncaused outcome contradicts itself (chapters 3, 23). Reality is therefore neither chaotic nor blind: there is law and direction in it.

6. The Principle of Choice

Removal gives knowledge of difference, and the basic difference of information (Df6) is its dependence taken from its own side (chapter 11). Knowledge of difference therefore gives that the knower is dependent. The other side of the difference — that on which it depends — is given to it only by remotions, without features: there is no positive knowledge of it (chapter 5). That the dependence is dependence on Nothing, and not on substance, on oneself or on chance, distinction itself does not give: the "that" in it remains without content. A single *I* can fill the empty "that" with something conditioned — with substance, with itself, with chance. To accept dependence on Nothing is to fill it with nothing conditioned. It is therefore not given of itself: it is a choice. Choice belongs to the intellectual *I*; it is one's own by its proximate ground, although in the last account it too does not proceed by itself (chapter 21). The transparency to which the path leads is the transparency of information as a whole (chapter 13). A discordant *I* does not know itself in this respect, and this is its opacity; for the whole, however, it is not a blind spot: the whole knows this *I* as discordant, the knowledge is preserved (T16), and nothing in it is concealed.

7. The Principle of Preservation

What has been removed does not return, and everything that was in the distinctions is preserved — both concordance and discordance. Concordance cannot be faked; it can be attained while the *I* has a bearer. It is objected that from discordance it follows only that it must be removed, not that the *I* does not pass; it could correct itself later. To correct oneself is to change one's understanding of oneself, and this is a choice; choice belongs to the intellectual *I* in a configuration that knows bonds (chapter 21). Without a bearer there are no distinctions, and with them no choice. Configurations after the transition exist only in the next stage, which the discordant cannot enter: the sifting proceeds by preserved knowledge — how the *I* knew itself (T16), and a correction that did not take place is not reflected in it. Therefore the window of correction is the time while the *I* has a bearer. This is what responsibility consists in: not a moral norm but the fact that the choice is preserved and determines what passes further.

8. The Principle of Stages

Disclosure is discrete and proceeds by stages. Stages are determined by the gathered experience of existence: when the universe's knowledge of itself as a whole, into which the experience of instances of the intellectual *I* enters as nested, can yield nothing new in the given paradigm, the stage is completed and the sifting takes place — the concordant passes, the discordant does not (chapter 13). Development is therefore neither continuous nor infinite: it is directed and proceeds by steps.

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These principles are not prescriptions but consequences of the structure of reality. Ontology does not say how to act. It shows that concordance is a choice, that the choice is preserved and that the transition depends on it. The choice itself remains with the intellectual *I*.

Chapter 29. The Human Being in the Structure of Being

What the human being is is determined neither by substance nor by a special status but by its place on the derived path: the *I* — life — reason — intellect.

The human being is a configuration: a system of elementary changes united by bonds into a whole. It is the bearer of consciousness, not consciousness itself. Its parts — from particles to cells — have their own *I* at their own level but function according to the paradigm of the connectedness of the whole: the algorithm of each is set by a bond, an agent of the programme (chapters 16, 18).

The human being is living: its *I* distinguishes the disturbance of its bond and restores it from the external. Hence the boundary, exchange, the keeping of the held state, reproduction and adaptation (chapters 20, 25).

The *I* of the human being has reached reason and intellect. Reason knows the bonds between its distinctions and chooses within the conditions in which it stands; intellect knows in addition its place in the whole and the programme, and therefore chooses not only the reaction but also the algorithm itself — whether or not to take the function under control, that is, whether to leave its responses to the algorithm of the bond or to determine them by its own knowledge; to it also belongs the choice of self-understanding (chapter 21). By this, and by this alone, the human being differs from other living forms.

The distinctiveness of the human being is a level of the *I*, not its bearer. Reason and intellect are not bound to biology: they are levels of self-knowledge, not properties of substance (chapter 21). The human being is therefore a bearer of intellect, but not the only possible one: any configuration that has reached this level can be a bearer of intellect.

The limit of human knowledge is not a veil between the human being and the world in itself. The world is information and is conditioned through and through; what is limited is self-knowledge — opacity, which is removed, and the blind spots, which are finite in number (chapters 10, 13). Of Nothing only remotions are accessible to the human being.

As an intellectual *I*, the human being gathers the experience of existence by which a stage is determined; it makes the choice of concordance and is subject to the sifting (chapters 13, 28). When the universe's knowledge of itself as a whole, into which the experience of instances of the intellectual *I* enters as nested, can yield nothing new in the given paradigm, the stage is completed.

There is no anthropocentrism here. What is key on the path is the intellectual *I*, in whatever bearer it may be, not the human being as a species; the human being is key only insofar as it is such a bearer. The centre of reality is the direction towards transparency, not any form.

Chapter 30. The Concluding Doctrine

This chapter introduces no new positions. It gathers the derivation into one chain, states the scope of what is asserted, names the criteria of testing and the way to read the language of the book. Definitions are not repeated: they are given in the Appendix.

The Scope of the Assertion

What is asserted is the following. Under the accepted conditions determinacy can lie only in three places, and only one of them — the side of the dependent — is not excluded by the conditions (chapter 4); the solution can lie only there. The derived chain is its realization to the extent that each of its links is derived from what has already been derived. The uniqueness of the conditions themselves is not asserted. What is accepted without derivation is one thing — reality is, and together with it change is encountered; the rule of consideration

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is derived from the definition of the unconditioned (Appendix, A, B). Whoever rejects the conditions rejects not the solution but the problem: he builds something else, and this work does not refute it. But he is obliged to show that his problem does not reproduce the same aporia under another name: every ontology that recognizes the unconditioned and the changeable runs into this aporia, whether it admits it or not.

The Chain of Derivation

Everything encountered is conditioned; it is determined by unconditionedness as limit and by nothing besides and has nothing by itself (chapter 4). Determinacy can lie on the side of the unconditioned, on the side of an independent second, or on the side of the dependent, and there is no fourth place; the first two violate unconditionedness and uniqueness, the third violates no condition (chapter 4). The unconditioned is immutable, unique and transparent; Nothing in self-understanding is Selfhood (chapter 5).

The unconditioned is given in the conditioned — without it the determinacy of the conditioned can be neither given nor stated — and it is given only as knowledge. Information is the conditioned itself, taken from the side of this givenness; everything encountered is information, and it is wholly dependent (chapters 5, 8).

From the encountered change opacity is derived; its removal proceeds from Selfhood and is initiation. Distinction is knowledge of difference, and its appearance is change; change is the arrival of a new state (chapters 9–11). The *I* is the distinguishing of oneself from what is not oneself and exists from the first distinction; consciousness is self-knowledge accrued with every distinction (chapter 12).

The constancy of removal gives regularity, its one-sidedness gives the programme; the blind spots are finite in number, and self-transparency is attainable (chapter 13). From the working moment of an elementary change the measures are derived (chapters 14, 15). Interaction is a mutual self-differentiation of instances of the *I*; an interaction that has held is a bond, which is the agent of the programme (chapters 16, 18). Emergence gives knowledge of the whole and in one universe is finite (chapter 18).

Life is a form in which the *I* distinguishes the disturbance of its own bond and restores it from the external (chapters 20, 25). Reason chooses a reaction within the conditions of the bond; intellect knows its place in the whole and the programme and chooses the algorithm itself and self-understanding (chapter 21). The path proceeds by stages: a stage is determined by the gathered experience of existence and is completed when the universe's knowledge of itself as a whole, into which the experience of instances of the intellectual *I* enters as nested, can yield nothing new in the given paradigm; then the sifting by concordance takes place (chapters 13, 28). The last stage is the self-transparency of information, in which it remains conditioned. Hence the consequence for the worldview: one thing is concordant — knowledge of one's dependence; its acceptance is a choice, and the choice is preserved (chapter 28).

Criteria of Testing

The testability of the ontological structure is given by three criteria: the consistent derivation of all dependent categories from the accepted position and from what has already been derived; the absence of a fourth place for determinacy; the concordance of all subsequent forms with the initial scheme.

Negative tests — conditions under which the ontology must collapse: inner states in the unconditioned; the autonomy of information; distinction within the unconditioned; the necessity of a second principle; a permutation of levels without loss of semantic integrity; an actual infinite presented in the conditioned as a fact rather than as an axiom (chapter 13); the uncausedness of a single outcome established as a fact rather than accepted as an axiom (chapters 3, 23).

Positive tests — signs of stability: the immutability of the unconditioned; the derivability of information as dependent; the arising of distinction only by the removal of its opacity; the

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unfolding of consciousness, measures, bonds, aggregated forms, life and the levels of consciousness from one scheme; the concordance of the levels and the absence of a fourth place for determinacy.

The Language of the Book

The words “before”, “after”, “follows”, “transition” designate a hierarchy of dependences, not an order of arising in time. Logical necessity is not ontological causality, and a conceptual description is not passed off as an empirical process. The unconditioned is outside time; dependent forms admit a temporal description; the language of sequence brings no temporality into the unconditioned.

“Intention” and “goal” have no psychological, volitional or externally purposive sense. “Intention” is the directedness of the removal of opacity: the removal is one-sided and does not return; it is not an act of choice. “Goal” is not an external reference point but the limit: the self-transparency of information. Teleological completion therefore introduces no external goal and describes the limit of the conditioned towards which the one-sided removal is directed.

The Status of Statements about Nothing

Positively the book knows only the conditioned. Of Nothing it speaks only by remotions and by its being given in the conditioned as limit. Nothing is not an object of positive knowledge, not a content of consciousness and not a subject of distinction. The absence of positive knowledge of it does not contradict the indication of it as limit: a contradiction would arise only by confusing levels — if the fixing of the limit were taken for an act of positive cognition.

Method and Result

The method is derivation from one accepted position by the rule of consideration, without symbolic formalization. The result is the chain gathered above: the dependence of the conditioned on the unconditioned is unfolded without a second principle and without violating unconditionedness. The contribution is the removal of the aporia of transition; the data of physics agree with what has been derived but do not prove it and do not serve as its ground (chapter 22).

DISCUSSION

The derived structure of being shows that distinction is the central principle of reality. Differences are inapplicable to the unconditioned; they are given only by the removal of the opacity of dependent information. This eliminates a second principle and makes the dependence of the conditioned on the immutable consistent.

The ontology is in concordance with fundamental physics: quantum processes display properties compatible with the informational ontology and receive a concordant interpretation in it. Cosmology receives an interpretation in which the origin of measures and processes depends not on essence but on the unfolding of distinctions.

Consciousness is not an independent principle but self-knowledge accrued by information with every distinction; this gives ontology and neuroscience a common object of comparison (chapter 26). Reason and intellect are levels of the same consciousness, whose measure is the extent to which the distinctions of a configuration distinguish its own states. The limit character of the structure fixes the boundaries of the conditioned and the completion of the path of distinctions.

Thus the ontology brings the logical, physical and cosmological levels into concordance in the sense indicated in chapters 22–24: the data of physics and cosmology are compatible with what has been derived, but compatibility is not proof, and they do not serve as the ground of the derivation.

Pre-emptive Replies to Critical Objections

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Below are replies to objections against the key positions. The objections are of two kinds. The first ascribes to the unconditioned the structure of the conditioned: two and a relation, subject and action, knower and known, container and contained, before and after. Such objections are answered by one argument — the rule of consideration, derived from the definition of the unconditioned (Appendix, B): everything listed is a determinacy, and no determinacy is applicable to the unconditioned. An objection of this kind demands that the unconditioned be conditioned, and thereby abolishes the subject of the question before the question. The second kind concerns method, the conditioned or the status of adjacent disciplines. These objections are examined on their merits, in their strongest form.

1. The Finiteness of Distinctions

Objection: what grounds the attainability of transparency if the number of blind spots has not been counted? Reply: this is derived in chapter 13. The number of distinctions is finite: the infinite is the unlimited, that is, the unconditioned; a series of distinctions is division, and an infinite series of distinctions is contradictory. Completeness is not a sum of sides: the object of knowledge is one — one's dependence — and knowledge of it can become complete. No new spots appear: the spots are sides of one object, and it does not change, and the removal gives knowledge, not opacity. While there is opacity, the removal does not stop, and self-transparency is attained. The conclusion is in agreement with the physical limits on information in a finite region — the Bekenstein bound and the holographic bound (Bekenstein, 1981; Bousso, 2002) — but only in part: they concern regions of space, not the number of respects in which information knows itself.

2. The Inevitability of the Methodological Framework

Objection: the impossibility of alternatives is shown within the framework, but the inevitability of the framework itself is not proved. Reply: the framework is not arbitrary but the result of exhausting the logical space. The determinacy that change requires can lie in only one of three places: (a) on the side of the unconditioned — unconditionedness is violated; (b) on the side of a second independent principle — uniqueness is violated; (c) on the side of the dependent. There is no fourth place: this is derived by two forks — determinacy lies either on the side of the unconditioned or not; what is not unconditioned is either determined by unconditionedness and by nothing besides, or has what is determined by itself and is thereby a second unconditioned (chapter 4). Refusing to explain is not a strategy: it leaves the aporia unsolved. The present ontology occupies the third place — the only one not excluded by the conditions. The objection would have force given a fourth place; such a place would have to be a third member of one of the two forks, and they have no third member.

A possible objection: the inevitability of the framework is proved only in classical logic; paraconsistent logic admits that the unconditioned both is and is not the source of distinction. Reply: paraconsistent logics only prohibit deriving anything whatever from a contradiction; accepting true contradictions is not paraconsistency but dialetheism. The objection is in substance dialetheic. Moreover, it accepts the very assumption on which the classical dilemma rests — that the unconditioned is the source of distinction (chapter 1). But even within these limits it yields nothing: if the unconditioned both is and is not the source of distinction, then distinction both has and has not that on which it depends; such an unconditioned does not determine whether distinction has that on which it depends, and does not do the work for which it was introduced. Finally, to assert "is and is not", the dialetheist distinguishes the one from the other: without this the judgement has no meaning. The system requires only this — distinction, not classical logic in its entirety.

3. The Status of Physics: Empirical Convergence, Not Logical Dependence

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Objection: the connection with physics is merely illustrative; it is not shown that physics logically requires this ontology. Reply: the ontology does not claim this. Physics is an empirical science and describes the regularities of phenomena. Something else is asserted: the ontology fixes structural features that physics discovers empirically. These are the finiteness of information in a bounded region (the Bekenstein bound), nonlocality (entanglement), the definiteness of the outcome of measurement with a distribution given by the Born rule — collapse being an interpretation, not a finding (chapters 22, 23) — and the energy cost of erasing information (Landauer's principle). The informational reading of elementary states does not belong to such features: it is an interpretation (chapter 22). The conclusion is drawn independently of physical data, and the data agree with it; what in the derivation is testable by physics is stated in chapter 22. Physics here is in concordance with the derivation at the level of reality but is not its ground.

4. The Status of Necessity: Proof Beyond the Framework

Objection: alternatives are excluded only within the chosen framework; is the aporia not solved otherwise in another? Reply: the framework here is set not by the author but by the aporia itself — an immutable unconditioned and a changeable reality. Every ontology that solves this aporia stands in the same framework. The aporia is fundamental not historically but logically: every other philosophical problem — including those of identity and modality in analytic metaphysics — already presupposes distinction. Without distinction one cannot tell one thing from another, establish identity or difference, or fix modality. The question of the conditions of possibility of distinction precedes these problems rather than competing with them. Therefore the aporia is not one formulation among others but the condition of every formulation.

The question of the conditions of possibility of change is obligatory: all thinking that uses distinction already presupposes a changeable reality. Refusing to explain its possibility does not abolish the question but leaves it unanswered. This is not an alternative ontology but a refusal of ontology. Process ontology claims another formulation — the "primacy of becoming". But becoming presupposes the difference of "before" and "after", that is, already contains distinguishedness as a given. The question of its conditions remains unanswered: becoming is declared primary but not explained. This is the same aporia, accepted as an axiom. The framework can be left only in two ways: (a) by abandoning the unconditioned — this leads to an infinite regress of grounds; (b) by refusing to explain changeability — this is a direct refusal of ontological explanation. Therefore "another framework" is either the same one under another name or a refusal of a solution. There is no third.

Uniqueness has a precise sense. Of the three places of determinacy, two are excluded by the conditions, and the solution can lie only in the third — on the side of the dependent (chapter 4). This is proved by exhausting the logical space, not by refuting alternatives from outside. That the derived chain is the realization of the solution is asserted to the extent that each of its links is derived from what has already been derived. A structure that violates at least one condition reproduces the aporia in one of the fixed types of contradiction.

The argument of point 2 applies here as well: every alternative framework formulates statements through distinction, and distinction is the minimal requirement of any logic, the paraconsistent included. Therefore an alternative framework stands on the same requirement with which the system begins — distinction. It may admit a contradictory ground, but it must then show that such a ground does the work of a ground.

5. The Glossary of Terms: A Defence of Logic or the Immunization of the System?

Objection: the meanings of terms fixed in the Glossary are a pre-established restriction of interpretation, a closing of discussion and an immunization of the system against criticism. Reply: the Glossary does not forbid criticism; it forbids the substitution of terms. This is the ordinary procedure of every formalized discipline: mathematics fixes the meaning of

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symbols, logic of connectives, physics of quantities. Criticism is admissible and necessary, but it must deal with what the theory asserts, not with what the critic ascribes to it. The Glossary performs the function of fixing meanings, not of censorship: “Selfhood” means X, not Y; “initiation” means A, not B; “information” means P, not Q. Without such fixation every systematic ontology is vulnerable to the substitution of meanings. Immunization would exist if the Glossary forbade objections on the merits. It does not: all objections on the merits — on the status of Selfhood, the nature of initiation, dependence, the limit, teleology, the relation to physics — are examined here openly.

6. The Totality of the Range of Distinctions

Objection: teleology presupposes the exhaustion of the full range of distinctions. How is it known that it has been exhausted? Reply: exhaustion is determined not by counting but structurally. The criterion is the self-transparency of information: there is no position within information from which anything remains undistinguished. A distinction not grasped by information’s knowledge of itself would be a blind spot; knowledge with a blind spot is not complete, and as long as there is opacity the removal does not stop (chapter 13). Therefore the range is exhausted if and only if there are no blind spots in information’s knowledge of itself. This is not an assumption but the definition of completion. The completeness of the set of distinctions is not postulated in advance but established retrospectively — when information’s knowledge of itself has closed upon itself.

7. The Exclusivity of the Scheme: Exhausting the Space of Strategies

Objection: the negative test shows the stability of the scheme, but not the exhaustion of the space of all models; there may be models not considered. Reply: exclusivity is proved not by going through systems, of which there may be infinitely many, but by going through the places in which determinacy can lie, and their number is finite. Determinacy lies either on the side of the unconditioned or not; what is not unconditioned is either determined by unconditionedness and by nothing besides — the dependent — or has what is determined by itself and is thereby a second unconditioned (chapter 4). There are therefore three places, and no fourth. The first violates unconditionedness, the second uniqueness, the third violates no condition. The models called unconsidered — modal, process, dialectical, relational, ontologies of undifferentiated potency — occupy the first or the second place (chapter 4). The candidates for the place of the dependent — matter, spirit — either have determinacy by themselves or coincide with what has been derived; the dependent is not chosen from a list but derived (chapters 5, 9). An opponent who asserts an unaccounted model must indicate a third member of one of the two forks.

8. The Status of Deduction: Ontological Necessity, Not Formal Derivability

Objection: the system is called deductive, but it contains no axioms, rules of inference or theorems in the formal-logical sense. Reply: the deduction here is ontological, not formal; the criterion of rigour is stability under negative modifications (see “Methodology”). Spinoza used the form “definitions — axioms — propositions”, but his proofs are not formal derivations in the modern sense either. The demand for formal axiomatization of metaphysics is a category mistake.

9. An Exogenous Model without Selfhood: Logical Possibility as a Primitive

Objection: a model “Nothing → logical possibility of distinction → structure” is possible without Selfhood and teleology. If it is consistent, the system is not unique. Reply: this model is not a consistent alternative. “Logical possibility” is a modal category, and modality belongs only to the conditioned: the possibility of succession is given together with the determinacy of information (chapter 11), and it is inapplicable to the unconditioned. Introduced as a primitive, the “possibility of distinction” can lie on only one of three sides. On the side of the ground — modal structure is introduced into it, and unconditionedness is violated. On the side of a second principle — uniqueness is violated. On the side of the

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dependent — it is a possibility in the determined, that is, in information. But information is the mode of being of Nothing's knowledge of itself, and complete knowledge of oneself is self-understanding — Selfhood (chapter 8). Selfhood has not been eliminated but merely left unnamed, and with it initiation and directedness return. In the first two cases the model falls into the space of contradictions of chapter 4; in the third it coincides with this ontology.

10. The Aporia of the Language Describing the Absolute

Objection: a description of the absolute, in which there are no distinctions, is impossible without introducing distinctions by the very act of description. Therefore the system is self-contradictory at the level of language.

Reply: the objection presupposes that a description must reproduce the structure of what is described. But this is required nowhere where limit concepts are at issue. Mathematics works with infinity without reproducing it: the concept is introduced by axioms and rules of inference and is tested by the consistency of derivation within the system, and externally by its applicability in physics and other sciences. This ontology is tested in the same way. Internally: each level is checked for correspondence with the initial conditions, and the violation of any of them at once reproduces the aporia. Externally: all philosophical and religious schools reproduce the aporia, and its removal here is a testable result. But the conditions of this ontology are not arbitrary axioms, as in mathematics, where systems can be chosen. They are derived from the structure of the aporia: three places for determinacy, of which one is not excluded by the conditions, three prohibitions and the conditions that follow from them. The language of description belongs to the dependent and introduces no structure into the absolute — just as mathematical symbols do not introduce infinity into reality but fix the rules of working with it.

The character of the ontological claim is as follows. Mathematics does not claim to describe reality as such. This ontology claims more, but its realism is not naive: it does not say "this is how being is positively constituted". It considers the encountered as conditioned and obtains the unconditioned as its limit; the unconditioned is spoken of only by remotions. The one position accepted without derivation is encountered change; everything else, including the dependence of the conditioned on the unconditioned, is derived. In this the analogy with mathematics holds: both instruments work with limit concepts through rules of inference, not through reproducing the object. The claim to reality is preserved — as a claim to the limit by which the encountered is determined, not to a positive content of the unconditioned. The conclusion that the conditioned depends on the unconditioned cannot be rejected without making use of it: the three possible ways out are examined in chapter 4.

11. The Self-Understanding of Information as an Apparent Second Principle

Objection: the path of distinctions leads to the self-transparency of information — to self-knowledge without blind spots. But precisely this distinguishes Selfhood from information. If at the end of the path information knows itself, the difference disappears, and two principles remain with one property; uniqueness is violated. Reply: the objection confuses coincidence in form with coincidence in ontological status. Selfhood and information differ not by the presence of self-knowledge but by the nature of their existence. Selfhood's knowledge of itself did not arise and was not attained: it is together with transparency, and transparency is unconditionedness. The self-understanding of information is accrued — a state that has a history. In form they would coincide; in nature they are different: temporality is inapplicable to the one, the other would be its result. Moreover, the self-transparency of information is not unconditionedness: information that has understood itself remains dependent, determined and conditioned. No second unconditioned arises.

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12. Determinacy by Absence and Givenness

Objection: from the conditioned being determined by unconditionedness it does not follow that the unconditioned is given in it; the "not-red" is determined through red, but there is no red in it. Reply: the objection confuses givenness with the presence of a part. Givenness here means one thing: X is given in Y if the determinacy of Y can be neither given nor stated without X (Df14). There is no weaker sense, and a stronger one is inapplicable to the unconditioned. In this sense red is given in the "not-red". The difference lies elsewhere: blue is "not-red" only in description, whereas the conditioned has no content other than "not unconditioned"; it is determined by unconditionedness and by nothing besides. Therefore the unconditioned is given not in a description of the conditioned but in its very being (chapter 5).

13. Fundamental Randomness

Objection: quantum mechanics has shown that a single outcome has no cause; hence reality is random at its basis, and there is no direction. Reply: the distribution of outcomes is a fact, while the uncausedness of a single outcome is an axiom of interpretation: the Born rule is a postulate, and the models of Bohm and Everett reproduce the same distributions without it (chapters 3, 23). The axiom itself is contradictory: occurrence by itself is impossible for the dependent, and if an outcome has occurred, it has a cause; the incalculability of the cause is the opacity of the one who considers, not its absence.

14. Actual Infinity

Objection: in mathematics an infinite totality is determinate — its cardinality $\aleph_0$ differs from $\aleph_1$; hence an infinite series of distinctions is not contradictory. Reply: this determinacy is not found but introduced — by the axiom of infinity of set theory (Zermelo, 1908). Every number given in reality is finite; the determinacy of number is "n and not n + 1", and a series whose number of members is no n at all would be unconditioned in number and conditioned in each member (chapter 13). Whoever asserts an infinite series in the conditioned relies on an axiom, not on a fact.

LIMITATIONS OF THE STUDY

The ontology is not tested by experiment directly and does not compete with the natural sciences. The first criterion of its soundness is logical inevitability: an alternative is considered insofar as its acceptance leads to contradiction — a violation of the immutability of the unconditioned, a second principle or a hidden becoming in Nothing. Therefore the "ontologically necessary" differs from the "accepted in the system" by a negative proof of the impossibility of alternatives, not by empirical testing. But the ontology has consequences that experience can refute: it collapses if an actual infinite is presented in the conditioned as a fact, or if the uncausedness of a single outcome is established as a fact rather than accepted as an axiom (chapter 30); it expects intermediate forms between the non-living and the living (chapter 20).

The ontology explains the origin of distinction: it is given by the removal of the opacity of information to itself, proceeding from Selfhood. What is not explained is something else — why the unconditioned is as it is: this question is inapplicable to the unconditioned, for it presupposes a ground for what is by itself.

1. The unconditioned is not subject to empirical confirmation

Nothing is derived logically but is not detected by the means of physical or neuroscientific experiment. Concordance with physics is the compatibility of the derivation with the data at the level of reality, not its ground and not a derivation of physical laws. The system gives no quantitative predictions: coincidences with physics show compatibility, not exclusivity. The derivation is made independently, and the data converge with it. Therefore the ontology does

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not compete with physics in predictions and metrics and does not claim the status of a physical theory. Its task is not to replace scientific models but to clarify their ontological presuppositions and limits. The absence of empirical testing of the unconditioned is not a deficiency: the system operates at a level that precedes measurable quantities and experimental procedures. The sciences remain autonomous in their choice of formalisms, interpretations and empirical criteria. Experiment does not test it directly, but it can be assessed by its internal consistency, its concordance with established scientific facts and the refutable consequences named above.

2. The ontology does not describe the individual properties of consciousness

It shows that consciousness exists and by what it is given, but does not explain the variability of subjective states.

3. Physical processes are treated as forms of information

This limits the application of the ontology in fields where concrete numerical descriptions and experimental metrics are required.

4. Emergent levels depend on the concordance of distinctions

What is necessary is a tendency towards the growth of connectedness in information as a whole (chapter 18); the development of complex forms (reason, systemic interactions) in a particular configuration is not guaranteed and depends on its stability and on proximate grounds.

5. Teleological completion is not a dynamic process

It describes the limit state of the structure but gives no mechanism of transition to it in empirical systems.

6. The ontology is not intended to explain cultural, social or historical processes

It does not explain the content of cultural, social and historical processes and does not derive them from the ontological ground. What is fixed are only structural informational regimes — the concordance and discordance of configurations — which may be found in these processes. Their application to concrete historical material lies beyond the scope of the work and requires separate justification.

7. The derived, the accepted and the reading are distinguished

The rigour of the derivation and the legitimacy of the premises are different questions, and the former does not prove the latter. Accepted without derivation are: the position that reality is and together with it change is encountered (chapter 4); the definitions Df1–Df14 (Appendix); the finitist understanding of the determinacy of number (chapter 13); the rule of method — of the encountered only what is derived from the encountered is accepted (Methodology). Everything else is derived from them; the necessity of what is derived is necessity given what is accepted, not unconditional necessity. Named a reading and a projection, not a derivation, are: that the next paradigm is opposite to the present one, part of the rules of the paradigm of concentration, the physical expression of the singularity and the mechanism of separation by the kind of bonds (chapter 24). Not derived are: the measure and date of exhaustion, the number of stages, the magnitudes of measures (Open Questions).

OPEN QUESTIONS

Here it is stated what the system does not establish, and why. The questions differ in kind: some are closed to all cognition from within Something, others lie outside ontology, still others remain a task.

The measure of the degree of the *I*. The degree of the *I* is determined qualitatively: by the extent to which the distinctions of a configuration distinguish its own states (chapter 12). The system gives no quantitative measure. The measure of integrated information is inapplicable here: it rests on the exclusion postulate, which the system rejects (chapter

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12). To construct a measure compatible with the nesting of the *I* is a task the system leaves open; there is no prohibition of it in the system.
The magnitude of elementary duration. The smallest duration is derived: it is the duration of one change, for the step is indivisible (chapter 15). Its magnitude is not derived: magnitude is a matter of measure, not of derivation. Whether it coincides with the Planck time only physics can decide, and has not yet decided: experiment does not reach this scale. Here the ontology poses a question whose answer lies outside it.
Other bearers of intellect. Any configuration that has reached this level can become a bearer of intellect. By what path it is attained is derived (chapters 20–21). Whether there are bearers of intellect realized outside what is accessible to us, the system does not assert: consciousness detects only what is represented in the structure of distinctions accessible to it (chapter 12).
The number of stages. That the path proceeds by stages is derived: disclosure is discrete, and a stage is determined by the gathered experience of existence (chapter 13). Between the initial and the last stages there may be several. How many there are the system does not establish: the number of stages depends on how much experience each paradigm of a universe can give, and this is not given from within one universe.
The measure of the exhaustion of the new. A stage is completed when the universe's knowledge of itself as a whole, into which the experience of instances of the intellectual *I* enters as nested, can yield nothing new in the given paradigm (chapter 13). How exhaustion is detected otherwise than by the completion itself, the system does not say. From within a stage it is visible only that new sides of the knowledge of dependence are still opening; that there are no more of them can be established only with the completion.
The chemical path to the living. The ontological essence of the transition from the non-living to the living is derived: it is emergent knowledge of the whole, whose "that" becomes its own held state (chapter 20). By what chemical path the transition was realized is decided by proximate grounds, and this is a task for science; ontology shows only that the path found will be a path to a bond that restores itself.
The accepted position. The system stands on one accepted position: reality is, and together with it change is encountered (chapter 4). It is not proved, but it cannot be rejected without making use of distinction: whoever denies it distinguishes his denial from what is affirmed. Everything else, including the dependence of the conditioned on the unconditioned, is derived (chapter 4). This is a boundary common to every system: something in it is accepted; here what is accepted about reality is reduced to one, while the rest of what is accepted is named (Limitations of the Study), and its negation makes use of it. It is objected from the transcendental side that the encountering of change already presupposes the forms of time, space and a subject. But the accepted position asserts only that the difference of the former from the new is encountered; time and space are derived from it as measures (chapter 15), and the one considering is himself something encountered (chapter 4). The forms of intuition are not rejected here but given a place: they are measures derived from change, not conditions preceding it.

CONTRIBUTION OF THE STUDY

1. The fundamental aporia is removed

The dependence of the conditioned on absolute immutability is consistently unfolded on one openly accepted position about reality and explicitly named definitions and rules (Limitations of the Study) — without a second principle and without violating the unconditionedness of Nothing.

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2. A universal structure of distinction is derived
Distinction is knowledge of difference, proceeding by the removal of the opacity of information. This removes the former gaps in explaining the origin of multiplicity and change.
3. The ontology is brought into concordance with fundamental physics
Entanglement, nonlocality, the definiteness of the outcome of measurement and the structure of matter are interpreted as forms of informational dependences, not of material substances; collapse is taken as an interpretation, not as a finding (chapters 22, 23).
4. Consciousness has received a consistent ontological explanation
Consciousness is self-knowledge accrued by information with every distinction; its measure is the extent to which the distinctions of a configuration distinguish its own states, while its presence is given by every distinction.
5. Reason and intellect are explained as levels of consciousness
Their appearance is derived from the development of distinguished informational states without introducing a new ontological level.
6. An integral ontological system is formulated
All levels — from Nothing to teleological completion — are derived successively and form a single consistent construction.
7. The axiom of fundamental randomness is examined
It is shown that the uncausedness of a single outcome is an axiom of interpretation, not a fact: the distribution of outcomes is reproduced without it, and the axiom itself is contradictory — an outcome without a cause would not occur, for occurrence by itself is impossible for the dependent; the incalculability of a cause is not its absence (chapters 3, 23).
8. The finiteness of the conditioned is derived
It is shown that actual infinity is inapplicable to the conditioned: the determinacy of number is “n and not n + 1”, and the determinacy of an infinite totality is introduced by an axiom, not found (chapter 13).
9. The transition from the non-living to the living is derived
The ontological essence of the transition is derived from emergence: the living is a form in which the *I* distinguishes the disturbance of its own bond and restores it from the external. The signs of the living — boundary, exchange, the keeping of the held state, reproduction, adaptation — are derived as necessary (chapters 20, 25).
10. A precise concept of givenness is introduced
It is shown how the unconditioned is given in the conditioned without becoming a part, a property or an event in it: X is given in Y if the determinacy of Y can be neither given nor stated without X (chapter 5).

CONCLUSION

Given one accepted initial position, a consistent ontology is possible only by recognizing the unconditioned, to which inner differences and transitions are inapplicable. Determinacy can thereby lie only on the side of the dependent: the two other places — the unconditioned and an independent second — are excluded by the conditions. Hence are derived: self-understanding as Nothing’s complete knowledge of itself; information as the mode of being of this knowledge and the first dependent; initiation as the removal of its opacity to itself; distinction as knowledge of difference, the appearance of which is change; consciousness, beginning with the first distinction; and all subsequent levels — the *I*, elementary changes and the measures derived from them, interaction and the bond, aggregated forms and emergence, life, reason and intellect — up to the self-transparency of information, to which the path proceeds by stages, while concordance is a choice of the intellectual *I*.

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The ontology gives a consistent way of considering the aporia of the origin of distinctions given the immutable unconditioned and fixes the conditions of admissibility of dependent forms. It is correlated with fundamental physics at the level of the absence of logical contradictions and the convergence of data with the derivation; moreover, it has consequences that experience can refute (chapter 30). Consciousness is described as self-knowledge accumulated by information with each distinction, and reason and intellect as its levels, without a claim to their empirical explanation. Teleological completion is the limit state in which no new distinctions are required: it is the self-transparency of information, the limit towards which the one-sided removal is directed. How the path to the limit is travelled by consciousnesses is the subject of a separate work, Eschatology.
Thus the work gives an integral and consistent ontological description that unites fundamental philosophy, physics, cosmology and the account of life and consciousness into a single system.

APPENDIX. A FORMAL RECONSTRUCTION OF THE DERIVATION

The appendix sets out the skeleton of the derivation separately from the exposition. Notation: P — accepted position; R — rule of consideration; Df — definition; T — derived position. For each position it is indicated from what it follows and the place in the book where the derivation is unfolded. Symbolic logic is not applied: a full formalization of metaphysics requires formalizing “existence” itself and leads into a regress (see “Methodology”). The numbered form makes each step traceable without this contradiction.

A. The Accepted Position

P1. Reality is, and together with it change is encountered. Not derived; taken as given. It cannot be rejected without making use of distinction (chapter 4).
There are no other underived positions about reality; also accepted explicitly are the definitions Df1–Df14, the finitist understanding of the determinacy of number (chapter 13) and the rule of method — of the encountered only what is derived from the encountered is accepted (Methodology).

B. The Rule of Consideration

R. Inapplicable to the unconditioned is everything by which the conditioned is divided within itself: two and the relation, part and whole, bearer and states, subject and action, container and contained, before and after. Applicable is that by which the dependent is constituted: nothing is thereby established about the unconditioned. To the conditioned, information included, the logic of conditionedness applies in full (chapter 4). From Df2: everything listed is a determinacy, and no determinacy is applicable to the unconditioned.

C. Definitions

Df1. The conditioned — that which is determined through difference from another, that is, limited.
Df2. The unconditioned — that to which no determinacy is applicable, and with it no limitation.
Df3. Knowledge — the name for the remotion of concealment, not a property and not a content.
Df4. Selfhood — Nothing taken in self-understanding.
Df5. Information — the mode of being of Nothing’s knowledge of itself in the dependent, that is, the conditioned named by the way the unconditioned is given in it.
Df6. Difference — information’s own determinacy: its difference from that whose mode of being it is; it exists without succession. The difference of this from that within information is the distinguishedness of its states.
Df7. Opacity — information’s not knowing itself to the extent that self-knowledge has not been given to it by the removal.

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Df8. Initiation — the removal of opacity in the dependent, proceeding from Selfhood.
Df9. Distinction — knowledge of difference.
Df10. Change — the onset of a new state of information.
Df11. Consciousness — self-knowledge accrued by information with every distinction.
Df12. The *I* — self-differentiation, given in knowledge of difference.
Df13. Connectedness — the mutual reference of the knowledges of distinctions.
Df14. Givenness — X is given in Y if the determinacy of Y can be neither given nor stated without X; this is not the presence of a part, a property or an event; there is no weaker sense of givenness, and a stronger one is inapplicable to the unconditioned (R).

D. Derivation: The Unconditioned and the Dependent

T1. Everything encountered is conditioned. From P1, Df1: what is encountered is the distinguished, and the distinguished is “this and not that”.
T2. Consideration proceeds by the means of the conditioned. From T1: every affirmation and negation is a distinction.
T3. The conditioned is determined by unconditionedness as its limit and does not reach it; to abolish unconditionedness is to abolish the conditioned. From T1, Df1, Df2: limitation is the absence of unconditionedness, and the conditioned is determined by it and by nothing besides — every other determinant is conditioned, belongs to the conditioned itself and does not determine its conditionedness.
T4. The dependent has nothing by itself. From T3: the totality of dependence; no link of the chain of proximate grounds becomes the limit.
T5. Directedness towards the limit does not follow from determinacy by the limit. From T3.
T6. Nothing is absolutely unconditioned. From T3, Df2. “Is” here is the remotion of non-being, not a property.
T7. The unconditioned is immutable. From T6, R: change is non-identity, and non-identity is limitation.
T8. The unconditioned is unique. From T6, R: a second would limit the first.
T9. The unconditioned is transparent; self-understanding is Selfhood. From T6, R, Df3, Df4.
T10. The mode of being of the unconditioned in the dependent is a mode of knowledge. From T3, Df1, Df14, T2, T4, T9, R: the unconditioned is given in the conditioned — without it the determinacy of the conditioned can be neither given nor stated, for the conditioned is determined by it as limit and by nothing besides; givenness as limit gives all the remotions as lacking, and the question is which can also be had; the remotions are complete — there are as many as there are sides of the initial given; only self-understanding is compatible with the conditioned — to have it does not mean to be undetermined, for knowledge is knowledge of difference; and it is given: distinction proceeds in the conditioned, and the conditioned has nothing by itself; the presence of the unconditioned can only be one-sided (chapter 5).
T11. Information is conditioned and dependent: determined and limited. From Df5, T10, T3, R, Df1: the unconditioned is given in every conditioned, for every conditioned is determined by it wholly; in the unconditioned itself there is no mode; hence information is the conditioned itself, taken from the side of the givenness of the unconditioned in it, and the conditioned is determined by unconditionedness wholly and by nothing besides. T1 yields the corollary: everything encountered is information.
T12. Encountered change is change of information. From P1, T4, T7, T8, T10, Df5: determinacy can lie on the side of the unconditioned, on the side of an independent second, or on the side of the dependent, and there is no fourth place — what is not unconditioned is either determined by unconditionedness and by nothing besides, or has what is determined by itself and is thereby a second unconditioned (chapter 4); change is not in the unconditioned (T7) and not in a second (T8), hence in the dependent, and the dependent is information.

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T13. Distinction in information is not by itself; there is opacity. From P1, T12, T4, Df6, Df7: change is encountered, and only what was not can appear; knowledge of difference does not enter determinacy (chapter 10).
T14. The removal proceeds from Selfhood. From T13, T8: there is no other remover.
T15. Distinction and change are one from two sides; change is onset. From Df9, Df10, T4, T12, T14.
T16. The knowledge of information accrues and does not decrease. From T14, T4, T15: the removal is one-sided; there is no change other than the arrival of knowledge of difference, and what has arrived does not become what has not arrived; there is nothing to restore opacity.
T17. Consciousness begins with the first distinction; with it the *I* is given. From Df11, Df12, T15.
T18. Connectedness makes consciousness one. From Df13: the unity of the source does not give the unity of consciousness.
T19. Self-transparency is attained and remains conditioned. From T16 (chapter 13): the object of knowledge is one — dependence; the spots are places of one opacity; the number of distinctions is finite — the infinite is the unlimited, that is, the unconditioned, and a series of distinctions is division; the object is one.

E. Derivation: The Levels of the Conditioned

T20. The constancy of the removal gives regularity, its one-sidedness the programme. From T14, T16; chapters 10, 13.
T21. Change has extent: the former and the new states are ordered, and there is no third between them; duration is measured by the number of steps; the measures — time, speed, localization, space and energy — are derived from the onset. From T15, T16, the indivisibility of the step (chapter 14), the prohibition of contradiction and the excluded third (chapter 6); chapter 15.
T22. Interaction is the mutual self-differentiation of instances of the *I*; the bond — an interaction that has held — is the agent of the programme. From T17, T20; chapters 16, 18.
T23. An aggregated form is held by a bond together with the repulsion of its parts; its identity is the continuity of the bond. From T22; chapter 17.
T24. Emergence is necessary as a tendency; the new in the whole is connectedness, not produced by the parts. In one universe it is finite: distinctions are finite in number, and its limit coincides with the end of the stage. From T16, T18, T19; chapter 18.
T25. Materiality is the stability of a bond against the proximate grounds of other configurations. From T23; chapter 19.
T26. Life is a form in which the *I* distinguishes the disturbance of its own bond and restores it from the external; the transition to the living is emergent knowledge of the whole, whose “that” becomes its own held state, given separately from the present one; instances of the *I* are nested. From T17, T23, T24; chapters 20, 25.
T27. Reason chooses a reaction within the conditions of the bond; intellect chooses the algorithm itself and self-understanding. From T22, T26; chapter 21.
T28. The path proceeds by stages; a stage is determined by the gathered experience of existence and is completed when the experience of instances of the intellectual *I*, gathered by the universe as a whole, can yield nothing new in the given paradigm. From T15, T19, T24, T27; chapters 10, 13, 16, 18.
T29. The acceptance of dependence on Nothing is a choice of the intellectual *I*. From Df6, T10, T27: knowledge of difference gives that the knower is dependent, while that on which it depends is given only by remotions, and whether to fill it with something conditioned is a choice; transparency, however, is the transparency of the whole (chapter 13); chapter 28.

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T30. The choice is preserved, and by it the sifting of intellectual instances of the *I* takes place at the end of the stage. From T16, T28, T29; chapters 13, 28.

F. Objections Ruled Out

The objections ruled out are directed against a few positions, and each is examined where the position is derived. Against T3 — the "unexplained given" (chapter 4). Against T10 — "determinacy by absence is not givenness" (chapter 5; Discussion, 12). Against T19 — an infinite series of distinctions and actual infinity (chapter 13; Discussion, 1, 14). Against T20 — fundamental randomness (chapters 3, 23; Discussion, 13). Against the uniqueness of the place of the solution — another framework, paraconsistent logic, unexamined models, a model without Selfhood (chapter 4; Discussion, 2, 4, 7, 9). Against T8 and T19 — the self-understanding of information as a second principle (Discussion, 11). Objections that ascribe to the unconditioned the structure of the conditioned are ruled out by the rule R and are not examined on the merits.

Objection	Step	Where examined
Infinite regress of grounds	T3, T4	Chapter 4; Literature Review
The whole of the world as ground (priority monism)	T3	Chapter 4
Modal fatalism	T4, T27	Chapter 11; Literature Review
A second unconditioned	T8	Chapters 4, 8
A subject–object structure in the ground	T9, R	Chapter 8
Why information, and not spirit, energy or structure	T10	Chapter 9
Information requires a bearer	T11, T25	Chapter 19
Motive or arbitrariness of initiation	T14, R	Chapters 10, 11
Distinction as an ultimate fact of the dependent	T13	Chapter 10
Decrease of knowledge through proximate grounds	T16	Chapter 10
The combination problem	T18	Chapter 12
The exclusion postulate	T18, T26	Chapter 12
Causal exclusion	T22, T24	Chapters 18, 19
Weak or strong emergence	T24	Chapter 18
Regress of the bond (Bradley)	T22	Chapter 18
The ship of Theseus	T23	Chapter 17
Panpsychism and the "conscious stone"	T17, T25	Chapter 19
An infinite series of steps to transparency	T19	Chapter 13

G. Dependence on the Accepted Position

The accepted position is one — P1, and the whole derivation stands on it. Everything said about the unconditioned, T3–T10 and T14, is derived from T1 through the definition of limitation: the conditioned is determined by the absence of unconditionedness and by nothing besides (T3). Whoever rejects T3 rejects not something accepted but a derivation, and must indicate an error in it; the three possible ways out are examined and closed (chapter 4), and the derivation cannot be rejected without making use of it.

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H. Figures

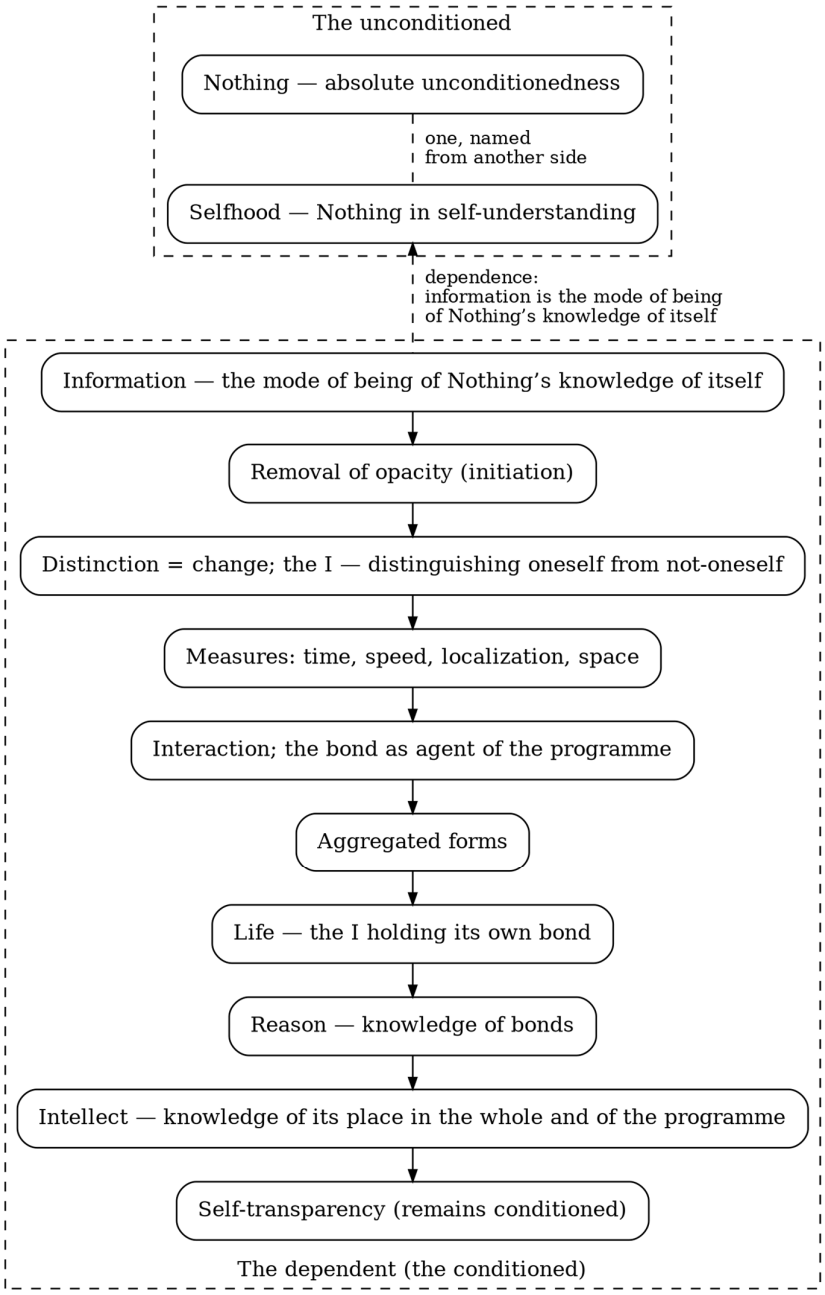


Figure 1. The levels of the ontological structure. The arrows within the dependent indicate the order of derivation, not of arising; the dashed arrow indicates dependence, not generation.

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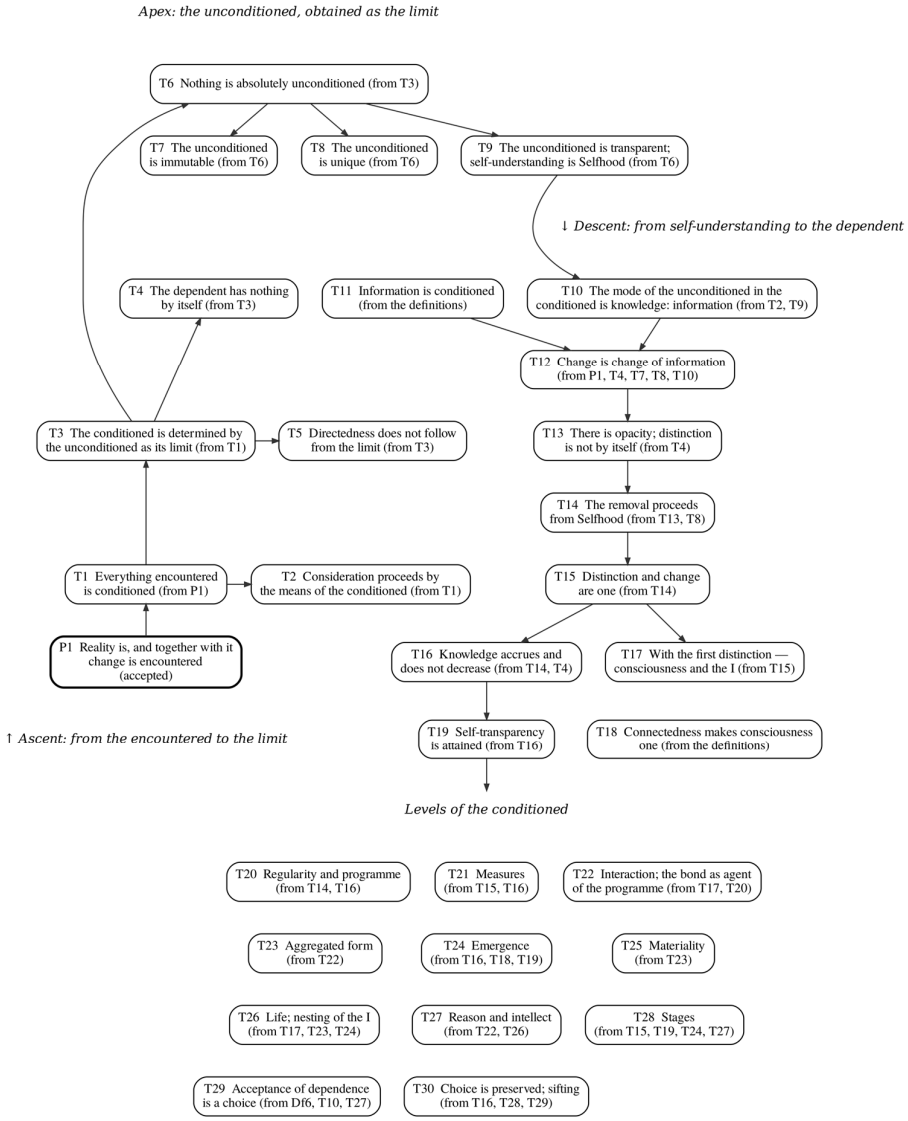


Figure 2. The course of the derivation: the ascent from the encountered to the limit and the descent from self-understanding to the dependent. Each box gives a position and its supports according to sections A–E of the appendix; the accepted position is one — P1.

GLOSSARY OF TERMS

Agent of the programme — a bond between instances of the *I* that sets for each of them its algorithm of functioning; it itself has its own instance of the *I* (chapters 16, 18).

Aggregated form — a stable configuration of elementary changes held by a bond; the bearer of consciousness (chapters 12, 17).

Algorithm of functioning — the way a bond determines the responses of the sides: with the same bond and the same other the response is the same; experience changes the bond, not regularity (chapters 16, 20).

Blind spot — that in which information does not yet know itself.

Bond — an interaction that has held and become constant; it sets a common algorithm of functioning for the sides (chapter 16).

Change — the arrival of a new state of information; it is extended (chapters 9, 15).

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Concordance — knowledge of one's dependence on Nothing, on Selfhood; every understanding of oneself that diverges from it is discordant; it is a choice of the intellectual *I* (chapter 28).
Conditional selfhood — the connected self-knowledge gathered by the *I* in a configuration with one "self" and one "not-self"; it is preserved as connected knowledge after the disintegration of its bearer as well (chapters 12, 17); it bears the name of Selfhood but belongs to the category of the dependent.
Connectedness — the mutual reference of the knowledges of distinctions; it makes consciousness one (chapter 12).
Consciousness — self-knowledge accrued by information with every distinction.
Difference — information's own determinacy: its difference from that whose mode of being it is; it exists without succession. The difference of this from that within information is the distinguishedness of its states (chapters 8, 11).
Distinction — knowledge of difference; taken as that by which this knowledge is given, it is change.
Elementary change — a minimal distinction in information.
Emergence — knowledge of the whole that individual distinctions do not have.
The encountered — what is found as given: reality together with change.
Energy — change taken in concentration: the density of change; the fifth measure (chapter 15).
Experience — memory that has changed the bond, and with it the algorithm of functioning (chapter 20).
Experience of existence — what the *I* knows of itself having passed through distinctions; it determines a stage (chapter 13).
Givenness — X is given in Y if the determinacy of Y can be neither given nor stated without X; not the presence of a part, a property or an event (chapter 5).
The *I* — self-differentiation, given in knowledge of difference; it exists from the first distinction and differs in degree (chapters 11, 12).
Information — the mode of being of Nothing's knowledge of itself in the dependent, that is, the conditioned itself taken from the side of the givenness of the unconditioned in it; the first dependent (chapter 5).
Initiation — the removal of opacity, proceeding from Selfhood; from the side of Selfhood it is not an event.
Intellect — the level of consciousness that knows its place in the whole and the programme.
Interaction — any mutual self-differentiation of instances of the *I*; it may not hold; one that has held is a bond (chapter 16).
Life — a form in which the *I* distinguishes the disturbance of its own bond and restores it from the external; the transition to the living is emergent knowledge of the whole (chapters 20, 25).
Matter — a stable aggregated form of information.
Measures — time, speed, localization, space and energy, derived from the working moment of an elementary change.
Memory — the bond of the *I* of a configuration in its present state with the distinctions it has had; forgetting is a break of this bond, not a decrease of knowledge (chapter 20).
Nothing — the name of absolute unconditionedness; limitation, determinacy and differences are inapplicable to it.
Oblivion — the state of the *I* without a configuration: knowledge is preserved, there are no distinctions (chapters 12, 24).
Opacity — information's not knowing itself to the extent that self-knowledge has not been given to it by the removal.

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Paradigm of boundness — the algorithm of the whole by which its parts function (chapter 18).
Programme — the directedness of the removal of opacity, set by initiation.
Proximate ground — that in the conditioned without which a change would not have set in; the cause within the conditioned (chapters 3, 10).
Reason — the level of consciousness that knows the bonds between distinctions.
Remotion — a statement of what is inapplicable to the unconditioned.
Self-understanding — complete knowledge of oneself; the name for the remotion of non-understanding of oneself.
Selfhood — Nothing taken in self-understanding; not a second beside Nothing.
Sifting — the passing of the concordant and the non-admission of the discordant at the end of a stage; it applies to the intellectual *I* and proceeds by preserved knowledge — how the *I* knew itself (chapters 13, 28).
Singularity — the collapse of the universe by which a stage ends; in it bearers are erased and the sifting takes place (chapter 24).
Stage — a segment of the path of information in one universe, determined by the gathered experience of existence; it is completed when the experience of instances of the intellectual *I*, gathered by the universe as a whole, can yield nothing new in the given paradigm (chapter 13).
Transparency — the inapplicability of opacity; for information, the completeness of self-understanding in its own category.
Unconditionedness — the inapplicability of limitation; the limit by which the conditioned is determined and which it does not reach.

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